MCV4U

Lines and Planes

Website coursebook

24 October 2026

This coursebook follows the classroom sequence in Chapters 8 and 9. Start with line equations in two dimensions, develop plane equations in three dimensions, and then solve intersections, systems and shortest-distance problems.

Work through the written lessons and examples, complete the homework for each section, then take the quiz and test for that unit before attempting the final examination. Every homework problem and assessment question has a typed solution. Written-response questions include model answers and marking criteria.

The online website contains the video lesson slots, interactive three-dimensional visualizer and calculators. The group's original video recordings will be added to those slots. This PDF preserves the written instructional and assessment content.

Open the website

https://mcv4u-lines-and-planes.hozantaxprivate.chatgpt.site

Assessment categories

Knowledge and Understanding (K), Application (A), Thinking (T), and Communication (C).

How to use solutions

Attempt a question first. Compare your method with the worked solution and check your conclusion in the original equations. Multiple-choice answers have a fixed key. Written marks are a guide for self-assessment and may differ from a teacher's awarded mark.

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1 Video lesson sequence

The four recordings follow the unit groups below. Each section has its own homework. Complete the homework for all sections in a unit, then its quiz and test. Open the website for recordings and interactive geometry.

Unit 8.1 Line equations

Sections 8.1, 8.2, 8.3. Complete Homework 8.1, Homework 8.2, Homework 8.3, followed by Quiz 8.1 and Test 8.1.

Unit 8.2 Plane equations and sketching

Sections 8.4, 8.5, 8.6. Complete Homework 8.4, Homework 8.5, Homework 8.6, followed by Quiz 8.2 and Test 8.2.

Unit 9.1 Intersections and systems

Sections 9.1, 9.2, 9.3, 9.4. Complete Homework 9.1, Homework 9.2, Homework 9.3, Homework 9.4, followed by Quiz 9.1 and Test 9.1.

Unit 9.2 Shortest distances

Sections 9.5, 9.6. Complete Homework 9.5, Homework 9.6, followed by Quiz 9.2 and Test 9.2.

1.1 Video 01 Line equations in two and three dimensions

Sections 8.1, 8.2, 8.3. Recording pending.

1. Derive vector form from a known point and parallel direction
2. Compare R^2 and R^3 and interpret the parameter
3. Construct lines from two points and write equivalent equations
4. Convert vector, parametric and symmetric forms
5. Compare coincident and distinct parallel lines
6. Use normals and Cartesian conversions in R^2
7. Handle zero components and check points in R^3

1.2 Video 02 Plane equations and sketches

Sections 8.4, 8.5, 8.6. Recording pending.

1. Derive vector and parametric plane equations
2. Construct planes from valid geometric information
3. Test point membership and independent directions
4. Find normals and convert equation forms
5. Compare plane orientations and calculate angles

6. Plot points and projections in R^3
7. Sketch planes with intercepts and coordinate-plane traces

1.3 Video 03 Intersections and systems

Sections 9.1, 9.2, 9.3, 9.4. Recording pending.

1. Classify line and plane relationships by solving and checking all coordinates.
2. Use elimination to distinguish unique, infinite and inconsistent solution sets.
3. Interpret the common solution of two or three plane equations geometrically.

1.4 Video 04 Shortest distance to lines and planes

Sections 9.5, 9.6. Recording pending.

1. Choose the distance method for the object and dimension.
2. Keep radical results exact before giving a decimal approximation.
3. Verify that a perpendicular foot lies on the target line or plane.

2 Lessons

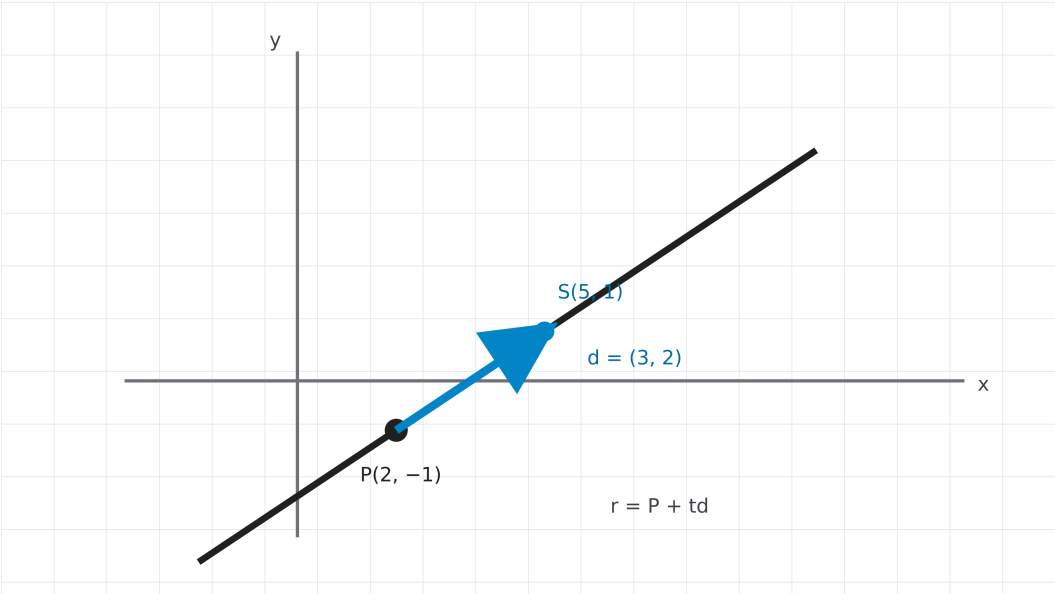
2.1 8.1 Vector and parametric equations of a line in R²

Derive a line equation from a known point and a parallel direction. Write vector and parametric forms, test points, and recognise equivalent equations.

Classroom reference: Chapter 8 slides, PDF pages 3, 5, 6, 9, 10, 11, 12, 13.

Assigned textbook practice: Textbook p. 433: 2, 3, 4, 5, 8, 9, 10, 13(a), 14

Start with a known point P_0 on the line and a nonzero direction vector parallel to it. The position-vector triangle gives $r = r_0 + P_0P$. Since P_0P is a multiple of the direction, $r = r_0 + tm$ reaches every point of the line.



A line in two dimensions uses one anchor and one direction

Why the equation works

1 Start with a line and a known point. Let P_0 be a known point on L, and let m be a nonzero vector parallel to L. Choose any other point P on L.

$$\vec{r}_0 = \overrightarrow{OP_0}, \qquad \vec{r} = \overrightarrow{OP}, \qquad \vec{m} \parallel L, \quad \vec{m} \neq \vec{0}$$

2 Follow the position-vector triangle. Going directly from O to P is the same displacement as going from O to P_0, then from P_0 to P.

$$\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P} \quad \Rightarrow \quad \vec{r} = \vec{r}_0 + \overrightarrow{P_0P}$$

3 Write the movement along the line. The displacement P_0P is parallel to m, so it is a scalar multiple of m. The multiplier may be positive, zero or negative.

$$\overrightarrow{P_0P} = t\vec{m}$$

4 Substitute and read the coordinates. Let t range over all real numbers to reach the whole line. One t produces both coordinates.

$$\vec{r} = \vec{r}_0 + t\vec{m} = [x_0, y_0] + t[a, b], \quad x = x_0 + at, \quad y = y_0 + bt, \quad t \in \mathbb{R}$$

Method

1. Identify a known point on the line and write its position vector.
2. Choose a nonzero parallel direction. If two distinct points are given, subtract the starting point from the ending point.
3. Write r = r_0 + tm, then read the x- and y-components using the same real parameter.
4. Check the known point with t=0. To test another point, require one parameter value to satisfy both coordinates.

Equations and notation

Vector equation

$$\vec{r} = [x_0, y_0] + t[a, b], \qquad t \in \mathbb{R}, \quad [a, b] \neq [0, 0]$$

$P_0(x_0,y_0)$ is a point on the line. The square-bracket vector [a,b] gives a parallel direction.

Parametric equations

$$x = x_0 + at, \qquad y = y_0 + bt$$

The same parameter controls both coordinates.

Direction through two points

$$\vec{d} = \overrightarrow{PQ} = [x_Q - x_P,\ y_Q - y_P]$$

Any non-zero scalar multiple gives the same line.

Perpendicular directions

$$[a, b] \cdot [-b, a] = 0$$

In R^2, swapping the components and changing one sign gives a perpendicular direction. Verify it with a zero dot product.

Common mistake

A line extends in both directions. The parameter is any real number unless a line segment or restricted motion is specified. Do not use a different parameter for each coordinate.

Check your understanding

1. Substituting t=0 returns the anchor.
2. A second known point must be produced by one shared parameter.
3. Different anchors or scaled directions can describe the same line.

Worked example

Write vector and parametric equations for the line through P(2,-1) and Q(8,3). Then determine whether S(5,1) lies on it.

1 Find the direction. Subtract P from Q and simplify by a non-zero factor.

$$\overrightarrow{PQ} = [6, 4] = 2[3, 2]$$

2 Write both forms. Use P as the starting point and [3,2] as the direction.

$$\vec{r} = [2, -1] + t[3, 2], \quad x = 2 + 3t, \quad y = -1 + 2t$$

3 Test S. The x-coordinate requires t=1. Check that value in y.

$$5 = 2 + 3t \Rightarrow t = 1, \qquad -1 + 2(1) = 1$$

4 Verify the original data. The simplified direction reaches Q when t=2.

$$[2, -1] + 2[3, 2] = [8, 3]$$

Conclusion

S lies on the line because one parameter, t=1, satisfies both coordinates.

Infinitely many equations for one line

Use A(1,4) and B(3,1) to write several vector equations of the same line.

1 Find a direction. Subtract the starting point from the ending point.

$$\overrightarrow{AB} = [3 - 1, 1 - 4] = [2, -3]$$

2 Start at either known point. A and B both lie on the line. Either can supply the starting position.

$$\vec{r} = [1, 4] + t[2, -3] \qquad \text{or} \qquad \vec{r} = [3, 1] + u[2, -3]$$

3 Reverse or scale the direction. A line extends in both directions. Any nonzero scalar multiple preserves its direction, though the parameter reaching a point may change.

$$\vec{r} = [1, 4] + v[-2, 3] \qquad \text{or} \qquad \vec{r} = [1, 4] + w[4, -6]$$

4 State the general rule. Choose any starting point on the line and any nonzero multiple of its direction. These choices produce infinitely many representations of one geometric line.

$$\vec{r} = \big([1,4] + a[2,-3]\big) + s\big(k[2,-3]\big), \quad a, s \in \mathbb{R}, \quad k \neq 0$$

Conclusion

Equivalent line equations must have parallel directions and a shared point. Different parameter values or starting points do not automatically mean different lines.

A point that fails the shared-parameter check

Determine whether Q(2,3) lies on r=[1,4]+t[2,-3].

1 Read the coordinates. The same t must give the required x and y together.

$$x = 1 + 2t, \qquad y = 4 - 3t$$

2 Use x. Set x equal to 2 and solve.

$$2 = 1 + 2t \quad \Rightarrow \quad t = \frac{1}{2}$$

3 Check y. The required y gives a different value of t.

$$3 = 4 - 3t \quad \Rightarrow \quad t = \frac{1}{3} \neq \frac{1}{2}$$

Conclusion

Q is not on the line. Matching each coordinate at a different parameter value does not establish point membership.

Continue with Homework 8.1 before the next section.

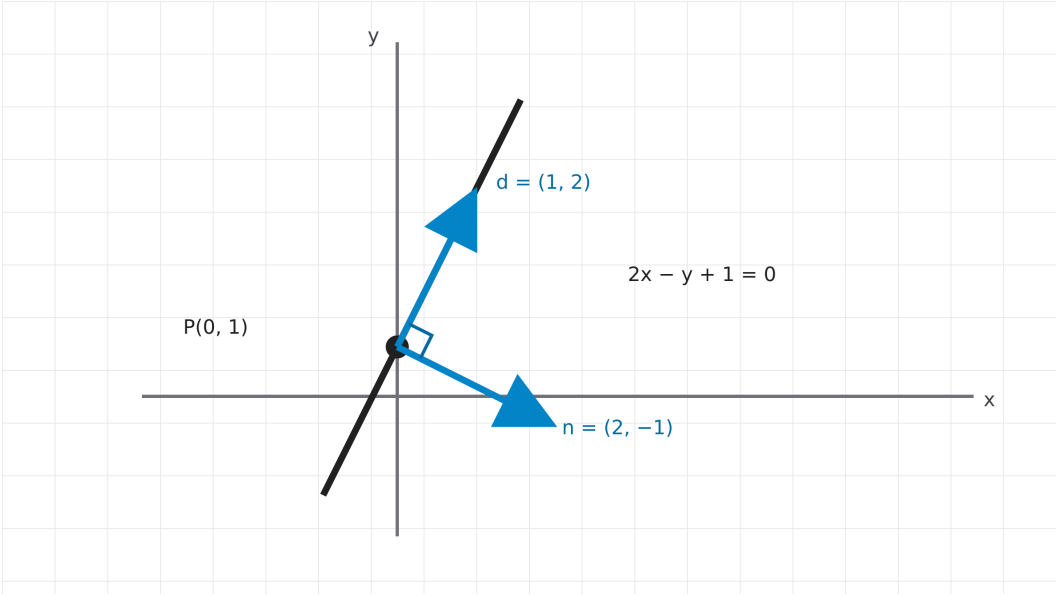
2.2 8.2 Cartesian equations and conversions in R²

Convert among Cartesian, vector, parametric and symmetric forms in R². Use normals to compare lines, handle horizontal or vertical directions, and calculate the smaller angle.

Classroom reference: Chapter 8 slides, PDF pages 14, 15, 17, 18, 19, 20, 21, 22, 23, 24, 25.

Assigned textbook practice: Textbook p. 443: 1(a,b,e), 3, 4, 5, 6, 7, 8, 9(b), 10(a,b), 11(b)

In a Cartesian equation, the coefficients give a normal vector. A line direction is perpendicular to that normal. Converting forms changes the notation, not the set of points.



A Cartesian line has perpendicular direction and normal vectors

Why the equation works

1 Choose a normal and a known point. Let n=[A,B] be a nonzero normal, and let $P_0(x_0,y_0)$ lie on the line. A displacement along the line is perpendicular to n.

$$\overrightarrow{P_0P} = [x - x_0, y - y_0], \qquad \vec{n} \cdot \overrightarrow{P_0P} = 0$$

2 Expand the dot product. Each normal component multiplies its corresponding coordinate displacement.

$$A(x - x_0) + B(y - y_0) = 0$$

3 Collect the constants. Define C from the known point. A and B cannot both be zero.

$$Ax + By + C = 0, \qquad C = -Ax_0 - By_0, \qquad [A, B] \neq [0, 0]$$

Method

1. Read the normal from the coefficients of x and y.
2. Choose a perpendicular direction and find a convenient point that satisfies the equation.
3. Build vector and parametric forms, or eliminate the parameter to recover Cartesian form.
4. For the smaller angle, use the absolute value of the dot product of two directions or two normals.

Equations and notation

Cartesian equation and directions

$$Ax + By + C = 0, \quad [A, B] \neq [0, 0], \quad \vec{n} = [A, B], \quad \vec{d} = [B, -A]$$

A normal is perpendicular to every displacement along the line.

From vector to Cartesian form

$$\vec{n} \cdot [x - x_0, y - y_0] = 0$$

Choose a non-zero normal perpendicular to the supplied direction.

Symmetric form in R²

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} \quad (ab \neq 0)$$

Solve both parametric equations for the same parameter. If a direction component is zero, keep that coordinate as a constant equation instead.

Vertical and horizontal lines

$$x = c \quad \text{or} \quad y = c$$

A zero direction component means that coordinate remains constant.

Smaller angle between lines

$$\theta = \cos^{-1}\left(\frac{|\vec{d_1} \cdot \vec{d_2}|}{\|\vec{d_1}\|\|\vec{d_2}\|}\right), \quad 0^\circ \leq \theta \leq 90^\circ$$

Both directions must be non-zero. Normals may be used in place of directions.

Common mistake

The coefficient vector [A,B] is a normal to Ax+By+C=0. Choose a line direction perpendicular to it. Vertical lines cannot be written with a finite slope, but vector and Cartesian forms still work.

Check your understanding

1. The anchor satisfies the Cartesian equation.
2. The direction has zero dot product with the normal.
3. Multiplying every term of a Cartesian equation by a non-zero constant preserves the line.

Worked example

Convert 2x-3y+6=0 to vector and parametric form, then verify the conversion.

1 Read a normal and choose a direction. The direction must be perpendicular to the normal.

$$\vec{n} = [2, -3], \qquad \vec{d} = [3, 2], \qquad \vec{n} \cdot \vec{d} = 6 - 6 = 0$$

2 Find an anchor. Set x=0 and solve for y.

$$-3y + 6 = 0 \Rightarrow P = (0, 2)$$

3 Write the line. Insert the anchor and direction.

$$\vec{r} = [0, 2] + t[3, 2], \qquad x = 3t, \quad y = 2 + 2t$$

4 Substitute to verify. Every parameter value must satisfy the original equation.

$$2(3t) - 3(2 + 2t) + 6 = 0$$

Conclusion

The Cartesian, vector, and parametric equations describe the same line.

Vector to parametric, symmetric and Cartesian forms

Convert r=[2,-1]+t[3,2] into the other line forms in R^2.

1 Read the components. Use the same real parameter in both coordinate equations.

$$x = 2 + 3t, \qquad y = -1 + 2t$$

2 Make t the subject. Both direction components are nonzero, so both divisions are defined.

$$t = \frac{x - 2}{3} = \frac{y + 1}{2}$$

3 Eliminate the parameter. Cross-multiply the symmetric equation, then expand and collect terms.

$$2(x - 2) = 3(y + 1) \quad \Rightarrow \quad 2x - 3y - 7 = 0$$

4 Check direction and location. The Cartesian coefficients give a normal perpendicular to the original direction, and the starting point satisfies the equation.

$$[2,-3]\cdot[3,2]=0, \qquad 2(2)-3(-1)-7=0$$

Conclusion

All four forms describe the same set of points. To reverse a symmetric conversion, set the common fraction equal to one parameter and rearrange each coordinate equation.

Coincident or distinct parallel lines

Compare L_1: 3x-4y-6=0 with L_2: 6x-8y+12=0.

1 Compare the normals. The second normal is twice the first, so the lines have parallel directions.

$$\vec{n}_2=[6,-8]=2[3,-4]=2\vec{n}_1$$

2 Test a point. Set y=0 in L_1 to obtain P(2,0), then substitute into L_2.

$$3x-6=0\Rightarrow x=2, \qquad 6(2)-8(0)+12=24\neq 0$$

3 Compare the complete equations. Doubling L_1 would give a constant of -12, not +12.

$$2L_1: \quad 6x-8y-12=0$$

Conclusion

The lines are distinct and parallel. In this comparison, distinct means different lines. Parallel directions alone do not prove that the lines coincide.

Exact magnitudes before the angle approximation

Find the smaller angle between lines with directions [-1,3] and [3,4].

1 Calculate the dot product and magnitudes. Keep the radical exact rather than substituting a rounded magnitude.

$$[-1,3]\cdot[3,4]=9, \quad \|[-1,3]\|=\sqrt{10}, \quad \|[3,4]\|=5$$

2 Use the smaller-angle formula. The absolute dot product selects the angle from 0° to 90°.

$$\theta=\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)\approx 55.3^\circ$$

Conclusion

The smaller angle is approximately 55.3°. Use degree mode for the final inverse-cosine calculation.

Continue with Homework 8.2 before the next section.

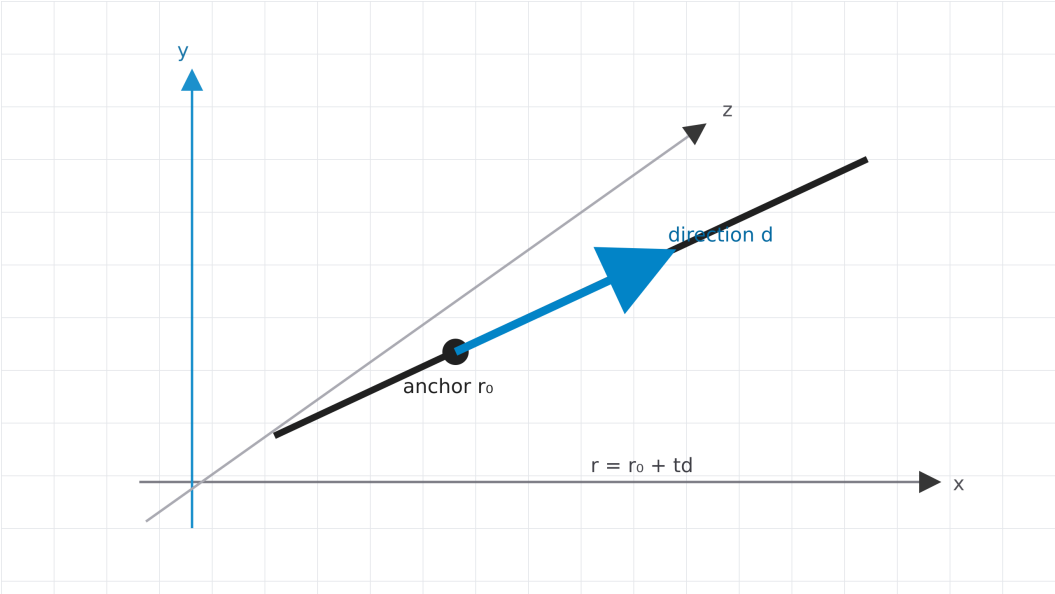
2.3 8.3 Vector parametric and symmetric equations in R³

Extend line equations to three coordinates, convert to symmetric form, and recognise equivalent representations and zero direction components.

Classroom reference: Chapter 8 slides, PDF pages 26, 27, 28, 29, 30, 31, 32, 33, 36, 37.

Assigned textbook practice: Textbook p. 449: 1(a,b,c), 3, 5(a,e,f), 6, 8(a), 9, 10(a,b), 15

A line in three-space still has only one independent direction and one parameter. Three coordinates do not mean three independent parameters.



A line starts at an anchor point and follows a direction vector

Why the equation works

1 Extend the point and direction rule. Adding a z-coordinate adds one component, but a line still has only one independent direction.

$$\vec{r} = [x_0, y_0, z_0] + t[a, b, c], \qquad [a, b, c] \neq [0, 0, 0]$$

2 Read the three coordinate equations. One parameter produces one point in space.

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct$$

3 Equate expressions for t. Use only nonzero direction components as denominators. Retain any fixed coordinates as separate equations.

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \qquad (abc \neq 0)$$

Method

1. Find an anchor and a non-zero direction in three dimensions.
2. Separate the vector equation into three parametric equations.
3. Solve the non-constant coordinate equations for the shared parameter to obtain symmetric form.
4. When a direction component is zero, write the corresponding coordinate as a separate constant equation.

Equations and notation

Vector and parametric forms

$$\vec{r} = [x_0, y_0, z_0] + t[a, b, c], \quad x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$$

Use one real parameter and a direction that is not the zero vector.

Symmetric form

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \qquad (abc \neq 0)$$

These expressions all equal the same parameter.

A zero direction component

$$b = 0 \quad \Rightarrow \quad y = y_0, \qquad \frac{x - x_0}{a} = \frac{z - z_0}{c} \quad (ac \neq 0)$$

If two components are zero, keep two constant-coordinate equations and the remaining coordinate free.

Equivalent line test

$$\vec{d_2} = k\vec{d_1}\ (k \neq 0), \qquad \vec{r}_{02} \in L_1$$

Parallel directions and one shared point establish that the lines coincide.

Common mistake

Never divide by a zero direction component in symmetric form. A single non-degenerate linear Cartesian equation in R³ describes a plane; a Cartesian line needs two independent plane constraints.

Check your understanding

1. The same parameter must reproduce x, y, and z.
2. Parallel directions alone do not prove that two equations describe the same line.
3. A negative scalar multiple reverses the parameter direction without changing the line.

Worked example

Find all three forms of the line through A(1,2,-1) and B(5,2,5). Explain how to handle the unchanged y-coordinate.

1 Find a direction. Subtract A from B and simplify.

$$\overrightarrow{AB} = [4, 0, 6] = 2[2, 0, 3]$$

2 Vector and parametric forms. Anchor at A and retain one shared parameter.

$$\vec{r} = [1, 2, -1] + t[2, 0, 3], \quad x = 1 + 2t, \quad y = 2, \quad z = -1 + 3t$$

3 Symmetric form. Only the x- and z-equations can be divided by their direction components.

$$\frac{x-1}{2} = \frac{z+1}{3}, \qquad y = 2$$

4 Check B. Use t=2 in all coordinates.

$$[1, 2, -1] + 2[2, 0, 3] = [5, 2, 5]$$

Conclusion

The line lies in the plane y=2, but it is only a line because x and z depend on one shared parameter.

A fixed coordinate in symmetric form

Write vector, parametric and symmetric equations through P(-2,3,5) and Q(-2,4,-1).

1 Find a direction. The identical x-coordinates give a zero x-component.

$$\overrightarrow{PQ} = [0, 1, -6]$$

2 Write vector and parametric forms. Start at P and use a single parameter.

$$\vec{r} = [-2, 3, 5] + t[0, 1, -6], \quad x = -2, \quad y = 3 + t, \quad z = 5 - 6t$$

3 Keep the fixed coordinate. Only y and z can be solved for t by division.

$$x = -2, \qquad \frac{y-3}{1} = \frac{z-5}{-6}$$

4 Verify Q. At t=1 all three coordinates agree with Q.

$$[-2, 3, 5] + [0, 1, -6] = [-2, 4, -1]$$

Conclusion

A zero direction component fixes that coordinate. If two components are zero, keep two fixed-coordinate equations and let the remaining coordinate vary.

Converting symmetric form back to vector form

Convert (x-1)/2=(y-3)/(-5)=(z-6)/4 into parametric and vector forms.

1 Name the common value. Set every fraction equal to the same parameter t.

$$\frac{x-1}{2}=t, \qquad \frac{y-3}{-5}=t, \qquad \frac{z-6}{4}=t$$

2 Rearrange each equation. Multiply by each direction component and add the starting coordinate.

$$x=1+2t, \qquad y=3-5t, \qquad z=6+4t$$

3 Group constants and parameter coefficients. Read a starting position and a direction from the coordinate equations.

$$\vec{r}=[1,3,6]+t[2,-5,4]$$

Conclusion

All three representations describe the same line. A single nondegenerate linear Cartesian equation in R³ would describe a plane, so a Cartesian description of a line requires two independent plane conditions together.

Continue with Homework 8.3 before the next section.

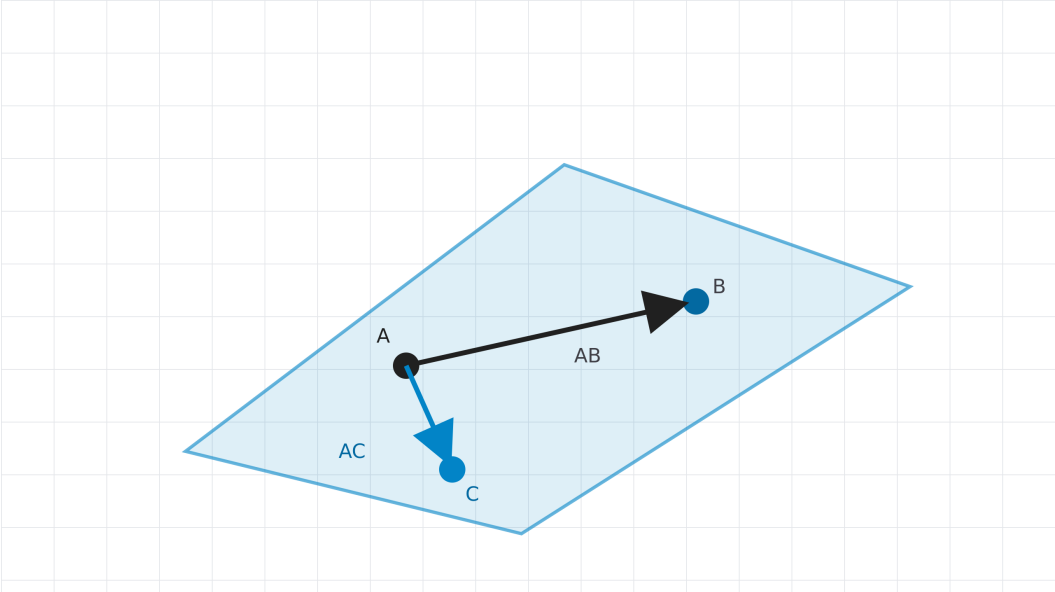
2.4 8.4 Vector and parametric equations of a plane

Construct planes from points and lines. Explain why two independent spanning directions are necessary and identify the four standard geometric definitions.

Classroom reference: Chapter 8 slides, PDF pages 34, 35, 38, 39, 40, 41, 42, 43, 44, 45.

Assigned textbook practice: Textbook p. 459: 1, 2, 4, 5, 6(b), 7, 9, 10, 11, 12, 13(a)

A plane requires an anchor and two independent directions. These directions can come from three non-collinear points, two intersecting lines, two distinct parallel lines, or a line and a point outside it.



A unique plane needs two independent directions

Why the equation works

1 Locate a known point. Let A lie on the plane and let a and b be two nonparallel directions within it. A general point P is reached from A by two independent movements.

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

2 Describe both movements. The real parameters s and t vary independently. Parallel directions would provide only one independent movement.

$$\overrightarrow{AP} = s\vec{a} + t\vec{b}, \qquad \vec{a} \times \vec{b} \neq \vec{0}$$

3 Substitute and read components. The starting position fixes the plane's location. Two independent directions cover its surface.

$$\vec{r} = \vec{r}_0 + s\vec{a} + t\vec{b}, \quad x = x_0 + sa_1 + tb_1, \quad y = y_0 + sa_2 + tb_2, \quad z = z_0 + sa_3 + tb_3$$

Method

1. Choose an anchor on the plane.
2. Build two in-plane vectors from the supplied points or lines.
3. Verify that the vectors are not parallel.
4. Write a vector equation with two independent parameters, then separate its coordinates.

Equations and notation

Vector and parametric plane

$$\vec{r} = \vec{p} + s\vec{a} + t\vec{b}, \quad \vec{a} \times \vec{b} \neq \vec{0}, \qquad x = p_1 + sa_1 + tb_1,\ y = p_2 + sa_2 + tb_2,\ z = p_3 + sa_3 + tb_3$$

Both parameters range independently over the real numbers.

Three non-collinear points

$$\vec{p} = \overrightarrow{OA}, \qquad \vec{a} = \overrightarrow{AB}, \qquad \vec{b} = \overrightarrow{AC}$$

The connecting vectors must not be scalar multiples.

Intersecting or parallel lines

$$\vec{r} = \vec{p} + s\vec{d_1} + t\vec{d_2} \quad \text{or} \quad \vec{r} = \vec{p} + s\vec{d} + t(\vec{q} - \vec{p})$$

For intersecting distinct lines, p is their intersection. For distinct parallel lines, p and q lie on different lines.

A line and an external point

$$\vec{r} = \vec{p} + s\vec{d} + t(\vec{P} - \vec{p})$$

The external point P must not lie on the given line.

Common mistake

Two spans must be independent. Three collinear points, coincident lines, or a line plus a point already on it do not determine a unique plane. Skew lines cannot lie in one plane.

Check your understanding

1. Two independent parameters are needed for a plane.
2. Each given point must be reproducible by some parameter pair.
3. Check the cross product of the spans is non-zero before claiming a unique plane.

Worked example

Find vector and parametric equations for the plane through A(1,0,2), B(2,1,2), and C(1,1,3).

1 Build two spans. Use A as the common starting point.

$$\vec{a} = \overrightarrow{AB} = [1, 1, 0], \qquad \vec{b} = \overrightarrow{AC} = [0, 1, 1]$$

2 Check independence. The vectors are not scalar multiples; their cross product is non-zero.

$$\vec{a} \times \vec{b} = [1, -1, 1] \neq \vec{0}$$

3 Write vector form. Add independent multiples of both spans to A.

$$\vec{r} = [1,0,2] + s[1,1,0] + t[0,1,1], \qquad s,t \in \mathbb{R}$$

4 Separate coordinates. Read the contribution of each parameter to each coordinate.

$$x = 1 + s, \qquad y = s + t, \qquad z = 2 + t$$

Conclusion

The parameter pairs (0,0), (1,0), and (0,1) reproduce A, B, and C respectively.

Two intersecting lines

Find a plane containing r=[2,-1,1]+s[1,2,0] and r=[2,-1,1]+t[0,1,3].

1 Identify their common point. Both lines pass through A(2,-1,1) when their parameters equal zero.

2 Use the two directions. The line directions are not parallel, so they are independent spans.

$$[1,2,0] \times [0,1,3] = [6,-3,1] \neq \vec{0}$$

3 Write vector and parametric forms. Use new independent parameters u and v for the plane.

$$\vec{r} = [2,-1,1] + u[1,2,0] + v[0,1,3], \quad x = 2 + u,\ y = -1 + 2u + v,\ z = 1 + 3v$$

Conclusion

Setting v=0 recovers the first line. Setting u=0 recovers the second line, so the entire pair lies in this plane.

Two distinct parallel lines

Find a plane containing r=[1,0,2]+u[1,1,0] and r=[1,1,3]+v[1,1,0].

1 Keep the common direction. Both lines have direction [1,1,0]. A second copy of it would add no new movement.

2 Join points on different lines. Use the connector from A(1,0,2) to C(1,1,3), which is not parallel to the line direction.

$$\overrightarrow{AC} = [0,1,1] \not\parallel [1,1,0]$$

3 Write the plane. The common direction and connector provide two independent spans.

$$\vec{r} = [1,0,2] + s[1,1,0] + t[0,1,1]$$

4 Check both lines. At t=0 the plane equation gives the first line. At t=1 it gives the second line, with s as its free line parameter.

Conclusion

Distinct parallel lines determine one plane. Coincident lines would supply only one independent direction and would not determine a unique plane.

A line and an external point

Find a plane containing r=[1,2,0]+s[2,0,1] and Q(0,3,2).

1 Find a connector. Use A(1,2,0) on the line. Q has y=3 while every line point has y=2, so Q is outside the line.

$$\overrightarrow{AQ} = [-1, 1, 2]$$

2 Write the plane. Use the line direction and the connector as independent spans.

$$\vec{r} = [1, 2, 0] + u[2, 0, 1] + v[-1, 1, 2]$$

3 Read and verify the coordinates. The pair (0,1) reaches Q. Setting v=0 recovers the whole original line.

$$x = 1 + 2u - v, \qquad y = 2 + v, \qquad z = u + 2v$$

Conclusion

The external-point condition supplies the second independent direction. A point already on the line would not determine a unique plane with it.

Continue with Homework 8.4 before the next section.

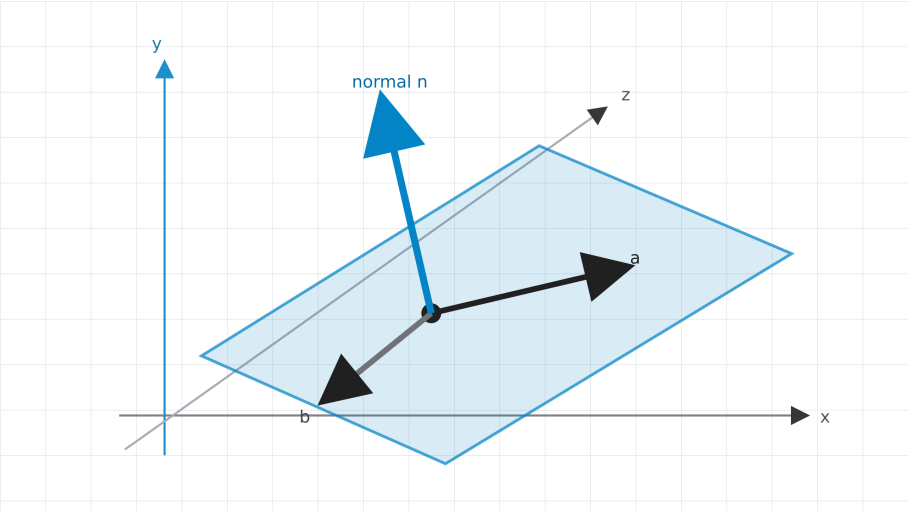
2.5 8.5 Cartesian equations and properties of planes

Derive Cartesian form from a point and normal, convert plane equations in both directions, compare plane orientations, and calculate the smaller angle using exact magnitudes.

Classroom reference: Chapter 8 slides, PDF pages 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58.

Assigned textbook practice: Textbook p. 468: 1, 3, 5, 6(a), 7, 8, 9(a), 10, 11, 12, 13, 15

A plane normal is perpendicular to every displacement inside the plane. The Cartesian coefficients record this normal directly.



Two in-plane directions span the plane and the normal points out of it

Why the equation works

1 Use perpendicularity. Let n=[A,B,C] be a nonzero normal and let $P_0(x_0,y_0,z_0)$ lie on the plane. Every displacement from P_0 to a point P on the plane is perpendicular to n.

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$

2 Expand the components. Multiply matching normal and displacement components.

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

3 Collect constants. The coefficients record the normal and D fixes the plane's position.

$$Ax + By + Cz + D = 0, \qquad D = -Ax_0 - By_0 - Cz_0$$

Method

1. Obtain a non-zero normal directly or by crossing two independent spans.
2. Insert a known point into point-normal form and expand.
3. To recover a vector equation, choose one point and two independent vectors perpendicular to the normal.
4. Use the normals to classify plane orientation and calculate the smaller angle between planes.

Equations and notation

Point-normal and Cartesian forms

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \quad \Longleftrightarrow \quad Ax + By + Cz + D = 0, \qquad \vec{n} = [A, B, C] \neq \vec{0}$$

A normal may be scaled by any non-zero factor if the complete equation is scaled too.

Normal from spans

$$\vec{n} = \vec{a} \times \vec{b}$$

A non-zero cross product is perpendicular to both independent in-plane directions.

Parallel and perpendicular planes

$$\vec{n}_1 \times \vec{n}_2 = \vec{0} \quad \text{or} \quad \vec{n}_1 \cdot \vec{n}_2 = 0$$

The first condition gives parallel orientation. The second gives perpendicular planes.

Smaller angle between planes

$$\theta = \cos^{-1}\left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\|\vec{n}_2\|}\right)$$

Use the absolute value so the reported plane angle lies between 0 and 90 degrees.

Common mistake

Parallel normals show that planes are parallel or coincident, but the constants decide which. A zero normal does not define a plane.

Check your understanding

1. The normal has zero dot product with both spans.
2. The anchor satisfies the final Cartesian equation.
3. When converting back to vector form, the new spans must be independent and perpendicular to the normal.

Worked example

Convert r=[1,0,2]+s[1,1,0]+t[0,1,1] to Cartesian form, then recover a vector form from that equation.

1 Find the normal. Cross the two spans.

$$\vec{n} = [1,1,0] \times [0,1,1] = [1,-1,1]$$

2 Use the anchor. Write point-normal form and expand.

$$(x-1) - y + (z-2) = 0 \quad \Rightarrow \quad x - y + z = 3$$

3 Choose free coordinates. Starting from the Cartesian equation, let y=u and z=v.

$$x = 3 + u - v, \qquad y = u, \qquad z = v$$

4 Recover vector form. Separate constants and parameter contributions.

$$\vec{r} = [3,0,0] + u[1,1,0] + v[-1,0,1]$$

Conclusion

Both vector equations describe the same plane x-y+z=3, even though their anchors and spans differ.

Directly from a point and a normal

Find a Cartesian equation through A(2,-1,3) with normal [1,2,-1].

1 Use point-normal form. Subtract the known point from the general point, keeping the signs in brackets.

$$[1,2,-1] \cdot [x-2, y+1, z-3] = 0$$

2 Expand and simplify. Collect the variable terms and constants.

$$(x-2) + 2(y+1) - (z-3) = 0 \quad \Rightarrow \quad x + 2y - z + 3 = 0$$

3 Verify the point. Substitute the original coordinates.

$$2 + 2(-1) - 3 + 3 = 0$$

Conclusion

The equation has the required normal and contains A. No cross product is needed when the normal is already supplied.

Eliminating both plane parameters

Convert x=1+s, y=s+t, z=2+t directly into Cartesian form.

1 Isolate both parameters. The x- and z-equations give separate expressions for s and t.

$$s = x - 1, \qquad t = z - 2$$

2 Substitute into the remaining coordinate. Replace both parameters in the y-equation.

$$y = (x - 1) + (z - 2)$$

3 Rearrange. The remaining coordinate relationship is Cartesian form.

$$x - y + z = 3$$

Conclusion

Elimination and the cross-product method produce the same plane equation. Two independent parameters are removed together.

Normals, constants and the smaller angle

Compare Π₁: x+2y-z=4 and Π₂: -2x-4y+2z=-8. Then find the smaller angle between x+y+z=2 and x+z=1.

1 Compare complete equations. Every coefficient and the right side of Π₁ is multiplied by -2, so Π₂ is the same plane.

$$[-2, -4, 2] = -2[1, 2, -1], \qquad -8 = -2(4)$$

2 Distinguish a parallel plane. With the same left side but right side +8 instead, the planes would be distinct and parallel.

3 Read the angle normals. The missing y term gives a zero y-component in the second normal.

$$\vec{n}_1 = [1, 1, 1], \qquad \vec{n}_2 = [1, 0, 1]$$

4 Keep the exact ratio. Use the absolute dot product and exact magnitudes before the inverse-cosine approximation.

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{3}\sqrt{2}}\right) = \cos^{-1}\left(\frac{2}{\sqrt{6}}\right) \approx 35.3^\circ$$

Conclusion

Parallel normals need a constant check for coincidence. Perpendicular normals have dot product zero. The smaller plane angle equals the smaller angle between the normals.

Continue with Homework 8.5 before the next section.

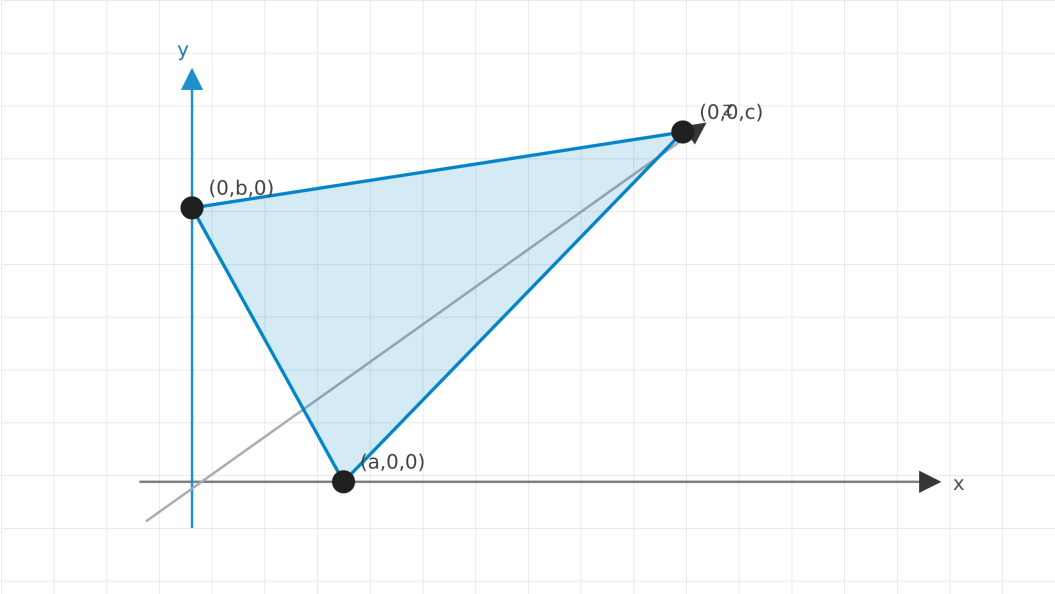
2.6 8.6 Sketching points and planes in R³

Plot ordered triples and use axis intercepts, coordinate-plane traces, or spanning directions to show a plane's orientation.

Classroom reference: Chapter 8 slides, PDF pages 2.

Sketching is listed in the Chapter 8 learning goals. This lesson develops point plotting, coordinate-plane traces and special plane sketches alongside the supplied plane-equation methods.

Plot an ordered triple by following its x-, y- and z-displacements. For a plane, use labelled intercepts, coordinate-plane traces or two independent spans to show its orientation. The drawn patch represents part of an infinite plane.



Three axis intercepts establish the orientation of a plane

Why the equation works

1 Draw labelled coordinate axes. Choose and label the positive x-, y- and z-directions. The slanted axes in a sketch represent three-dimensional space.

2 Locate a point in coordinate order. Move along x, then parallel to y, then parallel to z. Retain light construction lines to show the coordinate displacements.

$$(0,0,0) \longrightarrow (x,0,0) \longrightarrow (x,y,0) \longrightarrow (x,y,z)$$

3 Choose a plane-sketch method. Use three non-collinear intercepts when available. Otherwise find coordinate-plane traces or two independent spans. Label key points and show that the plane continues beyond the drawn patch.

Method

1. Draw labelled x-, y- and z-axes. Locate a point by moving along x, parallel to y, then parallel to z.
2. For a plane, set two coordinates to zero to find each available axis intercept.
3. Set one coordinate to zero for a trace. Use traces or independent spans when an intercept is missing or all intercepts meet at the origin.
4. Label key points and trace lines. Draw a patch with the correct orientation and state that the plane continues beyond it.

Equations and notation

Axis intercepts when defined

$$Ax + By + Cz = D: \quad (D/A, 0, 0),\ (0, D/B, 0),\ (0, 0, D/C)$$

Each fraction is used only when its denominator is non-zero.

Coordinate-plane traces

$$z = 0:\ Ax + By = D; \qquad y = 0:\ Ax + Cz = D; \qquad x = 0:\ By + Cz = D$$

A trace is the intersection of the plane with a coordinate plane.

A plane parallel to an axis

$$x + y = 2$$

The z-coordinate is free. Sketch the line x+y=2 in the xy-plane and extend it in the z-direction.

A plane through the origin

$$x + y + z = 0: \quad \vec{r} = s[1, -1, 0] + t[1, 0, -1]$$

The intercept triangle collapses at the origin, so use traces or two independent spans.

Common mistake

A missing coefficient does not always mean that an entire axis is excluded. If the corresponding intercept equation becomes 0=0, the whole axis lies in the plane; if it becomes a contradiction, the plane misses that axis.

Check your understanding

1. Substitute each labelled intercept into the original equation.
2. A plane patch is not a bounded triangle; the plane extends indefinitely.
3. A coefficient of zero means translation in that coordinate direction leaves the equation unchanged.

Worked example

Sketch 2x+3y+6z=6 and explain what would change when sketching x+y=2.

1 Find the x-intercept. Set y=z=0.

$$2x = 6 \Rightarrow A = (3, 0, 0)$$

2 Find the y- and z-intercepts. Set the other two coordinates to zero each time.

$$B = (0, 2, 0), \qquad C = (0, 0, 1)$$

3 Draw and verify the patch. Join the three non-collinear intercepts and label the axes. The full plane extends beyond the triangle.

$$2(3) = 6, \qquad 3(2) = 6, \qquad 6(1) = 6$$

4 Handle a missing intercept. For x+y=2, the z-axis would require 0=2, so there is no z-intercept. The plane is parallel to the z-axis.

$$\vec{r} = [2, 0, 0] + s[-1, 1, 0] + t[0, 0, 1]$$

Conclusion

Use an intercept triangle when three non-collinear intercepts exist. Otherwise use traces or spanning directions.

Plotting positive and negative coordinates

Plot P(2,-1,3), then identify the projection of R(-1,2,-2) onto the xy-plane.

1 Move along x and y. For P, move 2 along positive x, then 1 parallel to negative y.

$$O(0, 0, 0) \longrightarrow (2, 0, 0) \longrightarrow (2, -1, 0)$$

2 Move vertically. Move 3 along positive z to reach P.

$$(2, -1, 0) \longrightarrow P(2, -1, 3)$$

3 Project R. A projection onto the xy-plane retains x and y and sets z to zero.

$$R(-1, 2, -2) \longrightarrow H(-1, 2, 0)$$

Conclusion

P lies above the xy-plane because z=3. R lies below it because z=-2. The projection H lies in z=0.

The three coordinate-plane traces

Find the traces of 2x+3y+6z=6 and use them to support a sketch.

1 Trace in the xy-plane. Set only z to zero. The resulting line joins the x- and y-intercepts.

$$z = 0: \quad 2x + 3y = 6$$

2 Trace in the xz-plane. Set only y to zero.

$$y = 0: \quad 2x + 6z = 6$$

3 Trace in the yz-plane. Set only x to zero.

$$x = 0: \quad 3y + 6z = 6$$

Conclusion

An axis intercept sets two coordinates to zero. A coordinate-plane trace sets one coordinate to zero and gives an entire line.

A missed axis and a contained axis

Explain the different relationships between the z-axis and the planes x+y=2 and x+y=0.

1 Test the z-axis in the first plane. Every point of the z-axis has x=y=0. Substitution gives a contradiction.

$$x + y = 2: \quad 0 = 2$$

2 Test the z-axis in the second plane. The same substitution now gives an identity for every z.

$$x + y = 0: \quad 0 = 0$$

3 Describe the sketches. Sketch each trace in the xy-plane and extend it parallel to z. The first misses the z-axis; the second contains it.

Conclusion

A missing coefficient makes that coordinate free. Whether the corresponding axis lies in the plane depends on the remaining constant equation.

A plane through the origin

Sketch x+y+z=0 when all axis intercepts meet at O.

1 Find independent traces. The xy-trace is y=-x. The xz-trace is z=-x.

$$z = 0: \ y = -x, \qquad y = 0: \ z = -x$$

2 Choose trace directions. Both directions satisfy the equation and are not scalar multiples.

$$\vec{a} = [1, -1, 0], \qquad \vec{b} = [1, 0, -1]$$

3 Draw the patch through O. Use the directions to show orientation and retain all axis labels.

$$\vec{r} = s[1, -1, 0] + t[1, 0, -1]$$

Conclusion

One collapsed intercept point cannot show the plane's orientation. Two independent trace directions provide the missing geometric information.

Continue with Homework 8.6 before the next section.

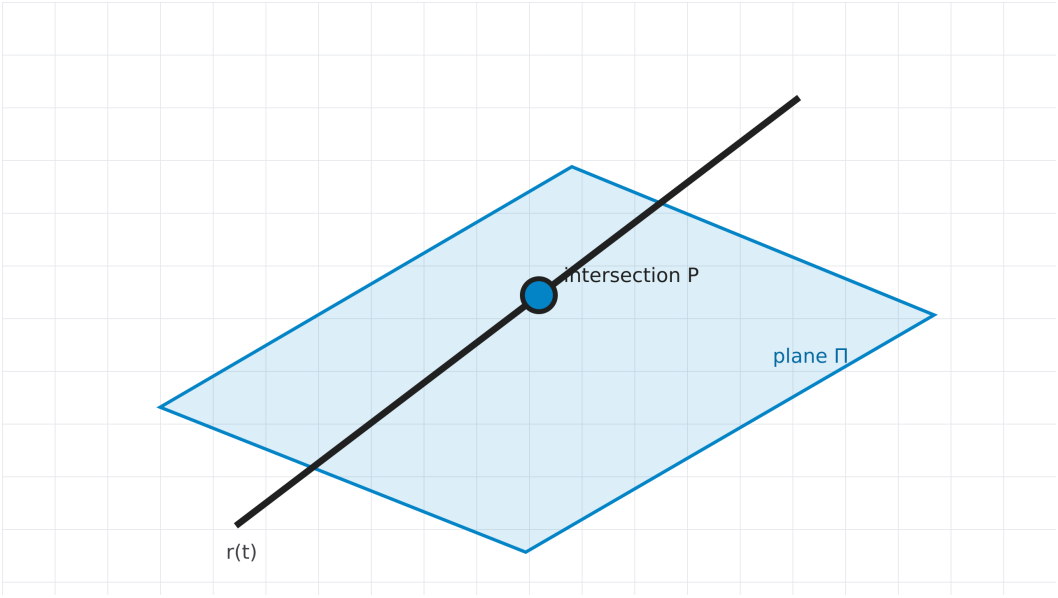
2.7 9.1 Intersections of lines and planes

Classify a line against a plane and compare two lines in R^3, including intersecting, parallel, coincident, and skew cases.

Classroom reference: Chapter 9 slides, PDF pages 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

Assigned textbook practice: Textbook p. 496: 4(a), 5(b), 7, 8(a), 9(a,b), 10, 14(a), 15(a)

An intersection point must satisfy every equation involved. A line-plane problem uses substitution; a two-line problem uses independent parameters and coordinate equations.



The shared point is found by substituting the line into the plane

Method

1. For a line and plane, substitute all parametric coordinates into the plane and solve for the parameter.
2. Interpret one parameter value, an identity, or a contradiction as one point, a contained line, or no intersection.
3. For two lines, use different parameters and equate corresponding coordinates.
4. Check every coordinate and compare directions before classifying the lines.

Equations and notation

A line and a plane

$$\vec{r} = \vec{p} + t\vec{d}, \quad \vec{n} \cdot \vec{r} = k \quad \Rightarrow \quad (\vec{n} \cdot \vec{d})t = k - \vec{n} \cdot \vec{p}$$

A non-zero parameter coefficient gives one intersection. With a zero coefficient, compare the constants.

Contained or disjoint

$$0t = 0 \Rightarrow L \subset \Pi; \qquad 0t = c \ (c \neq 0) \Rightarrow L \cap \Pi = \varnothing$$

The second case is a line parallel to and outside the plane.

Two-line intersection system

$$\vec{p} + s\vec{d_1} = \vec{q} + t\vec{d_2}$$

Solve the coordinate equations for s and t and verify all three.

Two-line classification

$\vec{d_1} \parallel \vec{d_2}:$ coincident or distinct parallel; $\vec{d_1} \nparallel \vec{d_2}:$ intersecting or skew

For parallel directions, test whether an anchor lies on the other line. For non-parallel directions, solve the coordinate system.

Common mistake

Two non-parallel lines need not intersect in R³. If their coordinate system has no solution and their directions are not parallel, the lines are skew. Solving only two coordinate equations is not enough.

Check your understanding

1. The final point must satisfy both objects.
2. An identity means all parameter values work, not just t=0.
3. Skew lines are neither parallel nor intersecting and cannot lie in one plane.

Worked example

Find where r=(1,0,2)+t(1,2,-1) meets x+y+z=5. Then find where r=(0,0,0)+s(1,1,0) meets r=(1,0,0)+u(-1,1,0).

1 Substitute the first line. Write its coordinates in the plane equation.

$$(1 + t) + 2t + (2 - t) = 5 \Rightarrow t = 1$$

2 Find and check the point. Return to the line, then check the plane.

$$P = (2, 2, 1), \qquad 2 + 2 + 1 = 5$$

3 Set up the two-line system. Use separate parameters for the two lines.

$$s = 1 - u, \qquad s = u, \qquad 0 = 0$$

4 Solve and verify. The first two equations give s=u=1/2, and the third coordinate agrees.

$$Q = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Conclusion

The first line crosses its plane at P. The two lines intersect at Q because the same parameter pair satisfies all three coordinate equations.

Continue with Homework 9.1 before the next section.

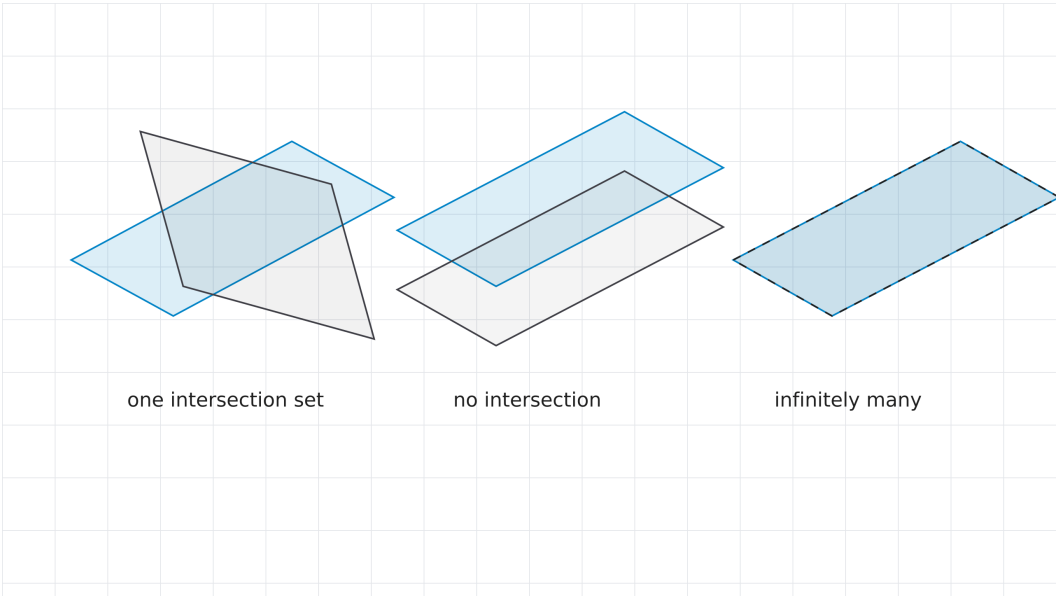
2.8 9.2 Solving systems of linear equations

Use elimination and substitution for up to three equations in three unknowns. Distinguish a unique solution, infinitely many solutions, and an inconsistent system.

Classroom reference: Chapter 9 slides, PDF pages 13, 14, 15, 16, 17.

Assigned textbook practice: Textbook p. 508: 5, 12(a,b,c)

Elimination replaces equations with equivalent restrictions. A contradiction means no common point; an identity removes a repeated restriction; free variables describe all remaining solutions.



Algebraic outcomes match geometric relationships

Method

1. Label the equations and eliminate one variable from two equation pairs.
2. Solve the resulting smaller system, keeping signs and constants aligned.
3. Back-substitute and verify every original equation.
4. If elimination produces an identity or contradiction, state the solution family or explain why no solution exists.

Equations and notation

Unique solution

$$x = a, \qquad y = b, \qquad z = c$$

A consistent system has a unique solution when all three variables are determined.

An identity

$$0 = 0$$

This equation adds no restriction. Use the remaining equations to determine whether free variables remain.

A contradiction

$$0 = c \qquad (c \neq 0)$$

The system is inconsistent and has no solution.

An infinite solution family

$$x + y + z = 3,\ y = 1 \quad \Rightarrow \quad (x, y, z) = (2 - t, 1, t)$$

One free variable gives a line of solutions; two free variables can give a plane.

Common mistake

The equation 0=0 does not prove that every triple is a solution. Keep the other non-redundant equations and express any free variables with parameters.

Check your understanding

1. Multiply every term, including the right side, during elimination.
2. Consistent means at least one solution, including an infinite solution set.
3. A proposed solution must satisfy every original equation.

Worked example

Solve x+y+z=6, 2x-y+z=3, and x+2y-z=2 by elimination.

1 Eliminate x. Subtract twice equation 1 from equation 2, and equation 1 from equation 3.

$$-3y - z = -9, \qquad y - 2z = -4$$

2 Eliminate y. Add three times the second reduced equation to the first.

$$(-3y - z) + 3(y - 2z) = -9 + 3(-4) \Rightarrow -7z = -21 \Rightarrow z = 3$$

3 Back-substitute. Use the reduced equation to find y, then the first original equation to find x.

$$y - 6 = -4 \Rightarrow y = 2, \qquad x + 2 + 3 = 6 \Rightarrow x = 1$$

4 Verify. Check all three original equations.

$$1 + 2 + 3 = 6, \qquad 2 - 2 + 3 = 3, \qquad 1 + 4 - 3 = 2$$

Conclusion

The system is consistent with the unique solution (1,2,3).

Continue with Homework 9.2 before the next section.

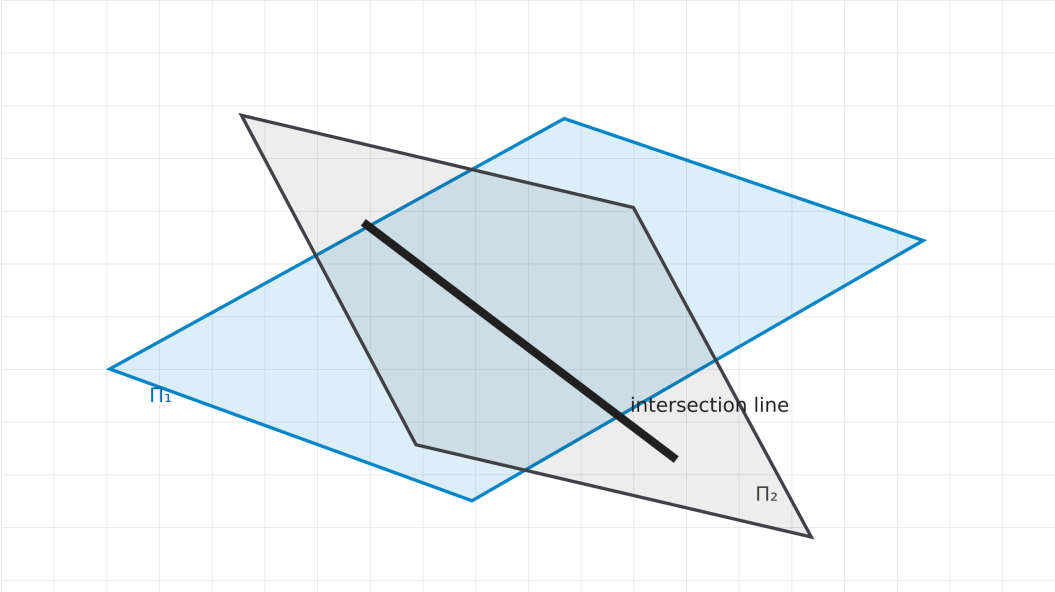
2.9 9.3 Intersections of two planes

Compare plane normals and solve the common system to identify an intersection line, coincident planes, or distinct parallel planes.

Classroom reference: Chapter 9 slides, PDF pages 18, 19, 20, 21, 22, 23, 24, 25, 26.

Assigned textbook practice: Textbook p. 516: 5(a,b), 6 (find the solution when one exists), 9, 11(a)

Two non-parallel planes meet along a line. The line direction is perpendicular to both normals, but one point on both planes is also needed.



Non-parallel planes meet along a line

Method

1. Compare the normals and the complete equations.
2. For non-parallel planes, eliminate one variable and choose a free parameter.
3. Write a vector equation for the common points.
4. Verify the anchor satisfies both planes and the direction is perpendicular to both normals.

Equations and notation

Intersection direction

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 \neq \vec{0}$$

A non-zero cross product establishes non-parallel planes and a line of intersection.

Parallel normals

$$\vec{n}_2 = k\vec{n}_1 \quad (k \neq 0)$$

Scale the entire first equation, including its constant, before comparing with the second.

Coincident or distinct

$$\Pi_1 : Ax+By+Cz = D, \quad \Pi_2 : kAx+kBy+kCz = E, \qquad E = kD \Rightarrow \Pi_1 = \Pi_2, \quad E \neq kD \Rightarrow \Pi_1 \cap \Pi_2 = \varnothing$$

Here k is non-zero. Matching constants give the same plane. Different constants give distinct parallel planes with no intersection.

Common mistake

A cross product gives only the direction of the intersection line. It does not supply the anchor point. Proportional normals alone do not distinguish coincident from distinct parallel planes.

Check your understanding

1. The line anchor lies on both planes.
2. Its direction has zero dot product with both normals.
3. One free parameter describes a line, not a single intersection point.

Worked example

Find a vector equation for the intersection of x+y+z=3 and x-y+z=1.

1 Find y. Subtract the second equation from the first to eliminate x and z.

$$2y = 2 \Rightarrow y = 1$$

2 Choose a free variable. Substitute y=1, then let z=t.

$$x + z = 2 \Rightarrow x = 2 - t, \quad y = 1, \quad z = t$$

3 Write the intersection line. Separate the constant point and direction.

$$\vec{r} = (2, 1, 0) + t(-1, 0, 1), \qquad t \in \mathbb{R}$$

4 Check its direction. Cross the two normals and compare.

$$(1, 1, 1) \times (1, -1, 1) = (2, 0, -2) = -2(-1, 0, 1)$$

Conclusion

The planes intersect in the line r=(2,1,0)+t(-1,0,1).

Continue with Homework 9.3 before the next section.

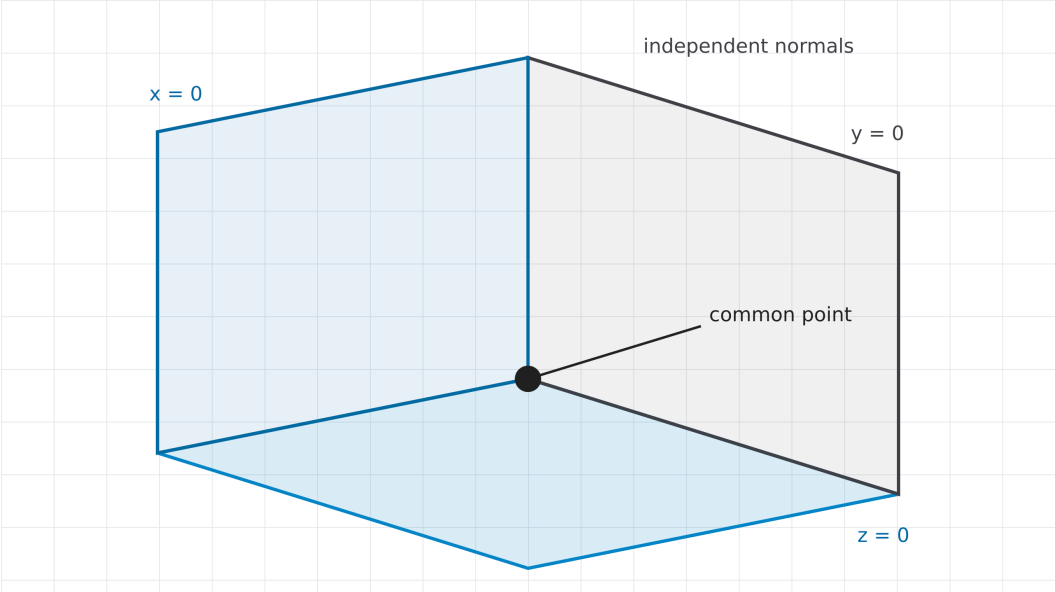
2.10 9.4 Intersections of three planes

Solve a three-plane system and distinguish a common point, common line, common plane, or no common intersection.

Classroom reference: Chapter 9 slides, PDF pages 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37.

Assigned textbook practice: Textbook p. 531: 8, 9, 12, 13

The answer is the set of points shared by all three planes. Pairwise intersections do not guarantee one common point or line across the complete system.



Three independent plane constraints can determine one common point

Method

1. Use elimination to reduce the three Cartesian equations.
2. Look for a contradiction first.
3. If the system is consistent, count the remaining free variables and express the full solution set.
4. Describe the common geometry and verify it in all three original planes.

Equations and notation

A common point

$$x = a, \quad y = b, \quad z = c$$

Three independent restrictions determine one ordered triple.

A common line

$$\vec{r} = \vec{p} + t\vec{d}$$

One free variable remains after consistent elimination.

A common plane

$$\vec{r} = \vec{p} + s\vec{a} + t\vec{b}$$

Two free variables remain when all three equations describe the same plane.

No common intersection

$$0 = c \quad (c \neq 0)$$

A contradiction rules out every common solution, even if some plane pairs intersect.

Common mistake

Three planes can intersect pairwise in three parallel lines while sharing no common point. Do not conclude consistency from pairwise intersections or from non-parallel normals alone.

Check your understanding

1. Use all three equations in the final verification.
2. A redundant equation is not automatically a contradiction.
3. Always state whether the common set is a point, a line, a plane, or empty.

Worked example

Determine the common intersection of x+y+z=6, x-y+z=2, and x+y-z=0.

1 Compare equations 1 and 2. Subtract equation 2 from equation 1.

$$2y = 4 \Rightarrow y = 2$$

2 Compare equations 1 and 3. Subtract equation 3 from equation 1.

$$2z = 6 \Rightarrow z = 3$$

3 Find x. Substitute into the first equation.

$$x + 2 + 3 = 6 \Rightarrow x = 1$$

4 Verify the common point. All three original equations hold.

$$1 + 2 + 3 = 6, \quad 1 - 2 + 3 = 2, \quad 1 + 2 - 3 = 0$$

Conclusion

The three planes have the unique common point (1,2,3), so the system is consistent.

Continue with Homework 9.4 before the next section.

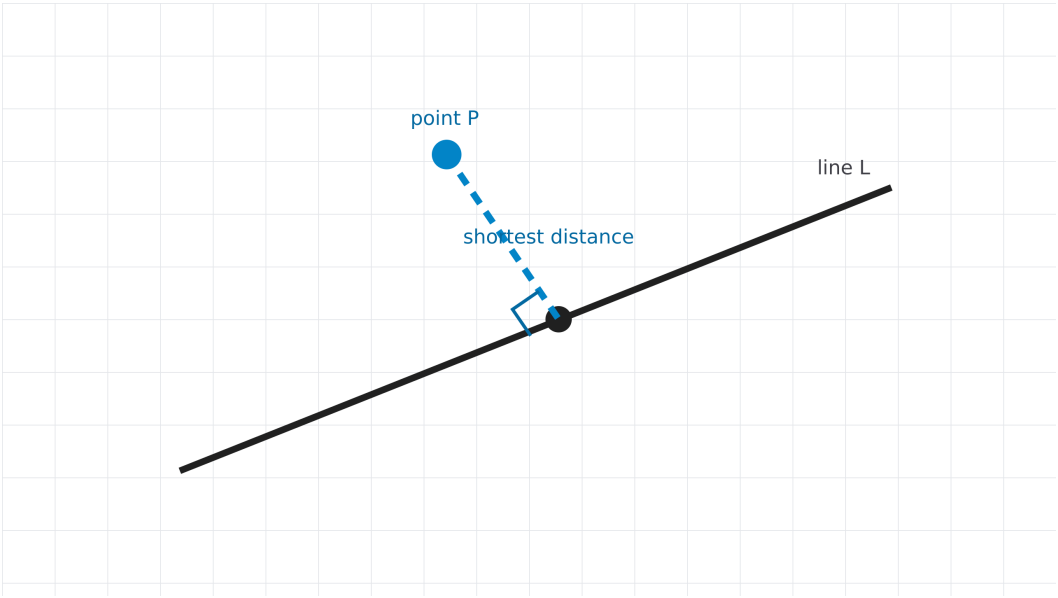
2.11 9.5 Distance from a point to a line in R² and R³

Measure the shortest perpendicular distance using a Cartesian line equation in R² or displacement and direction vectors in R³.

Classroom reference: Chapter 9 slides, PDF pages 38, 39, 40, 41, 42, 43, 44, 45, 46.

Assigned textbook practice: Textbook p. 540: 1(a,b), 2(b), 3(b,c), 5(a), 6(a,b), 7(a)

The displacement from an anchor to the point contains parallel and perpendicular parts. Only the perpendicular part determines the shortest distance.



Distance is measured along the shortest perpendicular segment

Method

1. Identify the dimension and choose the appropriate line representation.
2. In R², substitute the point into the Cartesian line formula.
3. In R³, form the anchor-to-point displacement and use a cross product or projection.
4. Simplify the exact length and state the units; approximate only if requested.

Equations and notation

Point to line in R²

$$d(P, L) = \frac{|Ax_P + By_P + C|}{\sqrt{A^2 + B^2}}, \qquad L : Ax + By + C = 0$$

The denominator is the normal magnitude. A and B cannot both be zero.

Point to line in R³

$$d(P, L) = \frac{\|\overrightarrow{AP} \times \vec{d}\|}{\|\vec{d}\|}$$

A lies on the line, and the non-zero vector d is its direction. The numerator is the parallelogram area.

Projection and foot of the perpendicular

$$t_H = \frac{\overrightarrow{AP} \cdot \vec{d}}{\vec{d} \cdot \vec{d}}, \quad H = A + t_H \vec{d}, \quad d(P, L) = \|P - H\|$$

This method works in both two and three dimensions.

Extension distance between parallel R² lines

$$L_1 : Ax + By + C_1 = 0, \quad L_2 : Ax + By + C_2 = 0 \quad \Rightarrow \quad d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

First make the x- and y-coefficients identical. Alternatively choose a point on one line and find its distance to the other.

Common mistake

The distance from the external point to an arbitrary line anchor is usually longer than the required distance. The final answer is a non-negative scalar, not a vector.

Check your understanding

1. The perpendicular displacement has zero dot product with the line direction.
2. Distance is zero exactly when the point lies on the line.
3. Scaling the direction vector leaves the distance unchanged.

Worked example

Find the distance from P(1,2) to 3x+4y-5=0. Then find the distance from Q(1,0,2) to r=t(1,1,0).

1 Use the R² formula. Substitute P and the Cartesian coefficients.

$$d(P, L) = \frac{|3(1) + 4(2) - 5|}{\sqrt{3^2 + 4^2}} = \frac{6}{5}$$

2 Build the R³ displacement. Use the origin as the second line's anchor.

$$\overrightarrow{AQ} = (1, 0, 2), \qquad \vec{d} = (1, 1, 0)$$

3 Use the cross product. Its magnitude divided by the direction magnitude gives the perpendicular height.

$$\overrightarrow{AQ} \times \vec{d} = (-2, 2, 1), \qquad d(Q, L) = \frac{\sqrt{9}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

4 Check with the perpendicular foot. Project onto the direction and calculate the remaining displacement.

$$t_H = \frac{1}{2}, \quad H = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad \|Q - H\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \frac{3\sqrt{2}}{2}$$

Conclusion

The exact distances are 6/5 units in R^2 and $3\sqrt{2}/2$ units in R^3.

Continue with Homework 9.5 before the next section.

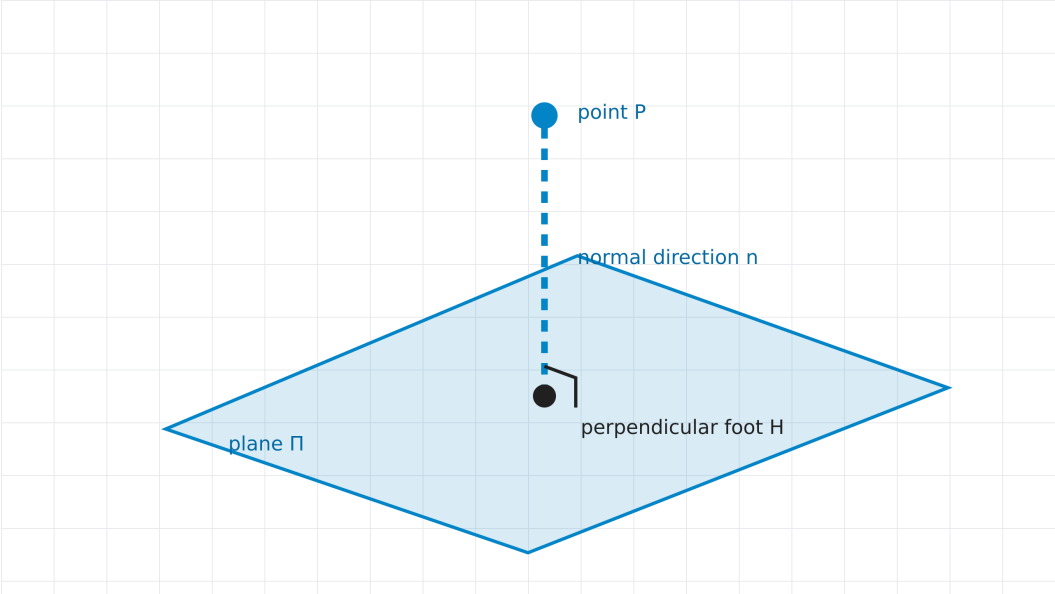
2.12 9.6 Distance from a point to a plane

Use the plane normal to measure a perpendicular distance, find the foot of the perpendicular, and interpret zero or positive distances geometrically.

Classroom reference: Chapter 9 slides, PDF pages 47, 48, 49, 50, 51, 52, 53, 54.

Assigned textbook practice: Textbook p. 549: 2(a,d,e), 3(a), 4(c), 5, 7, 8(a)

The shortest path from a point to a plane is parallel to the plane normal. The scalar equation measures a signed normal displacement before the absolute value turns it into a length.



The shortest path from a point to a plane follows its normal

Method

1. Write the plane as Ax+By+Cz+D=0 and read its non-zero normal.
2. Substitute the point into the numerator and take the absolute value.
3. Divide by the normal magnitude and simplify exactly.
4. If the nearest point is required, move along the normal until the plane equation is satisfied.

Equations and notation

Point to plane

$$d(P,\Pi) = \frac{|Ax_P + By_P + Cz_P + D|}{\sqrt{A^2 + B^2 + C^2}}$$

The normal (A,B,C) must be non-zero.

Projection form

$$d(P,\Pi) = \frac{|\overrightarrow{AP} \cdot \vec{n}|}{\|\vec{n}\|}$$

A is any known point on the plane. This is the magnitude of the scalar projection onto its normal.

Foot of the perpendicular

$$H = P - \frac{Ax_P + By_P + Cz_P + D}{A^2 + B^2 + C^2}(A, B, C)$$

Retain the sign in this expression so the movement is in the correct direction.

Parallel plane extension

$$Ax + By + Cz = D_1, \quad Ax + By + Cz = D_2 \quad \Rightarrow \quad d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

First rewrite both planes with exactly the same normal coefficients.

Extension distance between skew lines

$$L_1 : \vec{r} = \vec{A} + s\vec{d_1}, \quad L_2 : \vec{r} = \vec{B} + t\vec{d_2}, \qquad d = \frac{|(\vec{B} - \vec{A}) \cdot (\vec{d_1} \times \vec{d_2})|}{\|\vec{d_1} \times \vec{d_2}\|}$$

Use this when the directions are not parallel. The cross product is a common perpendicular direction. Intersecting non-parallel lines give distance zero.

Common mistake

Use the complete Cartesian expression, including D, and take the absolute value of the entire numerator. The numerator alone is not the distance unless the normal is a unit vector.

Check your understanding

1. A point on the plane has distance zero.
2. The nearest point satisfies the plane equation.
3. The vector from the nearest point to the external point is parallel to the normal.

Worked example

Find the distance from P(3,1,2) to 2x-y+2z-3=0 and find the nearest point H on the plane.

1 Read the normal and substitute. Keep the entire signed numerator together.

$$\vec{n} = (2, -1, 2), \quad 2(3) - 1 + 2(2) - 3 = 6, \quad \|\vec{n}\| = 3$$

2 Calculate the distance. Take the absolute value and divide by the normal magnitude.

$$d = \frac{|6|}{3} = 2$$

3 Find the perpendicular foot. Move from P along the normal by the signed projection factor.

$$H = (3,1,2) - \frac{6}{9}(2,-1,2) = \left(\frac{5}{3}, \frac{5}{3}, \frac{2}{3}\right)$$

4 Verify the geometry. Check that H lies on the plane and PH has length 2.

$$2\left(\frac{5}{3}\right) - \frac{5}{3} + 2\left(\frac{2}{3}\right) - 3 = 0, \qquad \|P - H\| = 2$$

Conclusion

The point is exactly 2 units from the plane, and H=(5/3,5/3,2/3) is the foot of the perpendicular.

Continue with Homework 9.6 before the next section.

3 Homework and worked solutions

3.1 Homework 8.1 Vector and parametric lines in R²

Build a line from a point and direction, generate points, and justify membership.

Unit 8.1 covers Sections 8.1, 8.2, 8.3. Complete every section homework before Quiz 8.1 and Test 8.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Build and interpret a vector equation

Categories: K, A, C

A line passes through $A(3,-2)$ and is parallel to $\vec{m}=[2,5]$. Derive its vector equation, write parametric equations, and find the points for $t=0,1,-1$.

Hint

The displacement from A to any point of the line is a scalar multiple of m.

Worked solution

1 Derive the form. A position vector reaches A first, then moves along the line.

$$\vec{r}=\overrightarrow{OA}+\overrightarrow{AP}=[3,-2]+t[2,5],\quad t\in\mathbb{R}$$

2 Read components. Use the same parameter in both coordinates.

$$x=3+2t,\qquad y=-2+5t$$

3 Generate and check. Substitute each parameter into both coordinates.

$$t=0:(3,-2),\quad t=1:(5,3),\quad t=-1:(1,-7)$$

Final answer

$\vec{r}=[3,-2]+t[2,5]$, with points $(3,-2),(5,3),(1,-7)$.

Problem 2 A line through two points

Categories: K, A

Find vector and parametric equations through $P(-1,4)$ and $Q(5,-2)$. Use a simplified integer direction and verify Q.

Hint

Subtract P from Q, then divide both direction components by a common nonzero factor.

Worked solution

1 Direction. Subtract the starting point from the ending point.

$$\overrightarrow{PQ}=[6,-6]=6[1,-1]$$

2 Equations. Anchor at P and use the simplified direction.

$$\vec{r}=[-1,4]+t[1,-1],\quad x=-1+t,\quad y=4-t$$

3 Verify Q. The simplified direction needs six parameter steps to reach Q.

$$t = 6: \quad [-1, 4] + 6[1, -1] = [5, -2]$$

Final answer

$\vec{r} = [-1, 4] + t[1, -1]$, $x = -1 + t,\ y = 4 - t$. Q occurs at $t = 6$.

Problem 3 Fractional coordinates and equivalent directions

Categories: A, T, C

The line contains $A(\frac{1}{2}, -3)$ and $B(\frac{3}{4}, \frac{1}{2})$. Find an integer direction and a vector equation. Explain why scaling the direction preserves the line.

Hint

Clear the denominators in the displacement vector, not in the starting point.

Worked solution

1 Subtract coordinates. Keep the fractional differences exact.

$$\overrightarrow{AB} = [\tfrac{1}{4}, \tfrac{7}{2}]$$

2 Scale the direction. Multiply both components by four.

$$4\overrightarrow{AB} = [1, 14], \quad \vec{r} = [\tfrac{1}{2}, -3] + t[1, 14]$$

3 Check the same line. Scaling changes the parameter assigned to B, while preserving the set of points.

$$t = \tfrac{1}{4}: \quad [\tfrac{1}{2}, -3] + \tfrac{1}{4}[1, 14] = [\tfrac{3}{4}, \tfrac{1}{2}]$$

Final answer

Use $[1, 14]$. Every nonzero multiple remains parallel, so the new equation describes the same line.

Problem 4 One parameter for both coordinates

Categories: A, T, C

For $L : \vec{r} = [2, -1] + t[3, 2]$, determine whether $Q(8, 3)$ and $R(5, 2)$ lie on L. Justify both decisions.

Hint

Find t using x and check that value in y.

Worked solution

1 Test Q. Its x-coordinate determines the candidate parameter.

$$8 = 2 + 3t \Rightarrow t = 2, \qquad -1 + 2(2) = 3$$

2 Test R. The x-coordinate requires a different candidate.

$$5 = 2 + 3t \Rightarrow t = 1, \qquad -1 + 2(1) = 1 \neq 2$$

3 State the conclusion. Membership requires one shared parameter, not separate choices for x and y.

$$Q \in L, \qquad R \notin L$$

Final answer

Q lies on L at $t = 2$. R does not lie on L.

Problem 5 Construct a perpendicular line

Categories: A, T

Find a vector equation through $P(6, 5)$ perpendicular to $L : \vec{r} = [4, 1] + s[-3, 2]$. Verify the perpendicularity.

Hint

In R², swap direction components and change one sign.

Worked solution

1 Choose a new direction. A perpendicular direction must have zero dot product with the given direction.

$$[-3, 2] \cdot [a, b] = -3a + 2b = 0$$

2 Select and verify. A convenient nonzero solution is [2,3].

$$[-3, 2] \cdot [2, 3] = -6 + 6 = 0$$

3 Use the required point. The point fixes the location of the new line.

$$\vec{r} = [6, 5] + t[2, 3], \quad t \in \mathbb{R}$$

Final answer

$\vec{r} = [6, 5] + t[2, 3]$. The directions are perpendicular because their dot product is zero.

3.2 Homework 8.2 Cartesian equations and conversions in R²

Connect normals, directions, Cartesian equations, slope, and line relationships.

Unit 8.1 covers Sections 8.1, 8.2, 8.3. Complete every section homework before Quiz 8.1 and Test 8.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Cartesian to vector and parametric form

Categories: K, A

Convert $2x - 3y + 6 = 0$ to vector and parametric equations. Verify the result by substitution.

Hint

Read a normal from the coefficients, then choose a perpendicular direction.

Worked solution

1 Normal and direction. The coefficients give a normal, not a line direction.

$$\vec{n} = [2, -3], \quad \vec{d} = [3, 2], \quad \vec{n} \cdot \vec{d} = 0$$

2 Anchor and equations. Set x=0 to obtain a convenient point.

$$P(0, 2), \quad \vec{r} = [0, 2] + t[3, 2], \quad x = 3t, \quad y = 2 + 2t$$

3 Verify all points. The original equation becomes an identity for every real t.

$$2(3t) - 3(2 + 2t) + 6 = 0$$

Final answer

$\vec{r} = [0, 2] + t[3, 2]$, $x = 3t,\ y = 2 + 2t$.

Problem 2 Vector to symmetric and Cartesian form

Categories: A, C

Convert $\vec{r} = [2, -1] + t[3, 2]$ to symmetric and Cartesian forms. Check the Cartesian normal.

Hint

First separate the coordinates and solve both for t.

Worked solution

1 Parametric form. Read corresponding components.

$$x = 2 + 3t, \qquad y = -1 + 2t$$

2 Eliminate t. Both expressions equal one shared parameter.

$$\frac{x - 2}{3} = \frac{y + 1}{2} \Rightarrow 2(x - 2) = 3(y + 1)$$

3 Cartesian form and check. Expand and verify perpendicularity.

$$2x - 3y - 7 = 0, \qquad [2, -3] \cdot [3, 2] = 0$$

Final answer

Symmetric: $\frac{x-2}{3} = \frac{y+1}{2}$. Cartesian: $2x - 3y - 7 = 0$.

Problem 3 A point and a normal

Categories: K, A

Find the Cartesian equation through $A(4, -2)$ with normal $[5, 3]$. Then state one direction vector.

Hint

Use the normal as the x and y coefficients and solve for the constant.

Worked solution

1 Set up the equation. The unknown constant determines the location.

$$5x + 3y + C = 0$$

2 Substitute A. Use the given point to determine C.

$$5(4) + 3(-2) + C = 0 \Rightarrow C = -14$$

3 Direction and check. Choose a vector perpendicular to the normal.

$$5x + 3y - 14 = 0, \quad \vec{d} = [3, -5], \quad [5, 3] \cdot [3, -5] = 0$$

Final answer

$5x + 3y - 14 = 0$, with direction $[3, -5]$.

Problem 4 Coincident or distinct parallel lines

Categories: T, C

Compare $L_1 : 3x - 4y - 6 = 0$ with $L_2 : 6x - 8y + 12 = 0$ and $L_3 : 6x - 8y - 12 = 0$. Justify both classifications.

Hint

Scale the complete equation, including the constant.

Worked solution

1 Compare normals. Both other normals are twice the first.

$$[6, -8] = 2[3, -4]$$

2 Check coincidence. Twice the entire first equation agrees with L3.

$$2L_1 : \quad 6x - 8y - 12 = 0 \Rightarrow L_1 = L_3$$

3 Distinguish L2. The variable coefficients match but the constants do not.

$$L_2: \quad 6x - 8y = -12, \qquad L_1: \quad 6x - 8y = 12$$

Final answer

L1 and L3 coincide. L1 and L2 are distinct parallel lines.

Problem 5 The acute angle between lines

Categories: A, C

Find the acute angle between lines with directions $\vec{d_1} = [-1, 3]$ and $\vec{d_2} = [3, 4]$. Give the exact cosine ratio before approximating.

Hint

Use the absolute dot product to obtain the smaller line angle.

Worked solution

1 Dot product. Multiply corresponding components and add.

$$\vec{d_1} \cdot \vec{d_2} = (-1)(3) + 3(4) = 9$$

2 Magnitudes. Retain exact radical values.

$$\|\vec{d_1}\| = \sqrt{10}, \quad \|\vec{d_2}\| = 5, \quad \cos\theta = \frac{9}{5\sqrt{10}}$$

3 Angle. Use inverse cosine in degree mode.

$$\theta = \cos^{-1}\left(\frac{9}{5\sqrt{10}}\right) \approx 55.3^\circ$$

Final answer

$\theta \approx 55.3^\circ$, with exact ratio $\cos\theta = \frac{9}{5\sqrt{10}}$.

3.3 Homework 8.3 Vector parametric and symmetric lines in R³

Use one shared parameter in three dimensions, including fixed coordinates and equivalent lines.

Unit 8.1 covers Sections 8.1, 8.2, 8.3. Complete every section homework before Quiz 8.1 and Test 8.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Two points with a fixed coordinate

Categories: K, A

Find vector, parametric, and symmetric equations through $A(1,2,-1)$ and $B(5,2,5)$. Verify B using your simplified direction.

Hint

The equal y-coordinates produce a zero direction component.

Worked solution

1 Direction. Subtract A from B and simplify.

$$\overrightarrow{AB} = [4,0,6] = 2[2,0,3]$$

2 Vector and parametric forms. Retain the fixed y-coordinate.

$$\vec{r} = [1,2,-1] + t[2,0,3], \quad x = 1+2t, \quad y = 2, \quad z = -1+3t$$

3 Symmetric form and verification. Do not divide by the zero y-component.

$$y = 2, \quad \frac{x-1}{2} = \frac{z+1}{3}, \qquad t = 2 \Rightarrow B(5,2,5)$$

Final answer

$\vec{r} = [1,2,-1] + t[2,0,3]$, with $y = 2$ and $\frac{x-1}{2} = \frac{z+1}{3}$.

Problem 2 Reverse a symmetric representation

Categories: K, A

Convert $\frac{x+2}{3} = \frac{y-1}{-2} = \frac{z-4}{5}$ to vector and parametric forms. Identify its anchor and direction.

Hint

Give all three fractions the same parameter.

Worked solution

1 Introduce t. The common value is the shared parameter.

$$\frac{x+2}{3} = t, \quad \frac{y-1}{-2} = t, \quad \frac{z-4}{5} = t$$

2 Rearrange coordinates. Multiply by denominators and isolate each coordinate.

$$x = -2+3t, \quad y = 1-2t, \quad z = 4+5t$$

3 Group components. Separate constant and parameter contributions.

$$\vec{r} = [-2, 1, 4] + t[3, -2, 5]$$

Final answer

Anchor $(-2, 1, 4)$, direction $[3, -2, 5]$, and $\vec{r} = [-2, 1, 4] + t[3, -2, 5]$.

Problem 3 Different equations for one line

Categories: A, T, C

Show that $L_1 : \vec{r} = [-1, 0, 4] + s[-1, 2, 5]$ and $L_2 : \vec{r} = [4, -10, -21] + u[-2, 4, 10]$ describe the same line.

Hint

Check parallel directions and one shared point separately.

Worked solution

1 Direction check. The second direction is a nonzero multiple of the first.

$$[-2, 4, 10] = 2[-1, 2, 5]$$

2 Point check. Use one parameter in all three coordinates of L1.

$$s = -5 : \quad [-1, 0, 4] - 5[-1, 2, 5] = [4, -10, -21]$$

3 Conclusion. Parallel directions and the shared anchor establish coincidence.

$$L_1 = L_2$$

Final answer

The equations describe the same geometric line. Their parameters and starting points differ.

Problem 4 Parallel directions are not enough

Categories: T, C

Classify $L_1 : \vec{r} = [1, 6, 1] + t[-1, 1, 2]$ and $L_2 : \vec{r} = [-3, 10, 12] + u[\frac{1}{2}, -\frac{1}{2}, -1]$. Show a complete point check.

Hint

After matching x and y, verify z before deciding coincidence.

Worked solution

1 Compare directions. The directions are parallel.

$$[\tfrac{1}{2}, -\tfrac{1}{2}, -1] = -\tfrac{1}{2}[-1, 1, 2]$$

2 Test the second anchor. Its x-coordinate requires t=4 in L1.

$$1 - t = -3 \Rightarrow t = 4, \quad y = 6 + 4 = 10, \quad z = 1 + 2(4) = 9$$

3 Conclude. The z-coordinate fails, so the parallel lines share no point.

$$[-3, 10, 9] \neq [-3, 10, 12] \Rightarrow \text{distinct parallel lines}$$

Final answer

The lines are distinct and parallel, because the second anchor is not on L1.

Problem 5 Two zero components and the zero vector

Categories: K, T, C

Describe $L : \vec{r} = [1, 2, 3] + t[0, 0, 4]$ without dividing by zero. Does $P(1, 2, -5)$ lie on L? Explain what changes if the direction is $[0, 0, 0]$.

Hint

Two coordinates stay fixed, while the nonzero component allows the third to vary.

Worked solution

1 Read coordinates. Only z depends on t.

$$x = 1, \quad y = 2, \quad z = 3 + 4t$$

2 Test P. The fixed coordinates agree and one value reaches the required z.

$$-5 = 3 + 4t \Rightarrow t = -2$$

3 Interpret the special cases. The free z-coordinate describes a line parallel to the z-axis. A zero direction generates only one point.

$$L : \ x = 1, \ y = 2, \ z \in \mathbb{R}; \qquad \vec{r} = [1, 2, 3] + t\vec{0} = [1, 2, 3]$$

Final answer

P lies on L at $t = -2$. The zero direction would produce only $(1, 2, 3)$, not a line.

3.4 Homework 8.4 Vector and parametric equations of a plane

Construct a plane from valid geometric information and verify independent directions.

Unit 8.2 covers Sections 8.4, 8.5, 8.6. Complete every section homework before Quiz 8.2 and Test 8.2.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 A plane through three points

Categories: K, A

Find vector and parametric equations through $A(1,0,2)$, $B(2,1,2)$, and $C(1,1,3)$. Verify all three points.

Hint

Use AB and AC from the same starting point.

Worked solution

1 Build spans. Subtract A from each other point.

$$\vec{a} = \overrightarrow{AB} = [1,1,0], \quad \vec{b} = \overrightarrow{AC} = [0,1,1]$$

2 Write equations. The spans are not scalar multiples, so use two independent parameters.

$$\vec{r} = [1,0,2] + s[1,1,0] + t[0,1,1], \quad x = 1+s,\ y = s+t,\ z = 2+t$$

3 Verify points. One parameter pair must work in all three coordinates.

$$(s,t) = (0,0) : A, \quad (1,0) : B, \quad (0,1) : C$$

Final answer

$\vec{r} = [1,0,2] + s[1,1,0] + t[0,1,1]$, with $s,t \in \mathbb{R}$.

Problem 2 Two intersecting lines

Categories: A, C

Find vector and parametric equations for the plane containing $L_1 : \vec{r} = [2,-1,1] + s[1,2,0]$ and $L_2 : \vec{r} = [2,-1,1] + t[0,1,3]$. Verify that each line lies in the plane.

Hint

The common starting point is the intersection, and the line directions supply the spans.

Worked solution

1 Check the data. Both lines contain P and their directions are not parallel.

$$P(2,-1,1), \quad \vec{a} = [1,2,0], \quad \vec{b} = [0,1,3]$$

2 Construct the plane. Use independent parameters u and v.

$$\vec{r} = [2,-1,1] + u[1,2,0] + v[0,1,3], \quad x = 2+u,\ y = -1+2u+v,\ z = 1+3v$$

3 Verify whole lines. Fix one plane parameter at zero and vary the other.

$$v = 0 \Rightarrow L_1, \qquad u = 0 \Rightarrow L_2$$

Final answer

$\vec{r} = [2, -1, 1] + u[1, 2, 0] + v[0, 1, 3]$.

Problem 3 Two distinct parallel lines

Categories: A, T, C

Find a plane containing $L_1 : \vec{r} = [1, 0, 2] + s[1, 1, 0]$ and $L_2 : \vec{r} = [1, 1, 3] + t[1, 1, 0]$. Explain where the second independent direction comes from.

Hint

Use the shared line direction and a connector between the two anchors.

Worked solution

1 Connector. Join A on L1 to B on L2.

$$\overrightarrow{AB} = [1, 1, 3] - [1, 0, 2] = [0, 1, 1]$$

2 Independence. The connector is not parallel to the shared direction.

$$[1, 1, 0] \times [0, 1, 1] = [1, -1, 1] \neq \vec{0}$$

3 Plane and checks. The second parameter selects the parallel line, while the first moves along it.

$$\vec{r} = [1, 0, 2] + u[1, 1, 0] + v[0, 1, 1]; \quad v = 0 : L_1, \quad v = 1 : L_2$$

Final answer

The connector supplies the second span, giving $\vec{r} = [1, 0, 2] + u[1, 1, 0] + v[0, 1, 1]$.

Problem 4 A line and an external point

Categories: A, T

Construct a plane containing $L : \vec{r} = [1, 2, 0] + s[2, 0, 1]$ and $Q(0, 3, 2)$. Give vector and parametric forms and verify both requirements.

Hint

Connect the line anchor A to Q.

Worked solution

1 Second span. The connector has a nonzero y-component, unlike the line direction.

$$\overrightarrow{AQ} = [-1, 1, 2] \nparallel [2, 0, 1]$$

2 Equations. Use A and the two independent spans.

$$\vec{r} = [1, 2, 0] + u[2, 0, 1] + v[-1, 1, 2], \quad x = 1 + 2u - v,\ y = 2 + v,\ z = u + 2v$$

3 Verify. The plane includes every point of L and also Q.

$$v = 0 \Rightarrow L, \qquad (u, v) = (0, 1) \Rightarrow Q$$

Final answer

$\vec{r} = [1, 2, 0] + u[2, 0, 1] + v[-1, 1, 2]$.

Problem 5 When three points fail to determine a unique plane

Categories: T, C

Do $A(0, 1, 0)$, $B(2, 1, 1)$, and $C(4, 1, 2)$ determine a unique plane? Justify and give two different planes containing all three points.

Hint

Check whether the connecting directions are scalar multiples.

Worked solution

1 Directions. Subtract A from B and C.

$$\overrightarrow{AB} = [2, 0, 1], \quad \overrightarrow{AC} = [4, 0, 2] = 2\overrightarrow{AB}$$

2 Interpret. Only one independent direction is supplied, so the points are collinear.

$$\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$$

3 Show non-uniqueness. Both equations hold for every listed point but describe different planes.

$$y = 1, \qquad x - 2z = 0$$

Final answer

No unique plane is determined. Infinitely many planes contain the common line, including $y = 1$ and $x - 2z = 0$.

3.5 Homework 8.5 Cartesian equations and plane properties

Use normals, convert plane forms, and justify plane orientations and angles.

Unit 8.2 covers Sections 8.4, 8.5, 8.6. Complete every section homework before Quiz 8.2 and Test 8.2.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Vector to Cartesian using a cross product

Categories: K, A

Convert $\vec{r} = [1,0,2]+s[1,1,0]+t[0,1,1]$ to Cartesian form. Verify that the normal is perpendicular to both spans.

Hint

Cross the two independent directions, then use the anchor.

Worked solution

1 Normal. Compute the cross product in a fixed order.

$$\vec{n} = [1,1,0] \times [0,1,1] = [1,-1,1]$$

2 Point-normal form. Use A(1,0,2).

$$[1,-1,1] \cdot [x-1, y, z-2] = 0 \Rightarrow x - y + z = 3$$

3 Verify the normal. Both span dot products are zero.

$$[1,-1,1] \cdot [1,1,0] = 0, \qquad [1,-1,1] \cdot [0,1,1] = 0$$

Final answer

$x - y + z = 3$, with normal $[1,-1,1]$.

Problem 2 Cartesian back to vector and parametric form

Categories: A, C

Convert $2x + y - z = 4$ to vector and parametric form using integer span vectors. Check that every generated point satisfies the Cartesian equation.

Hint

Let y and z be twice independent parameters to avoid fractional spans.

Worked solution

1 Choose free coordinates. Solve for x after setting y=2s and z=2t.

$$y = 2s, \quad z = 2t, \quad x = 2 - s + t$$

2 Group components. Separate constants and parameter contributions.

$$\vec{r} = [2,0,0] + s[-1,2,0] + t[1,0,2]$$

3 Verify. Substitution reduces to the required identity.

$$2(2-s+t)+2s-2t=4$$

Final answer

$\vec{r}=[2,0,0]+s[-1,2,0]+t[1,0,2]$. Parametric: $x=2-s+t,\ y=2s,\ z=2t$.

Problem 3 Cartesian equation from a point and normal

Categories: K, A, C

Find a Cartesian equation through $A(2,-1,3)$ with normal $[1,2,-1]$. Verify A and identify why the normal must be nonzero.

Hint

Dot the normal with the displacement from A to a general point.

Worked solution

1 Point-normal form. Subtract A component by component.

$$[1,2,-1]\cdot[x-2,y+1,z-3]=0$$

2 Expand. Keep signs inside brackets until multiplication is complete.

$$(x-2)+2(y+1)-(z-3)=0\Rightarrow x+2y-z+3=0$$

3 Check and interpret. A satisfies the equation. A zero normal would give no plane orientation.

$$2+2(-1)-3+3=0,\qquad \vec{n}=[1,2,-1]\neq\vec{0}$$

Final answer

$x+2y-z+3=0$. A nonzero normal is needed to define a plane.

Problem 4 Classify plane relationships

Categories: T, C

Compare $\Pi_1: x+2y-z=4$ with $\Pi_2: -2x-4y+2z=-8$, $\Pi_3: -2x-4y+2z=8$, and $\Pi_4: x+y+3z=1$.

Hint

Use a complete-equation scaling check for parallel normals and a dot product for perpendicular normals.

Worked solution

1 Coincident planes. Multiplying every term of Pi1 by -2 gives Pi2.

$$-2\Pi_1:\quad -2x-4y+2z=-8\Rightarrow \Pi_1=\Pi_2$$

2 Distinct parallel planes. Pi3 has the same left side as Pi2 but a different constant.

$$8\neq -8\Rightarrow \Pi_1\parallel\Pi_3 \text{ and } \Pi_1\neq\Pi_3$$

3 Perpendicular planes. The normals to Pi1 and Pi4 have zero dot product.

$$[1,2,-1]\cdot[1,1,3]=1+2-3=0\Rightarrow \Pi_1\perp\Pi_4$$

Final answer

Pi1 and Pi2 coincide. Pi1 and Pi3 are distinct parallel planes. Pi1 and Pi4 are perpendicular.

Problem 5 Smaller angle between planes

Categories: A, C

Find the smaller angle between $x+y+z=2$ and $x+z=1$. Keep an exact ratio before the final approximation.

Hint

Read normals directly from the coefficients, including the missing y term.

Worked solution

1 Normals. A missing variable has coefficient zero.

$$\vec{n}_1=[1,1,1],\quad \vec{n}_2=[1,0,1]$$

2 Exact ratio. Calculate the dot product and exact magnitudes.

$$\vec{n}_1\cdot\vec{n}_2=2,\quad \|\vec{n}_1\|=\sqrt{3},\quad \|\vec{n}_2\|=\sqrt{2},\quad \cos\theta=\frac{2}{\sqrt{6}}$$

3 Angle. The absolute dot product selects the smaller angle.

$$\theta=\cos^{-1}\left(\frac{2}{\sqrt{6}}\right)\approx 35.3^\circ$$

Final answer

The smaller angle is approximately 35.3°.

3.6 Homework 8.6 Sketching points and planes in R³

Draw labelled axes and use coordinates, intercepts, and traces to explain a sketch.

Unit 8.2 covers Sections 8.4, 8.5, 8.6. Complete every section homework before Quiz 8.2 and Test 8.2.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Plot a point and its projection

Categories: K, A, C

Sketch $P(2,-1,3)$ on labelled three-dimensional axes. Mark its projection H onto the xy-plane and describe the coordinate movements.

Hint

The xy-plane has z=0. Keep x and y when finding the projection.

Worked solution

1 First coordinate. Start at O and move two units in positive x.

$$O(0,0,0) \longrightarrow (2,0,0)$$

2 Second coordinate. Move one unit parallel to negative y.

$$(2,0,0) \longrightarrow H(2,-1,0)$$

3 Third coordinate. Move three units parallel to positive z and retain light construction lines.

$$H(2,-1,0) \longrightarrow P(2,-1,3)$$

Final answer

$H=(2,-1,0)$. P lies three units above the xy-plane. A correct sketch labels the axes, H, P, and coordinate movements.

Problem 2 A point below the xy-plane

Categories: A, C

Sketch $R(-1,2,-2)$ and its projection onto the xy-plane. Explain how each negative coordinate changes the construction.

Hint

A negative z-coordinate places the point below the xy-plane.

Worked solution

1 Horizontal construction. Move one unit in negative x, then two units parallel to positive y.

$$(0,0,0) \longrightarrow (-1,0,0) \longrightarrow H(-1,2,0)$$

2 Vertical movement. Move down two units parallel to z.

$$H(-1,2,0) \longrightarrow R(-1,2,-2)$$

3 Explain the signs. Negative x reverses the x movement. Negative z reverses the vertical movement.

$$z_R = -2 < 0, \qquad H = (-1,2,0)$$

Final answer

The projection is $(-1,2,0)$, and R lies below the xy-plane. Retain labelled construction lines in the sketch.

Problem 3 Intercepts and coordinate-plane traces

Categories: K, A, C

Sketch $2x + 3y + 6z = 6$. Find all three axis intercepts and all three coordinate-plane traces. Explain why the drawn triangle is not the whole plane.

Hint

An axis intercept sets two coordinates to zero. A trace sets only one to zero.

Worked solution

1 Intercepts. Set the other two coordinates to zero each time.

$$A(3,0,0), \quad B(0,2,0), \quad C(0,0,1)$$

2 Traces. Each restriction gives a line in the named coordinate plane.

$$z = 0 : 2x + 3y = 6, \quad y = 0 : 2x + 6z = 6, \quad x = 0 : 3y + 6z = 6$$

3 Complete the sketch. Plot A, B, and C on labelled axes and join them to show orientation. Indicate that the plane extends beyond the triangle.

Π is infinite in both independent in-plane directions

Final answer

Intercepts $(3,0,0), (0,2,0), (0,0,1)$. The three traces join the corresponding intercept pairs; the plane extends beyond the patch.

Problem 4 Missing coefficients and contained axes

Categories: T, C

Compare sketches of $x + y = 2$ and $x + y = 0$ in R³. Does either plane meet or contain the z-axis? Explain why z being absent does not settle this by itself.

Hint

Test a general point (0,0,z) on the z-axis in each equation.

Worked solution

1 Test the first plane. Substitute x=y=0.

$$0 + 0 = 2 \quad \text{is impossible}$$

2 Test the second plane. Every z-axis point satisfies the equation.

$$0 + 0 = 0 \quad \text{for every } z \in \mathbb{R}$$

3 Sketch and interpret. In each case, draw its xy-trace and extend parallel to z. The first misses the z-axis; the second contains it.

$$x + y = 2: \text{ no z-axis points}; \qquad x + y = 0 : \{(0, 0, z) : z \in \mathbb{R}\}$$

Final answer

Both planes are parallel to the z direction. The plane $x+y = 2$ misses the z-axis, while $x+y = 0$ contains the whole axis.

Problem 5 A plane through the origin

Categories: A, T, C

Sketch $x + y + z = 0$ using traces. Explain why its axis intercepts alone do not show the plane orientation, and give two independent spans.

Hint

All three axis intercepts coincide at O, so draw two independent trace directions.

Worked solution

1 Find traces. Set one coordinate to zero at a time.

$$z = 0 : y = -x, \qquad y = 0 : z = -x, \qquad x = 0 : z = -y$$

2 Choose spans. Use directions from two different traces.

$$\vec{a} = [1, -1, 0], \quad \vec{b} = [1, 0, -1], \quad \vec{a} \not\parallel \vec{b}$$

3 Describe the plane. Draw a patch through O and the two trace directions, extending beyond them.

$$\vec{r} = s[1, -1, 0] + t[1, 0, -1], \quad s, t \in \mathbb{R}$$

Final answer

All intercepts are O. The independent traces and spans show orientation, with vector form $\vec{r} = s[1, -1, 0] + t[1, 0, -1]$.

3.7 Homework 9.1 Intersections involving lines

Solve line-plane and line-line systems, then verify every coordinate.

Unit 9.1 covers Sections 9.1, 9.2, 9.3, 9.4. Complete every section homework before Quiz 9.1 and Test 9.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 A line crossing a plane

Categories: K, A

Find the intersection of $L : \vec{r} = [1, 0, 2] + t[1, 2, -1]$ and $\Pi : x + y + z = 5$. Verify the intersection in both objects.

Hint

Substitute all three coordinate expressions into the plane.

Worked solution

1 Substitute. The line has x=1+t, y=2t, and z=2-t.

$$(1 + t) + 2t + (2 - t) = 5 \Rightarrow 3 + 2t = 5$$

2 Find the point. Solve for the parameter and return to the line.

$$t = 1 \Rightarrow P(2, 2, 1)$$

3 Verify. The coordinates come from the line and satisfy the plane.

$$2 + 2 + 1 = 5$$

Final answer

The unique intersection is $P(2, 2, 1)$.

Problem 2 Contained or parallel and disjoint

Categories: A, T, C

Classify $L : \vec{r} = [1, 0, 1] + t[1, -1, 0]$ relative to $\Pi_1 : x + y + z = 2$ and $\Pi_2 : x + y + z = 3$. Explain the difference between an identity and a contradiction.

Hint

Substitution produces a constant expression because the direction is perpendicular to the normal.

Worked solution

1 Check the orientation. The normal-direction dot product vanishes.

$$[1, 1, 1] \cdot [1, -1, 0] = 0$$

2 Substitute into Pi1. No restriction remains on the parameter.

$$(1 + t) - t + 1 = 2 \Rightarrow 2 = 2 \quad \text{for every } t$$

3 Substitute into Pi2. The constant equality fails for every parameter.

$$(1+t)-t+1=3 \Rightarrow 2=3 \quad \text{is impossible}$$

Final answer

L is contained in Pi1. It is parallel and disjoint from Pi2. An identity allows all t, while the contradiction allows none.

Problem 3 Two intersecting lines

Categories: K, A

Find the common point of $L_1 : \vec{r} = s[1,1,0]$ and $L_2 : \vec{r} = [1,0,0] + u[-1,1,0]$. Use separate parameters and verify all coordinates.

Hint

Equate x, y, and z separately.

Worked solution

1 Coordinate system. One parameter pair must satisfy all equations.

$$s = 1-u, \qquad s = u, \qquad 0 = 0$$

2 Solve. Substitute u=s into the first equation.

$$2s = 1 \Rightarrow s = u = \tfrac{1}{2}$$

3 Verify both lines. Both produce the same ordered triple.

$$\tfrac{1}{2}[1,1,0] = [1,0,0] + \tfrac{1}{2}[-1,1,0] = [\tfrac{1}{2}, \tfrac{1}{2}, 0]$$

Final answer

The lines intersect at $(\frac{1}{2}, \frac{1}{2}, 0)$.

Problem 4 A third-coordinate check reveals skew lines

Categories: T, C

Determine the relationship between $L_1 : \vec{r} = s[1,1,0]$ and $L_2 : \vec{r} = [1,0,1] + u[-1,1,0]$. Show why solving only x and y is insufficient.

Hint

The z-coordinates must agree too, and the directions must be checked before using the word skew.

Worked solution

1 First two coordinates. They produce a candidate parameter pair.

$$s = 1-u, \quad s = u \Rightarrow s = u = \tfrac{1}{2}$$

2 Third coordinate. The lines lie at different fixed z-values.

$$0 = 1 \quad \text{is impossible}$$

3 Classify. The directions are not scalar multiples, so the disjoint lines are not parallel.

$$[1,1,0] \nparallel [-1,1,0] \Rightarrow \text{skew lines}$$

Final answer

The lines are skew. The x and y equations agree at a candidate pair, but z gives a contradiction.

Problem 5 A family of parallel planes

Categories: A, T, C

Classify $L : \vec{r} = [1,2,0] + t[1,1,-1]$ relative to $\Pi_k : 2x - y + z = k$ for every real k. Can any member intersect L at exactly one point?

Hint

Calculate the normal-direction dot product and then substitute the entire line.

Worked solution

1 Orientation. The same normal applies to every plane in the family.

$$[2,-1,1] \cdot [1,1,-1] = 2 - 1 - 1 = 0$$

2 Substitute. The parameter cancels and the left side is always zero.

$$2(1+t) - (2+t) - t = 0$$

3 Classify every k. The constant comparison decides whether all or no parameters work.

$$k = 0 \Rightarrow L \subset \Pi_k; \qquad k \neq 0 \Rightarrow L \cap \Pi_k = \varnothing$$

Final answer

L is contained when $k = 0$ and parallel-disjoint otherwise. No member has exactly one intersection with L.

3.8 Homework 9.2 Solving linear systems

Use elimination, identify free variables, and distinguish consistency from uniqueness.

Unit 9.1 covers Sections 9.1, 9.2, 9.3, 9.4. Complete every section homework before Quiz 9.1 and Test 9.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 A unique solution by elimination

Categories: K, A

Solve $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$. Show elimination and verify all original equations.

Hint

Eliminate x from two equation pairs first.

Worked solution

1 Reduce the system. Use equation 2 minus twice equation 1, then equation 3 minus equation 1.

$$-3y - z = -9, \qquad y - 2z = -4$$

2 Solve the reduced equations. Add three times the second reduced equation to the first.

$$-7z = -21 \Rightarrow z = 3, \quad y = 2, \quad x = 1$$

3 Verify. Check each original restriction.

$$1 + 2 + 3 = 6, \quad 2(1) - 2 + 3 = 3, \quad 1 + 2(2) - 3 = 2$$

Final answer

The system has the unique solution $(1, 2, 3)$.

Problem 2 An identity with a remaining line of solutions

Categories: A, T, C

Solve $x + y + z = 3$, $x - y + z = 1$, and $2x + 2z = 4$. Explain why the identity from elimination does not allow every triple.

Hint

The third equation repeats information from the first two.

Worked solution

1 Find a fixed coordinate. Subtract equation 2 from equation 1.

$$2y = 2 \Rightarrow y = 1$$

2 Identify redundancy. The remaining restriction is x+z=2, which agrees with equation 3.

$$2x + 2z = 4 \Rightarrow 0 = 0 \text{ after comparison}$$

3 Write the full family. Let z=t, retaining all non-redundant restrictions.

$$(x, y, z) = (2 - t, 1, t), \qquad t \in \mathbb{R}$$

Final answer

Infinitely many solutions form the line $(2 - t, 1, t)$. The identity adds no restriction, but the other equations still do.

Problem 3 A contradiction after elimination

Categories: K, T, C

Determine whether $x + y + z = 3$, $x - y + z = 1$, and $2x + 2z = 5$ are consistent. Show the contradiction.

Hint

Add the first two equations, then compare with the third.

Worked solution

1 Combine equations. The first pair forces a value of 2x+2z.

$$(x + y + z) + (x - y + z) = 3 + 1 \Rightarrow 2x + 2z = 4$$

2 Compare. Subtract the combined equation from the third.

$$(2x + 2z) - (2x + 2z) = 5 - 4 \Rightarrow 0 = 1$$

3 Interpret. A contradiction has no solution, so no triple satisfies the original system.

Final answer

The system is inconsistent and has no solution.

Problem 4 Two free variables after redundant equations

Categories: A, T, C

Give the complete solution set of $x - y + z = 3$, $2x - 2y + 2z = 6$, and $-x + y - z = -3$. State how many free variables remain.

Hint

All three equations express the same restriction.

Worked solution

1 Compare complete equations. The second is twice and the third is negative one times the first.

$$E_2 = 2E_1, \qquad E_3 = -E_1$$

2 Choose free coordinates. Let y=s and z=t, then solve the first equation.

$$x = 3 + s - t, \qquad y = s, \qquad z = t$$

3 Describe all solutions. Two independent parameters remain.

$$\vec{r} = [3, 0, 0] + s[1, 1, 0] + t[-1, 0, 1], \quad s, t \in \mathbb{R}$$

Final answer

The solution set is a plane, with two free variables. It is consistent but not uniquely determined.

Problem 5 Consistency depending on a constant

Categories: A, T, C

For which k is $x + y + z = 2$, $x - y + z = 0$, $2x + 2z = k$ consistent? Give the full solution set whenever it is consistent.

Hint

The first two equations force a value for the left side of the third.

Worked solution

1 Use the first pair. Subtract and add to find the retained restrictions.

$$y = 1, \qquad x + z = 1$$

2 Check the parameter. Twice x+z equals 2.

$$2x + 2z = 2 \Rightarrow k = 2$$

3 State both cases. Keep the free variable only in the consistent case.

$$k = 2 : (x, y, z) = (1 - t, 1, t); \qquad k \neq 2 : \text{ no solution}$$

Final answer

Consistent exactly when $k = 2$, with line of solutions $(1 - t, 1, t)$.

3.9 Homework 9.3 Intersections of two planes

Find the complete intersection set and verify its location and direction.

Unit 9.1 covers Sections 9.1, 9.2, 9.3, 9.4. Complete every section homework before Quiz 9.1 and Test 9.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 A line of intersection

Categories: K, A, C

Find a vector equation for the intersection of $x + y + z = 3$ and $x - y + z = 1$. Verify both the anchor and direction.

Hint

Subtract the plane equations, then choose one free coordinate.

Worked solution

1 Eliminate. Subtract the second equation from the first.

$$2y = 2 \Rightarrow y = 1, \qquad x + z = 2$$

2 Parameterize. Let z=t and separate the constant point and direction.

$$\vec{r} = [2, 1, 0] + t[-1, 0, 1]$$

3 Verify. The anchor lies in both planes and the direction is perpendicular to both normals.

$$2 + 1 = 3, \quad 2 - 1 = 1, \quad [1, 1, 1] \cdot [-1, 0, 1] = [1, -1, 1] \cdot [-1, 0, 1] = 0$$

Final answer

The common set is $\vec{r} = [2, 1, 0] + t[-1, 0, 1]$, for every real t.

Problem 2 Distinct parallel planes

Categories: K, T, C

Determine the intersection of $x + 2y - z = 4$ and $2x + 4y - 2z = 10$. Justify the geometric conclusion.

Hint

Match the complete first equation to the second before comparing constants.

Worked solution

1 Compare normals. The normals are parallel.

$$[2, 4, -2] = 2[1, 2, -1]$$

2 Scale the equation. Twice the first requires a different right side.

$$2x + 4y - 2z = 8$$

3 Eliminate. Subtract twice equation 1 from equation 2.

$$0 = 10 - 8 = 2 \Rightarrow \Pi_1 \cap \Pi_2 = \varnothing$$

Final answer

The planes are distinct and parallel, with no common points.

Problem 3 Coincident planes and their full common set

Categories: A, T, C

Determine the common set of $x + 2y - z = 4$ and $2x + 4y - 2z = 8$. Give a vector representation rather than a single point.

Hint

Only one independent restriction remains.

Worked solution

1 Compare complete equations. The second is exactly twice the first.

$$E_2 = 2E_1 \Rightarrow \Pi_1 = \Pi_2$$

2 Choose two free variables. Let y=s and z=t.

$$x = 4 - 2s + t$$

3 Represent the common plane. Every point of this plane belongs to both equations.

$$\vec{r} = [4, 0, 0] + s[-2, 1, 0] + t[1, 0, 1]$$

Final answer

The common set is the entire plane $\vec{r} = [4, 0, 0] + s[-2, 1, 0] + t[1, 0, 1]$.

Problem 4 Find and verify an intersection direction

Categories: A, C

Find the intersection of $x + y = 2$ and $y + z = 3$. Compare your line direction with the cross product of the normals and explain any sign difference.

Hint

Let y be the free parameter.

Worked solution

1 Solve coordinates. Choose y=t and use both equations.

$$x = 2 - t, \quad y = t, \quad z = 3 - t$$

2 Vector form. Separate the fixed point and direction.

$$\vec{r} = [2, 0, 3] + t[-1, 1, -1]$$

3 Check direction. Opposite nonzero directions describe the same line.

$$[1,1,0]\times[0,1,1]=[1,-1,1]=-[-1,1,-1]$$

Final answer

The intersection is $\vec{r}=[2,0,3]+t[-1,1,-1]$. Its direction is the negative of the normal cross product, which is equally valid.

Problem 5 A parameter-dependent plane pair

Categories: A, T, C

Classify the intersection of $\Pi_1: x+y+z=1$ and $\Pi_k: x+y+kz=2$ for every real k. Give a vector equation whenever an intersection line exists.

Hint

Subtract the equations before dividing by k-1.

Worked solution

1 Subtract. Keep the exceptional coefficient value visible.

$$(k-1)z=1$$

2 Exceptional case. At k=1 the reduced equation is impossible.

$$k=1\Rightarrow 0=1\Rightarrow \text{distinct parallel planes}$$

3 General case. For k not equal to 1, let y=t and use the first plane.

$$\vec{r}=[\tfrac{k-2}{k-1},0,\tfrac{1}{k-1}]+t[-1,1,0],\qquad k\neq 1$$

Final answer

No common points at $k=1$. For $k\neq 1$, the planes meet in the stated line.

3.10 Homework 9.4 Intersections of three planes

Interpret the common solution as a point, line, plane, or empty set.

Unit 9.1 covers Sections 9.1, 9.2, 9.3, 9.4. Complete every section homework before Quiz 9.1 and Test 9.1.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 One common point

Categories: K, A

Find the common intersection of $x+y+z=6$, $x-y+z=2$, and $x+y-z=0$. Verify the answer in all three planes.

Hint

Subtract equation pairs to isolate y and z.

Worked solution

1 Find y and z. Subtract equation 2 from 1, then equation 3 from 1.

$$2y=4 \Rightarrow y=2, \qquad 2z=6 \Rightarrow z=3$$

2 Find x. Use the first equation.

$$x+2+3=6 \Rightarrow x=1$$

3 Verify. Check the complete system.

$$1+2+3=6, \quad 1-2+3=2, \quad 1+2-3=0$$

Final answer

The unique common point is $(1,2,3)$.

Problem 2 Three planes sharing a line

Categories: A, T, C

Describe the common solution of $x+y+z=2$, $2x-y+z=1$, and $3x+2z=3$. Explain why the third equation does not produce a single point.

Hint

Compare the third equation with the sum of the first two.

Worked solution

1 Identify redundancy. The third equation is already implied by the first pair.

$$E_3=E_1+E_2$$

2 Solve the independent restrictions. Subtract equation 1 from equation 2 and let y=t.

$$x-2y=-1 \Rightarrow x=2t-1, \qquad z=3-3t$$

3 State the common geometry. One free parameter remains.

$$\vec{r} = [-1, 0, 3] + t[2, 1, -3]$$

Final answer

The common intersection is a line. The third plane contains the entire intersection of the first two.

Problem 3 Three equations describing one plane

Categories: K, T, C

Classify the common intersection of $x - y + z = 3$, $2x - 2y + 2z = 6$, and $-3x + 3y - 3z = -9$. Give its vector equation.

Hint

Compare all coefficients and constants, then retain two free parameters.

Worked solution

1 Compare equations. Every equation is a nonzero multiple of the first.

$$E_2 = 2E_1, \qquad E_3 = -3E_1$$

2 Parameterize. Let y=s and z=t.

$$x = 3 + s - t$$

3 Describe the common plane. There are two independent free variables.

$$\vec{r} = [3, 0, 0] + s[1, 1, 0] + t[-1, 0, 1]$$

Final answer

All three planes coincide, so their common set is the entire stated plane.

Problem 4 Pairwise intersections without a common point

Categories: A, T, C

The planes are $x = 0$, $y = 0$, and $x + y = 1$. Find each pairwise intersection and explain why no point belongs to all three.

Hint

The first two force x=y=0, which must still satisfy the third.

Worked solution

1 First pair. The intersection of x=0 and y=0 is the z-axis.

$$L_{12} : \vec{r} = [0, 0, 0] + t[0, 0, 1]$$

2 Other pairs. Use one fixed coordinate in the third equation.

$$L_{13} : \vec{r} = [0, 1, 0] + t[0, 0, 1], \quad L_{23} : \vec{r} = [1, 0, 0] + t[0, 0, 1]$$

3 Test all three. The first pair violates the third plane.

$$x = y = 0 \Rightarrow x + y = 0 \neq 1$$

Final answer

The pairs meet in three distinct parallel lines, but the three planes have no common point.

Problem 5 A family with a point or a line

Categories: A, T, C

Classify the common solution of $x + y + z = 3$, $x - y + z = 1$, and $x + y + kz = 3$ for every real k. Give the complete set in each case.

Hint

Subtract the first equation from the third and keep the exceptional k-value.

Worked solution

1 Reduce. The first pair gives y=1 and x+z=2.

$$y = 1, \quad x + z = 2, \quad (k - 1)z = 0$$

2 Non-exceptional case. For k not equal to 1, z is fixed.

$$k \neq 1 \Rightarrow z = 0, \quad x = 2, \quad y = 1$$

3 Exceptional case. At k=1 the third equation repeats the first.

$$k = 1 \Rightarrow \vec{r} = [2, 1, 0] + t[-1, 0, 1]$$

Final answer

For $k \neq 1$, the common set is the point $(2, 1, 0)$. For $k = 1$, it is the line $[2, 1, 0] + t[-1, 0, 1]$.

3.11 Homework 9.5 Distance from a point to a line

Use perpendicular distance, exact radicals, and foot verification in R² and R³.

Unit 9.2 covers Sections 9.5, 9.6. Complete every section homework before Quiz 9.2 and Test 9.2.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Cartesian distance and perpendicular foot in R²

Categories: K, A, C

Find the exact distance from $P(1,2)$ to $3x+4y-5=0$. Find the perpendicular foot H and verify it.

Hint

Use the normal [3,4]. The signed residual is 6 and its squared magnitude is 25.

Worked solution

1 Distance. Substitute P into the complete Cartesian expression.

$$d = \frac{|3+8-5|}{\sqrt{3^2+4^2}} = \frac{6}{5}$$

2 Foot. Subtract the signed normal displacement.

$$H = [1,2] - \frac{6}{25}[3,4] = [\tfrac{7}{25}, \tfrac{26}{25}]$$

3 Verify. H lies on the line and the connector is normal to it.

$$3(\tfrac{7}{25}) + 4(\tfrac{26}{25}) - 5 = 0, \quad \overrightarrow{HP} = \frac{6}{25}[3,4]$$

Final answer

$d = \frac{6}{5}$ units, with $H = (\frac{7}{25}, \frac{26}{25})$.

Problem 2 Cross-product distance in R³

Categories: A, T

Find the exact distance from $Q(1,0,2)$ to $L : \vec{r} = t[1,1,0]$. Verify it using a perpendicular foot.

Hint

Use the origin as the line anchor and divide the parallelogram area by the direction magnitude.

Worked solution

1 Cross product. The anchor-to-point displacement is [1,0,2].

$$[1,0,2] \times [1,1,0] = [-2,2,1]$$

2 Exact distance. Use the magnitude ratio.

$$d = \frac{\sqrt{4+4+1}}{\sqrt{1+1}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

3 Projection check. The foot uses the scalar projection parameter.

$$t_H = \frac{[1,0,2]\cdot[1,1,0]}{2} = \frac{1}{2}, \quad H = (\tfrac{1}{2},\tfrac{1}{2},0), \quad \|Q-H\| = \sqrt{\tfrac{1}{4}+\tfrac{1}{4}+4} = \frac{3\sqrt{2}}{2}$$

Final answer

The exact perpendicular distance is $\frac{3\sqrt{2}}{2}$ units.

Problem 3 Recognising zero distance

Categories: K, T, C

Find the distance from $P(3,1,2)$ to $L : \vec{r} = [1,1,0] + t[1,0,1]$. Explain the result geometrically.

Hint

Test membership before carrying out a long distance calculation.

Worked solution

1 Test the point. One value must work in every coordinate.

$$t = 2: \quad [1,1,0] + 2[1,0,1] = [3,1,2]$$

2 Check the displacement. The anchor-to-point vector is parallel to the direction.

$$\overrightarrow{AP} = [2,0,2] = 2[1,0,1], \quad \overrightarrow{AP} \times \vec{d} = \vec{0}$$

3 Interpret. P is already on the line, so no perpendicular separation remains.

$$d(P,L) = 0$$

Final answer

The distance is zero because P lies on L at $t = 2$.

Problem 4 Distance between parallel lines in R²

Categories: A, T, C

Find the exact distance between $L_1 : 3x - 4y - 6 = 0$ and $L_2 : 6x - 8y + 8 = 0$. Explain why the constants cannot be compared before normalising.

Hint

First make the x- and y-coefficients identical.

Worked solution

1 Normalise. Divide the complete second equation by two.

$$L_2 : 3x - 4y + 4 = 0$$

2 Distance. The two equations now use the same normal.

$$d = \frac{|4-(-6)|}{\sqrt{3^2+(-4)^2}} = \frac{10}{5} = 2$$

3 Verify with a point. The point (2,0) lies on L1 and gives the same distance to L2.

$$3(2) - 4(0) - 6 = 0, \qquad \frac{|3(2) - 4(0) + 4|}{5} = 2$$

Final answer

The lines are 2 units apart. Constants must be compared using the same normal coefficients.

Problem 5 Projection and the nearest point

Categories: A, T, C

Find the nearest point H on $L : \vec{r} = [0, 1, 0] + t[1, 0, 1]$ to $P(1, 2, 3)$, then calculate the exact distance. Verify perpendicularity.

Hint

Project the anchor-to-point displacement onto the line direction.

Worked solution

1 Projection parameter. Use A=(0,1,0) and w=P-A.

$$\vec{w} = [1, 1, 3], \quad t_H = \frac{\vec{w} \cdot [1, 0, 1]}{[1, 0, 1] \cdot [1, 0, 1]} = \frac{4}{2} = 2$$

2 Foot and distance. Return to the line, then measure the residual.

$$H = (2, 1, 2), \quad \overrightarrow{HP} = [-1, 1, 1], \quad d = \sqrt{3}$$

3 Verify. The residual is perpendicular to the line direction.

$$[-1, 1, 1] \cdot [1, 0, 1] = -1 + 1 = 0$$

Final answer

The nearest point is $H = (2, 1, 2)$, and the exact distance is $\sqrt{3}$ units.

3.12 Homework 9.6 Distance from a point to a plane

Use the normal projection, retain signed information for the foot, and check the geometry.

Unit 9.2 covers Sections 9.5, 9.6. Complete every section homework before Quiz 9.2 and Test 9.2.

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Distance and nearest point

Categories: K, A, C

Find the distance from $P(3,1,2)$ to $2x-y+2z-3=0$, and find the perpendicular foot H. Verify both conditions for H.

Hint

The normal is [2,-1,2], whose squared magnitude is 9.

Worked solution

1 Residual and distance. Substitute P into the Cartesian expression.

$$f(P)=2(3)-1+2(2)-3=6, \quad \|\vec{n}\|=3, \quad d=2$$

2 Foot. Keep the residual sign in the projection factor.

$$H=[3,1,2]-\frac{6}{9}[2,-1,2]=[\tfrac{5}{3},\tfrac{5}{3},\tfrac{2}{3}]$$

3 Verify. H lies in the plane and HP is parallel to its normal.

$$2(\tfrac{5}{3})-\tfrac{5}{3}+2(\tfrac{2}{3})-3=0, \qquad \overrightarrow{HP}=\tfrac{2}{3}[2,-1,2]$$

Final answer

$d=2$ units and $H=(\frac{5}{3},\frac{5}{3},\frac{2}{3})$.

Problem 2 An integer perpendicular foot

Categories: A, C

Find the exact distance from $P(3,1,4)$ to $x+2y+2z=4$. Find H and verify its location and normal connector.

Hint

Move from P by the signed residual divided by the squared normal magnitude.

Worked solution

1 Distance. The residual is 9 and the normal magnitude is 3.

$$d=\frac{|3+2(1)+2(4)-4|}{\sqrt{1+4+4}}=3$$

2 Foot. The signed projection factor is one.

$$H=[3,1,4]-\frac{9}{9}[1,2,2]=[2,-1,2]$$

3 Verify. Check the plane and parallel normal direction.

$$2 + 2(-1) + 2(2) = 4, \qquad P - H = [1, 2, 2]$$

Final answer

The distance is 3 units, and the perpendicular foot is $H = (2, -1, 2)$.

Problem 3 Zero distance and an absolute value

Categories: K, T, C

Find the distance from $P(1, -1, 2)$ to $2x + y - z + 1 = 0$. A student says a negative signed numerator would give a negative distance. Explain the error.

Hint

A point satisfying the Cartesian equation has no separation from the plane.

Worked solution

1 Substitute. Use the complete expression including the constant.

$$2(1) + (-1) - 2 + 1 = 0$$

2 Calculate. The numerator vanishes.

$$d = \frac{|0|}{\sqrt{2^2 + 1^2 + (-1)^2}} = 0$$

3 Explain the absolute value. A signed residual identifies a side of the plane. Its absolute value gives a non-negative length.

$$d(P, \Pi) = \frac{|f(P)|}{\|\vec{n}\|} \geq 0$$

Final answer

P lies in the plane, so its distance is zero. A negative residual never gives a negative distance.

Problem 4 Distance between parallel planes

Categories: A, T, C

Find the distance between $x - y + z = 3$ and $2x - 2y + 2z = 12$. Verify using a point on the first plane.

Hint

Rewrite both equations with exactly the same normal.

Worked solution

1 Normalise. Divide the entire second equation by two.

$$\Pi_2 : x - y + z = 6$$

2 Distance. Use the difference of constants with the common normal magnitude.

$$d = \frac{|6-3|}{\sqrt{1+1+1}} = \sqrt{3}$$

3 Point check. A=(3,0,0) lies in the first plane.

$$\frac{|3-0+0-6|}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

Final answer

The planes are exactly $\sqrt{3}$ units apart.

Problem 5 Distance from a vector plane

Categories: A, T, C

Find the exact distance from $P(3,1,4)$ to $\Pi : \vec{r} = [1,0,2] + s[1,1,0] + t[0,1,1]$. Find and verify the perpendicular foot.

Hint

First obtain a nonzero normal from the independent spans.

Worked solution

1 Normal and displacement. Cross the spans and subtract the known plane point from P.

$$\vec{n} = [1,-1,1], \quad \overrightarrow{AP} = [2,1,2], \quad \overrightarrow{AP} \cdot \vec{n} = 3$$

2 Distance and foot. Use the projection factor 3/3=1.

$$d = \frac{3}{\sqrt{3}} = \sqrt{3}, \quad H = [3,1,4] - [1,-1,1] = [2,2,3]$$

3 Verify the geometry. The plane has Cartesian equation x-y+z=3.

$$2 - 2 + 3 = 3, \qquad P - H = [1,-1,1] = \vec{n}$$

Final answer

The distance is $\sqrt{3}$ units, with perpendicular foot $H = (2,2,3)$.

4 Quizzes

4.1 Quiz 8.1 Line equations

Covers Sections 8.1, 8.2, 8.3.

Sections 8.1 to 8.3. A unit quiz covering the section homework, with worked feedback and written explanations.

10 questions 22 marks Suggested time: 35 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: K
For the R² line $\vec{r} = [1, -1] + t[2, 3]$, which vector is parallel to the line?

A. $[-3, 2]$

B. $[1, -1]$

C. $[2, -3]$

D. $[2, 3]$

Question 2 [1 mark]

Categories: A
Find the point on $\vec{r} = [1, -1] + t[2, 3]$ when $t = 2$.

A. $(5, -7)$

B. $(-3, 5)$

C. $(5, 5)$

D. $(3, 2)$

Question 3 [1 mark]

Categories: K
Which vector is a normal to the Cartesian line $2x - 3y = 5$?

A. $[-3, 2]$

B. $[2, -3]$

C. $[3, 2]$

D. $[2, 3]$

Question 4 [1 mark]

Categories: K, A
Which symmetric description represents $\vec{r} = [1, -1, 2] + t[0, 1, -2]$?
A. $x = 1, \quad \frac{y+1}{1} = \frac{z-2}{-2}$

B. $x = 1, \quad \frac{y-1}{1} = \frac{z+2}{-2}$

C. $y = -1, \quad \frac{x-1}{1} = \frac{z-2}{-2}$

D. $z = 2, \quad x = 1 + t,\ y = -1 - 2t$

Question 5 [1 mark]

Categories: T
Classify $L_1 : \vec{r} = [1, -1, 2] + t[2, -1, 3]$ and $L_2 : \vec{r} = [3, -2, 5] + u[-4, 2, -6]$.
A. Distinct parallel lines

B. Perpendicular intersecting lines

C. Skew lines

D. Coincident lines

Question 6 [1 mark]

Categories: A
The directions of two R² lines through $(1, 0)$ are $[1, 1]$ and $[1, -1]$. What is their smaller angle?
A. 45°

B. 60°

C. 90°

D. 0°

Part B Reasoning and communication

Question 7 [4 marks]

Categories: A, C
The line passes through $A(1, 2)$ and $B(7, 6)$. (a) Find a simplified direction. (b) Write vector and parametric equations. (c) Obtain Cartesian form. (d) Verify B using one parameter.
Show your working and state a complete conclusion.

Question 8 [4 marks]

Categories: A, T, C
For $A(1, -1, 2)$ and $B(1, 3, -6)$, (a) write vector and symmetric equations of AB, (b) explain the fixed coordinate, and (c) determine whether $Q(1, 1, -2)$ lies on the line.
Show your working and state a complete conclusion.

Question 9 [4 marks]

Categories: K, A, C
Find k so the R² lines $kx + 2y = 1$ and $3x + 1y = 2$ are perpendicular. Show the normal-vector test and justify your conclusion.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: T, C
Let $L : \vec{r} = [1, 0, 1] + t[1, 2, -1]$. Compare it with $M : \vec{r} = [4, 6, -2] + u[-2, -4, 2]$ and $N : \vec{r} = [4, 6, 0] + v[-2, -4, 2]$. Establish whether each pair is coincident or distinct parallel.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$[2, 3]$

The coefficient of the real parameter is the direction vector. The constant vector locates a starting point.

1 Reasoning. The coefficient of the real parameter is the direction vector. The constant vector locates a starting point.

$$\vec{r} = \vec{r}_0 + t\vec{d}, \qquad \vec{d} = [2, 3]$$

Question 2 solution

Correct answer: C

$(5, 5)$

Multiply both direction components by two, then add the corresponding starting coordinates.

1 Reasoning. Multiply both direction components by two, then add the corresponding starting coordinates.

$$[1, -1] + 2[2, 3] = [5, 5]$$

Question 3 solution

Correct answer: B

$[2, -3]$

The x and y coefficients form a nonzero normal. A direction must be perpendicular to it.

1 Reasoning. The x and y coefficients form a nonzero normal. A direction must be perpendicular to it.

$$\vec{n} = [2, -3], \qquad [2, -3] \cdot [3, 2] = 0$$

Question 4 solution

Correct answer: A

$x = 1, \quad \frac{y+1}{1} = \frac{z-2}{-2}$

The zero x-direction component makes x constant. Only the y and z equations can be divided by nonzero direction components.

1 Reasoning. The zero x-direction component makes x constant. Only the y and z equations can be divided by nonzero direction components.

$$x = 1, \quad y = -1 + t, \quad z = 2 - 2t$$

Question 5 solution

Correct answer: D

Coincident lines

The second direction is negative two times the first. Its starting point is reached on the first line at t=1, so the lines coincide.

1 Reasoning. The second direction is negative two times the first. Its starting point is reached on the first line at t=1, so the lines coincide.

$$[-4, 2, -6] = -2[2, -1, 3], \quad [1, -1, 2] + [2, -1, 3] = [3, -2, 5]$$

Question 6 solution

Correct answer: C

90°

The dot product is zero and both directions are nonzero, so the lines are perpendicular.

1 Reasoning. The dot product is zero and both directions are nonzero, so the lines are perpendicular.

$$[1, 1] \cdot [1, -1] = 0 \Rightarrow \theta = 90^\circ$$

Question 7 solution

Model response

A direction is $[3, 2]$. The equations are $\vec{r} = [1, 2] + t[3, 2]$, $x = 1 + 3t, y = 2 + 2t$, and $2x - 3y = -4$. The point B is reached when $t = 2$.

Marking criteria

1. Obtains a nonzero direction parallel to AB (1 mark).
2. Writes both forms with one shared parameter (1 mark).
3. Derives the correct Cartesian equation (1 mark).
4. Checks both coordinates of B (1 mark).

A direction is $[3, 2]$. The equations are $\vec{r} = [1, 2] + t[3, 2]$, $x = 1 + 3t, y = 2 + 2t$, and $2x - 3y = -4$. The point B is reached when $t = 2$.

1 Direction. Subtract A from B and divide by two.

$$\overrightarrow{AB} = [6, 4] = 2[3, 2]$$

2 Represent. Use A as the starting point.

$$\vec{r} = [1, 2] + t[3, 2], \quad x = 1 + 3t, \ y = 2 + 2t$$

3 Eliminate. Both expressions equal the same parameter.

$$\frac{x - 1}{3} = \frac{y - 2}{2} \Rightarrow 2x - 3y = -4$$

4 Verify. Use the same parameter in both coordinates.

$$t = 2: \quad [1, 2] + 2[3, 2] = [7, 6]$$

Question 8 solution

Model response

Use direction $[0, 1, -2]$. Then $\vec{r} = [1, -1, 2] + t[0, 1, -2]$ and $x = 1,\ (y + 1)/1 = (z - 2)/(-2)$. The x-coordinate stays fixed. Q lies on the line at $t = 2$; B is reached at $t = 4$.

Marking criteria

1. Obtains a valid simplified direction (1 mark).
2. Writes vector and symmetric forms (1 mark).
3. Explains why division by the zero component is invalid (1 mark).
4. Checks Q using one shared parameter (1 mark).

Use direction $[0, 1, -2]$. Then $\vec{r} = [1, -1, 2] + t[0, 1, -2]$ and $x = 1,\ (y + 1)/1 = (z - 2)/(-2)$. The x-coordinate stays fixed. Q lies on the line at $t = 2$; B is reached at $t = 4$.

1 Direction. Simplify the displacement without changing its orientation.

$$\overrightarrow{AB} = [0, 4, -8] = 4[0, 1, -2]$$

2 Vector form. Use A and a nonzero parallel direction.

$$\vec{r} = [1, -1, 2] + t[0, 1, -2]$$

3 Symmetric form. Retain x separately because its direction component is zero.

$$x = 1, \qquad \frac{y + 1}{1} = \frac{z - 2}{-2}$$

4 Membership. Check all coordinates with t=2.

$$[1, -1, 2] + 2[0, 1, -2] = [1, 1, -2]$$

Question 9 solution

Model response

The normals are $[k, 2]$ and $[3, 1]$. Their dot product must vanish, so $3k + 2 = 0$ and $k = \frac{-2}{3}$. Both normals remain nonzero, and in R² this establishes perpendicular line directions.

Marking criteria

1. Identifies both normal vectors (1 mark).
2. Uses the correct zero-dot-product condition (1 mark).
3. Solves for k exactly (1 mark).
4. Explains why the test applies to lines in R² (1 mark).

The normals are $[k, 2]$ and $[3, 1]$. Their dot product must vanish, so $3k + 2 = 0$ and $k = \frac{-2}{3}$. Both normals remain nonzero, and in R² this establishes perpendicular line directions.

1 Normals. Read the coefficients of x and y.

$$\vec{n}_1 = [k, 2], \quad \vec{n}_2 = [3, 1]$$

2 Perpendicularity. For lines in R², perpendicular normals give perpendicular line directions.

$$[k, 2] \cdot [3, 1] = 0$$

3 Solve. Expand and solve the scalar equation.

$$3k + 2 = 0 \Rightarrow k = \frac{-2}{3}$$

4 Check. Substitute the value into the dot product.

$$3\left(\frac{-2}{3}\right) + 2 = 0$$

Question 10 solution

Model response

All directions are parallel because $[-2, -4, 2] = -2[1, 2, -1]$. The anchor of M lies on L at $t = 3$, so M and L coincide. The same x and y force $t = 3$ for the anchor of N, but L gives $z = -2$ rather than zero. Thus N and L are distinct parallel lines.

Marking criteria

1. Compares all direction vectors (1 mark).
2. Verifies a common point for L and M (1 mark).
3. Shows why the anchor of N fails (1 mark).
4. States both relationships with complete reasoning (1 mark).

All directions are parallel because $[-2, -4, 2] = -2[1, 2, -1]$. The anchor of M lies on L at $t = 3$, so M and L coincide. The same x and y force $t = 3$ for the anchor of N, but L gives $z = -2$ rather than zero. Thus N and L are distinct parallel lines.

1 Directions. Compare scalar multiples.

$$\vec{d}_M = \vec{d}_N = -2\vec{d}_L$$

2 Shared point. Test the starting point of M.

$$t = 3: \quad [1, 0, 1] + 3[1, 2, -1] = [4, 6, -2]$$

3 Failed point. The anchor of N has the wrong z-coordinate.

$$x = 4 \Rightarrow t = 3, \quad z = 1 - 3 = -2 \neq 0$$

4 Classify. Parallel directions require a separate location check.

$$L = M, \qquad L \parallel N, \quad L \neq N$$

4.2 Quiz 8.2 Plane equations and sketching

Covers Sections 8.4, 8.5, 8.6.

Sections 8.4 to 8.6. A unit quiz covering the section homework, with worked feedback and written explanations.

10 questions 22 marks Suggested time: 35 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: K
A plane starts at $A(1,0,1)$. Which condition on its proposed spans $\vec{a},\vec{b}$ ensures they determine a unique plane?

A. $\vec{a}\times\vec{b}=\vec{0}$

B. $\vec{a}=2\vec{b}$

C. Both spans are zero

D. $\vec{a}\times\vec{b}\neq\vec{0}$

Question 2 [1 mark]

Categories: A
The plane $\vec{r}=[1,0,1]+s[1,0,1]+t[0,1,1]$ has which normal?

A. $[1,0,1]$

B. $[0,1,1]$

C. $[-1,-1,1]$

D. $[1,1,1]$

Question 3 [1 mark]

Categories: T
Which equation describes the same plane as $x+2y-z=4$?

A. $x-2y-z=4$

B. $-2x-4y+2z=-8$

C. $-2x-4y+2z=8$

D. $x+2y+z=4$

Question 4 [1 mark]

Categories: A
What is the perpendicular projection of $P(1,-2,3)$ onto the xy-plane?

A. $(1,-2,0)$

B. $(0,-2,3)$

C. $(1,0,3)$

D. $(1,-2,-3)$

Question 5 [1 mark]

Categories: K, A
Which equation is the xy-plane trace of $x+2y+z=4$?

A. $x+z=4, y=0$

B. $2y+z=4, x=0$

C. $x=0, y=0$

D. $2y+x=4, z=0$

Question 6 [1 mark]

Categories: A
Find the x-axis intercept of $x+y+z=4$.

A. $(0,0,4)$

B. $(4,4,0)$

C. $(4,0,0)$

D. $(0,4,0)$

Part B Reasoning and communication

Question 7 [4 marks]

Categories: A, T, C
The plane contains $A(1,0,1)$, $B(2,1,1)$ and $C(1,1,2)$. Find vector, parametric and Cartesian equations. Explain why the plane is unique and check C.
Show your working and state a complete conclusion.

Question 8 [4 marks]

Categories: A, C
Find vector and Cartesian equations of the plane containing $L: \vec{r}=[1,1,0]+s[2,0,1]$ and $Q(2,2,2)$. Verify both the whole line and Q.
Show your working and state a complete conclusion.

Question 9 [4 marks]

Categories: K, A, C
Find the smaller angle between $x+y+z=4$ and $x+z=1$. Show exact magnitudes and an exact cosine ratio before rounding to one decimal place.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, T, C
Sketch $2x+3y+6z=12$ using labelled axes. Find all three intercepts and all three coordinate-plane trace equations. Explain why the triangle does not bound the plane.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$\vec{a} \times \vec{b} \neq \vec{0}$

Two independent in-plane directions are required. Their cross product must be nonzero.

1 Reasoning. Two independent in-plane directions are required. Their cross product must be nonzero.

$$\vec{r} = \vec{r}_0 + s\vec{a} + t\vec{b}, \quad \vec{a} \times \vec{b} \neq \vec{0}$$

Question 2 solution

Correct answer: C

$[-1, -1, 1]$

Crossing the independent spans gives a vector perpendicular to both.

1 Reasoning. Crossing the independent spans gives a vector perpendicular to both.

$$[1, 0, 1] \times [0, 1, 1] = [-1, -1, 1]$$

Question 3 solution

Correct answer: B

$-2x - 4y + 2z = -8$

Multiplying the complete equation by the same nonzero constant preserves every solution.

1 Reasoning. Multiplying the complete equation by the same nonzero constant preserves every solution.

$$-2(x + 2y - z) = -2(4)$$

Question 4 solution

Correct answer: A

$(1, -2, 0)$

The xy-plane has z=0. The projection preserves x and y and removes only the z displacement.

1 Reasoning. The xy-plane has z=0. The projection preserves x and y and removes only the z displacement.

$$P(1, -2, 3) \mapsto H(1, -2, 0)$$

Question 5 solution

Correct answer: D

$2y + x = 4, z = 0$

Set z to zero while allowing x and y to vary subject to the remaining linear relation.

1 Reasoning. Set z to zero while allowing x and y to vary subject to the remaining linear relation.

$$z = 0 \Rightarrow x + 2y = 4$$

Question 6 solution

Correct answer: C

$(4, 0, 0)$

Set y=z=0 at an x-axis intercept.

1 Reasoning. Set y=z=0 at an x-axis intercept.

$$y = z = 0 \Rightarrow x = 4$$

Question 7 solution

Model response

The spans are $[1, 1, 0]$ and $[0, 1, 1]$, with nonzero cross product $[1, -1, 1]$. Thus $\vec{r} = [1, 0, 1]+s[1, 1, 0]+t[0, 1, 1]$, $x = 1 + s, y = s + t, z = 1 + t$ and $x - y + z = 2$. C corresponds to $(s, t) = (0, 1)$.

Marking criteria

1. Obtains two valid spans (1 mark).
2. Checks independence and obtains a normal (1 mark).
3. Writes all three requested forms (1 mark).
4. Verifies C and explains uniqueness (1 mark).

The spans are $[1, 1, 0]$ and $[0, 1, 1]$, with nonzero cross product $[1, -1, 1]$. Thus $\vec{r} = [1, 0, 1]+s[1, 1, 0]+t[0, 1, 1]$, $x = 1 + s, y = s + t, z = 1 + t$ and $x - y + z = 2$. C corresponds to $(s, t) = (0, 1)$.

1 Spans. Subtract A from B and C.

$$\overrightarrow{AB} = [1, 1, 0], \quad \overrightarrow{AC} = [0, 1, 1]$$

2 Independence and normal. The cross product is nonzero.

$$[1, 1, 0] \times [0, 1, 1] = [1, -1, 1] \neq \vec{0}$$

3 Equations. Use the known point, then read components and expand point-normal form.

$$\vec{r} = [1, 0, 1] + s[1, 1, 0] + t[0, 1, 1], \quad x - y + z = 2$$

4 Check C. Use one parameter pair and substitute into Cartesian form.

$$(s,t)=(0,1), \qquad 1-1+2=2$$

Question 8 solution

Model response

A connector from the line anchor to Q is $[1,1,2]$. Hence $\vec{r}=[1,1,0]+u[2,0,1]+v[1,1,2]$. A normal is $[-1,-3,2]$, giving $x+3y-2z=4$. The line is obtained when $v=0$; Q is obtained at $(u,v)=(0,1)$.

Marking criteria

1. Finds the connector and checks independence (1 mark).
2. Constructs vector form with two parameters (1 mark).
3. Finds the correct normal and Cartesian equation (1 mark).
4. Verifies the whole line and external point (1 mark).

A connector from the line anchor to Q is $[1,1,2]$. Hence $\vec{r}=[1,1,0]+u[2,0,1]+v[1,1,2]$. A normal is $[-1,-3,2]$, giving $x+3y-2z=4$. The line is obtained when $v=0$; Q is obtained at $(u,v)=(0,1)$.

1 Connector. Subtract the line anchor from Q.

$$\overrightarrow{AQ}=[1,1,2]$$

2 Construct. The connector is independent of the line direction.

$$\vec{r}=[1,1,0]+u[2,0,1]+v[1,1,2]$$

3 Cartesian form. Cross the spans and use the anchor.

$$[2,0,1]\times[1,1,2]=[-1,-3,2], \quad x+3y-2z=4$$

4 Verify. Substitution reproduces both required objects.

$$v=0\Rightarrow L, \quad (u,v)=(0,1)\Rightarrow Q$$

Question 9 solution

Model response

The normals are $[1,1,1]$ and $[1,0,1]$. Their dot product is two and their magnitudes are $\sqrt{3}$ and $\sqrt{2}$. Thus $\cos\theta=2/\sqrt{6}$ and $\theta\approx 35.3^\circ$.

Marking criteria

1. Identifies both normals (1 mark).
2. Calculates the dot product and exact magnitudes (1 mark).
3. Obtains the correct exact cosine ratio (1 mark).

4. Calculates and labels the smaller angle (1 mark).

The normals are $[1,1,1]$ and $[1,0,1]$. Their dot product is two and their magnitudes are $\sqrt{3}$ and $\sqrt{2}$. Thus $\cos\theta = 2/\sqrt{6}$ and $\theta \approx 35.3^\circ$.

1 Normals. Read coefficients, including the missing y term.

$$\vec{n}_1 = [1,1,1], \quad \vec{n}_2 = [1,0,1]$$

2 Dot product and magnitudes. Keep the radicals exact.

$$\vec{n}_1 \cdot \vec{n}_2 = 2, \quad \|\vec{n}_1\| = \sqrt{3}, \quad \|\vec{n}_2\| = \sqrt{2}$$

3 Smaller angle. Use the absolute dot product.

$$\cos\theta = \frac{|2|}{\sqrt{3}\sqrt{2}} = \frac{2}{\sqrt{6}}$$

4 Approximate. Use inverse cosine in degree mode only at the end.

$$\theta = \cos^{-1}(2/\sqrt{6}) \approx 35.3^\circ$$

Question 10 solution

Model response

The intercepts are $(6,0,0)$, $(0,4,0)$ and $(0,0,2)$. The traces are $2x+3y=12, z=0$, $2x+6z=12, y=0$ and $3y+6z=12, x=0$. Join the intercepts to indicate orientation and show that the plane extends beyond the triangle.

Marking criteria

1. Finds and labels all intercepts (1 mark).
2. Obtains all three trace equations (1 mark).
3. Draws a coherent labelled plane sketch (1 mark).
4. Explains that the triangle is only a finite patch (1 mark).

The intercepts are $(6,0,0)$, $(0,4,0)$ and $(0,0,2)$. The traces are $2x+3y=12, z=0$, $2x+6z=12, y=0$ and $3y+6z=12, x=0$. Join the intercepts to indicate orientation and show that the plane extends beyond the triangle.

1 Intercepts. Set two coordinates to zero at each axis.

$$A(6,0,0), \quad B(0,4,0), \quad C(0,0,2)$$

2 xy trace. Set z to zero.

$$z=0: \quad 2x+3y=12$$

3 Other traces. Set y=0 and x=0 separately.

$$y=0:\ 2x+6z=12, \qquad x=0:\ 3y+6z=12$$

4 Sketch and interpretation. Draw labelled axes, points and trace lines. The plane is infinite in every in-plane direction.

4.3 Quiz 9.1 Intersections and systems

Covers Sections 9.1, 9.2, 9.3, 9.4.

Sections 9.1 to 9.4. A unit quiz covering the section homework, with worked feedback and written explanations.

10 questions 22 marks Suggested time: 35 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: A
Find the intersection parameter for $\vec{r} = [0, 0, 1] + t[1, 1, 1]$ with $x + y + z = 7$.

A. $t = 1$

B. $t = 3$

C. Every real t

D. $t = 2$

Question 2 [1 mark]

Categories: T
Classify $L : \vec{r} = [1, 0, 0] + t[1, -1, 0]$ relative to $\Pi : x + y = 1$.

A. Parallel and disjoint

B. The line is perpendicular to the plane

C. The line is contained in the plane

D. Exactly one intersection

Question 3 [1 mark]

Categories: T
Classify $L : \vec{r} = [1, 0, 1] + t[1, -1, 0]$ relative to $\Pi : x + y + z = 3$.

A. The line is normal to the plane

B. Parallel and disjoint

C. Contained in the plane

D. Exactly one intersection

Question 4 [1 mark]

Categories: K, T
Classify $L_1 : \vec{r} = [1, 0, 0] + s[1, 0, 0]$ and $L_2 : \vec{r} = [0, 1, 1] + t[0, 1, 0]$.

A. Skew lines

B. Coincident lines

C. Distinct parallel lines

D. Intersecting perpendicular lines

Question 5 [1 mark]

Categories: A
Which vector is parallel to the intersection of $x + y + z = 1$ and $x - y + z = 1$?

A. $[1, 0, 1]$

B. $[0, 1, 0]$

C. $[1, 1, 1]$

D. $[1, 0, -1]$

Question 6 [1 mark]

Categories: A, T
What is the common solution set of $x + y = 1$, $y + z = 1$ and $x + z = 1$?

A. A plane

B. The empty set

C. $\{(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})\}$

D. An infinite line

Part B Reasoning and communication

Question 7 [4 marks]

Categories: A, C
Find the intersection of $L : \vec{r} = [1, 0, 1] + t[2, 1, -1]$ with $\Pi : x + y + z = 6$. Give the point and verify it in both objects.
Show your working and state a complete conclusion.

Question 8 [4 marks]

Categories: A, T, C
Determine whether $L_1 : \vec{r} = [1, 0, 1] + s[1, 2, -1]$ and $L_2 : \vec{r} = [3, 4, -1] + u[0, 1, 1]$ intersect. Solve using separate parameters and check all coordinates. State the full line relationship.
Show your working and state a complete conclusion.

Question 9 [4 marks]

Categories: A, T, C
Find a vector equation of the intersection of $\Pi_1 : x + y + z = 2$ and $\Pi_2 : x - y + z = 0$. Verify its anchor and direction against both planes.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: T, C
For $\Pi_1 : x + y + z = 1$, $\Pi_2 : x - y + z = 1$ and $\Pi_3 : 2x + 2z = k$, find every value of k for which the system is consistent. Describe the complete common solution set in every case.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$t = 2$

Substitute all three coordinate expressions into the plane before solving for the shared parameter.

1 Reasoning. Substitute all three coordinate expressions into the plane before solving for the shared parameter.

$$t + t + (1 + t) = 7 \Rightarrow 3t = 6 \Rightarrow t = 2$$

Question 2 solution

Correct answer: C

The line is contained in the plane

Substitution is an identity for every real t, so every point on the line belongs to the plane.

1 Reasoning. Substitution is an identity for every real t, so every point on the line belongs to the plane.

$$(1 + t) - t = 1$$

Question 3 solution

Correct answer: B

Parallel and disjoint

The parameter cancels, leaving a false statement. The direction is parallel to the plane but the line has no common point with it.

1 Reasoning. The parameter cancels, leaving a false statement. The direction is parallel to the plane but the line has no common point with it.

$$(1 + t) - t + 1 = 2 \neq 3$$

Question 4 solution

Correct answer: A

Skew lines

Their nonparallel directions do not guarantee intersection. The first has z=0 and the second z=1, so they cannot meet.

1 Reasoning. Their nonparallel directions do not guarantee intersection. The first has z=0 and the second z=1, so they cannot meet.

$$z_{L_1} = 0, \quad z_{L_2} = 1, \quad [1, 0, 0] \nparallel [0, 1, 0]$$

Question 5 solution

Correct answer: D

$[1, 0, -1]$

A line in both planes has direction perpendicular to both plane normals.

1 Reasoning. A line in both planes has direction perpendicular to both plane normals.

$$[1, 1, 1] \times [1, -1, 1] = [2, 0, -2] = 2[1, 0, -1]$$

Question 6 solution

Correct answer: C

$\{(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})\}$

Subtracting pairs gives x=y=z. Substituting into any original equation gives twice each coordinate equal to the constant.

1 Reasoning. Subtracting pairs gives x=y=z. Substituting into any original equation gives twice each coordinate equal to the constant.

$$x = y = z, \quad 2x = 1 \Rightarrow x = y = z = \frac{1}{2}$$

Question 7 solution

Model response

Substitution gives $1 + 1 + 2t = 6$, so $t = 2$. The intersection is $P(5, 2, -1)$. Its coordinates come from one shared line parameter and satisfy the plane because $5 + 2 + -1 = 6$.

Marking criteria

1. Forms the correct scalar equation (1 mark).
2. Solves the intersection parameter (1 mark).
3. Calculates all coordinates consistently (1 mark).
4. Verifies the point in both objects (1 mark).

Substitution gives $1 + 1 + 2t = 6$, so $t = 2$. The intersection is $P(5, 2, -1)$. Its coordinates come from one shared line parameter and satisfy the plane because $5 + 2 + -1 = 6$.

1 Substitute. Use all three line coordinates.

$$(1 + 2t) + t + (1 - t) = 6$$

2 Solve. Collect the parameter terms.

$$2t = 4 \Rightarrow t = 2$$

3 Point. Substitute the same value into x,y,z.

$$P = (5, 2, -1)$$

4 Verify. Check the original line and plane.

$$[1,0,1]+2[2,1,-1]=[5,2,-1], \quad 5+2+(-1)=6$$

Question 8 solution

Model response

The x equation forces $s=2$; the y equation gives $u=0$. Both z values are negative one, so the lines intersect at $(3,4,-1)$. Their directions are not parallel.

Marking criteria

1. Writes corresponding coordinate equations with separate parameters (1 mark).
2. Solves for the candidate parameter pair (1 mark).
3. Checks the z-coordinate explicitly (1 mark).
4. States and justifies the complete relationship (1 mark).

The x equation forces $s=2$; the y equation gives $u=0$. Both z values are negative one, so the lines intersect at $(3,4,-1)$. Their directions are not parallel.

1 x equation. Use separate parameters for the two lines.

$$1+s=3 \Rightarrow s=2$$

2 y equation. Use the value of s to solve u.

$$2s=4+u \Rightarrow u=0$$

3 z check. Check the third equation rather than assuming agreement.

$$1-s=-1=-1+u$$

4 Classify. The directions are not scalar multiples; use the coordinate check to distinguish an intersection from skew lines.

$$L_1 \cap L_2 = \{(3,4,-1)\}$$

Question 9 solution

Model response

Subtract the equations to obtain $y=1$. Then $z=1-x$. One intersection line is $\vec{r}=[0,1,1]+t[1,0,-1]$. The anchor satisfies both equations, and the direction has zero dot product with both normals.

Marking criteria

1. Eliminates a variable correctly (1 mark).
2. Introduces a free parameter and obtains all coordinates (1 mark).
3. Writes a correct intersection line (1 mark).

4. Checks both location and direction in both planes (1 mark).

Subtract the equations to obtain $y = 1$. Then $z = 1 - x$. One intersection line is $\vec{r} = [0, 1, 1] + t[1, 0, -1]$. The anchor satisfies both equations, and the direction has zero dot product with both normals.

1 Eliminate. Subtract the second equation from the first.

$$2y = 2 \Rightarrow y = 1$$

2 Free variable. Let x=t and solve z.

$$x = t, \quad y = 1, \quad z = 1 - t$$

3 Vector form. Separate constants from parameter coefficients.

$$\vec{r} = [0, 1, 1] + t[1, 0, -1]$$

4 Verify. The anchor lies on both planes and the direction is perpendicular to both normals.

$$0 + 1 + 1 = 2, \quad 0 - 1 + 1 = 0, \quad [1, 0, -1] \cdot [1, \pm 1, 1] = 0$$

Question 10 solution

Model response

Adding the first two equations gives $2x + 2z = 2$. Thus $k = 2$ makes the third equation redundant and the common set is $\vec{r} = [0, 0, 1] + t[1, 0, -1]$. For every other k the system is inconsistent. No value gives exactly one point because the third equation either repeats or contradicts the existing constraint.

Marking criteria

1. Identifies the dependent combination of equations (1 mark).
2. Obtains the exact consistency condition (1 mark).
3. Parameterizes the full common line (1 mark).
4. Classifies all other values and explains why a unique point is impossible (1 mark).

Adding the first two equations gives $2x + 2z = 2$. Thus $k = 2$ makes the third equation redundant and the common set is $\vec{r} = [0, 0, 1] + t[1, 0, -1]$. For every other k the system is inconsistent. No value gives exactly one point because the third equation either repeats or contradicts the existing constraint.

1 Compare equations. Add the first two constraints.

$$(x + y + z) + (x - y + z) = 2x + 2z = 2$$

2 Consistency. The third equation must have the same right side.

$$k = 2$$

3 Common line. Subtract the first two equations and parameterize the remaining relation.

$$y = 0, \quad z = 1 - x, \quad \vec{r} = [0, 0, 1] + t[1, 0, -1]$$

4 Other cases. A different right side produces a contradiction.

$$k \neq 2 \Rightarrow \Pi_1 \cap \Pi_2 \cap \Pi_3 = \varnothing$$

4.4 Quiz 9.2 Shortest distances

Covers Sections 9.5, 9.6.

Sections 9.5 to 9.6. A unit quiz covering the section homework, with worked feedback and written explanations.

10 questions 22 marks Suggested time: 35 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: A
Find the distance from $P(0,1)$ to $3x+4y=0$.

A. 1

B. $\frac{3}{5}$

C. 4

D. $\frac{4}{5}$

Question 2 [1 mark]

Categories: A
Find the distance from $P(0,0,1)$ to $L:\vec{r}=t[1,0,0]$.

A. 2

B. 3

C. 1

D. 0

Question 3 [1 mark]

Categories: K
The plane is $2x-y+2z=1$. What is the magnitude of a normal used in the Cartesian point-plane distance formula?

A. $\sqrt{5}$

B. 3

C. 5

D. 9

Question 4 [1 mark]

Categories: K
What is the distance from $P(1,0,0)$ to $x=1$?

A. 0

B. 1

C. 2

D. 3

Question 5 [1 mark]

Categories: T
A student multiplies $x+2y+2z=1$ by five before calculating the distance from a fixed point. What happens to the distance?

A. It becomes five times larger

B. It becomes one fifth as large

C. It becomes zero

D. It remains unchanged

Question 6 [1 mark]

Categories: K, T
For a point outside $\Pi: x+2y+2z=1$, which condition must its shortest connector to the plane satisfy?

A. Its endpoint may be outside the plane

B. Its direction has zero magnitude

C. Its direction is parallel to [1,2,2]

D. Its direction is parallel to the plane

Part B Reasoning and communication

Question 7 [4 marks]

Categories: A, T, C
For $P(1,2)$ and $L: 3x+4y=1$, find the exact distance and nearest point H. Verify that H lies on L and that PH is perpendicular to L.
Show your working and state a complete conclusion.

Question 8 [4 marks]

Categories: A, T, C
Find the exact distance from $P(2,1,3)$ to $L: \vec{r}=[1,1,0]+t[1,1,0]$. Use projection or the cross-product area method, then find and verify the perpendicular foot.
Show your working and state a complete conclusion.

Question 9 [4 marks]

Categories: A, C
For $P(1,1,4)$ and $\Pi: x+2y+2z=4$, find the exact distance and the nearest point H. Verify both plane membership and the normal direction of PH.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: T, C
A student claims that the shortest distance from $P(1,1,3)$ to the x-axis is its distance to the origin. Assess the claim by finding the correct foot, comparing the two exact lengths and explaining the geometric condition.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$\frac{4}{5}$

Divide the absolute Cartesian residual by the magnitude of the nonzero normal.

1 Reasoning. Divide the absolute Cartesian residual by the magnitude of the nonzero normal.

$$d = \frac{|3(0) + 4(1)|}{\sqrt{3^2 + 4^2}} = \frac{4}{5}$$

Question 2 solution

Correct answer: C

1

The line is the x-axis. The perpendicular foot is the origin for this particular point.

1 Reasoning. The line is the x-axis. The perpendicular foot is the origin for this particular point.

$$d = \sqrt{0^2 + 1^2} = 1$$

Question 3 solution

Correct answer: B

3

The normal is [2,-1,2], whose squared magnitude is four plus one plus four.

1 Reasoning. The normal is [2,-1,2], whose squared magnitude is four plus one plus four.

$$\|[2, -1, 2]\| = \sqrt{4 + 1 + 4} = 3$$

Question 4 solution

Correct answer: A

0

The point satisfies the plane equation, so it already lies on the plane.

1 Reasoning. The point satisfies the plane equation, so it already lies on the plane.

$$|1 - 1| = 0$$

Question 5 solution

Correct answer: D

It remains unchanged

Both the absolute residual and the normal magnitude gain the same factor of five.

1 Reasoning. Both the absolute residual and the normal magnitude gain the same factor of five.

$$\frac{|5(Ax_0 + By_0 + Cz_0 - D)|}{\|5\vec{n}\|} = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\|\vec{n}\|}$$

Question 6 solution

Correct answer: C

Its direction is parallel to [1,2,2]

The shortest connector is perpendicular to the plane, so it follows a nonzero normal direction and ends on the plane.

1 Reasoning. The shortest connector is perpendicular to the plane, so it follows a nonzero normal direction and ends on the plane.

$$\overrightarrow{HP} = \lambda[1, 2, 2], \qquad H \in \Pi$$

Question 7 solution

Model response

The residual is 10, so $d = 2$. The foot is $H(\frac{-1}{5}, \frac{2}{5})$. It satisfies $3x + 4y = 1$; $P - H = \frac{10}{25}[3, 4]$ is parallel to the line normal and perpendicular to direction $[4, -3]$.

Marking criteria

1. Calculates the exact distance using a nonzero normal (1 mark).
2. Finds the exact perpendicular foot (1 mark).
3. Checks that H lies on the line (1 mark).
4. Verifies the perpendicular connector (1 mark).

The residual is 10, so $d = 2$. The foot is $H(\frac{-1}{5}, \frac{2}{5})$. It satisfies $3x + 4y = 1$; $P - H = \frac{10}{25}[3, 4]$ is parallel to the line normal and perpendicular to direction $[4, -3]$.

1 Distance. Use the absolute residual and the normal magnitude.

$$d = \frac{|3(1) + 4(2) - 1|}{5} = 2$$

2 Foot. Remove the normal displacement from P.

$$H = [1, 2] - \frac{10}{25}[3, 4] = [\frac{-1}{5}, \frac{2}{5}]$$

3 Membership. Substitute the exact foot coordinates.

$$3(\frac{-1}{5}) + 4(\frac{2}{5}) = 1$$

4 Perpendicularity. The connector is a multiple of the normal.

$$[3, 4] \cdot [4, -3] = 12 - 12 = 0$$

Question 8 solution

Model response

The displacement from the anchor to P is $[1, 0, 3]$. Projection gives $t = 1/2$, so $H(3/2, \frac{3}{2}, 0)$. The residual is $[1/2, -1/2, 3]$, perpendicular to the line direction. Its magnitude is $\sqrt{38}/2$.

Marking criteria

1. Obtains the correct anchor displacement and direction (1 mark).
2. Uses projection or cross-product distance correctly (1 mark).
3. Calculates and verifies the perpendicular foot (1 mark).
4. Reports the exact distance with reasoning (1 mark).

The displacement from the anchor to P is $[1, 0, 3]$. Projection gives $t = 1/2$, so $H(3/2, \frac{3}{2}, 0)$. The residual is $[1/2, -1/2, 3]$, perpendicular to the line direction. Its magnitude is $\sqrt{38}/2$.

1 Displacement. Subtract the line anchor from P.

$$\vec{w} = [1, 0, 3], \quad \vec{d} = [1, 1, 0]$$

2 Projection parameter. Project w onto the line direction.

$$t = \frac{\vec{w} \cdot \vec{d}}{\vec{d} \cdot \vec{d}} = \frac{1}{2}$$

3 Foot and check. Use t=1/2 in the line, then check the residual.

$$H = [\tfrac{3}{2}, \frac{3}{2}, 0], \quad [\tfrac{1}{2}, -\tfrac{1}{2}, 3] \cdot [1, 1, 0] = 0$$

4 Exact length. Compute the magnitude without premature rounding.

$$d = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + 9} = \frac{\sqrt{38}}{2}$$

Question 9 solution

Model response

The normal is $[1,2,2]$ with magnitude three. The residual is 7, giving $d = \frac{7}{3}$. The foot is $H(\frac{2}{9}, \frac{-5}{9}, \frac{22}{9})$. Substitution gives four, and $P - H = \frac{7}{9}[1,2,2]$.

Marking criteria

1. Identifies the normal and evaluates the residual (1 mark).
2. Obtains the exact distance (1 mark).
3. Computes the perpendicular foot (1 mark).
4. Verifies membership and perpendicularity (1 mark).

The normal is $[1,2,2]$ with magnitude three. The residual is 7, giving $d = \frac{7}{3}$. The foot is $H(\frac{2}{9}, \frac{-5}{9}, \frac{22}{9})$. Substitution gives four, and $P - H = \frac{7}{9}[1,2,2]$.

1 Residual and normal. Evaluate the Cartesian expression at P.

$$r = 1 + 2 + 8 - 4 = 7, \quad \|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$$

2 Distance. Take the absolute scalar residual over normal magnitude.

$$d = \frac{7}{3} = \frac{7}{3}$$

3 Nearest point. Subtract the required normal displacement.

$$H = [1,1,4] - \frac{7}{9}[1,2,2]$$

4 Verify. The point lies on the plane and the connector is normal to it.

$$x_H + 2y_H + 2z_H = 4, \quad P - H = \frac{7}{9}\vec{n}$$

Question 10 solution

Model response

The foot is $H(1,0,0)$, giving $PH = \sqrt{10}$. The origin is on the line, but $PO = \sqrt{11}$ is longer. The correct connector has zero dot product with the x-axis direction. A point on the target is not necessarily the nearest point.

Marking criteria

1. Finds the correct perpendicular foot (1 mark).
2. Checks the zero dot product (1 mark).
3. Compares both exact connector lengths (1 mark).
4. Explains why target membership alone is insufficient (1 mark).

The foot is $H(1,0,0)$, giving $PH = \sqrt{10}$. The origin is on the line, but $PO = \sqrt{11}$ is longer. The correct connector has zero dot product with the x-axis direction. A point on the target is not necessarily the nearest point.

1 Foot. A point on the x-axis has y=z=0; choose the same x-coordinate as P.

$$H = (1, 0, 0)$$

2 Perpendicularity. Check the connector against the line direction.

$$[0, 1, 3] \cdot [1, 0, 0] = 0$$

3 Compare lengths. Keep both magnitudes exact.

$$PH = \sqrt{10} < \sqrt{11} = PO$$

4 Conclusion. Reject the claim. The shortest connector must meet the target at a right angle, which the connector to the origin does not.

5 Unit tests

5.1 Test 8.1 Line equations

Covers Sections 8.1, 8.2, 8.3.

Sections 8.1 to 8.3. An independent unit test with extended written reasoning and complete marking guidance.

12 questions 32 marks Suggested time: 60 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: K
For the R² line $\vec{r} = [3, -1] + t[2, 3]$, which vector is parallel to the line?

A. $[-3, 2]$

B. $[3, -1]$

C. $[2, -3]$

D. $[2, 3]$

Question 2 [1 mark]

Categories: A
Find the point on $\vec{r} = [3, -1] + t[2, 3]$ when $t = 2$.

A. $(7, -7)$

B. $(-1, 5)$

C. $(7, 5)$

D. $(5, 2)$

Question 3 [1 mark]

Categories: K
Which vector is a normal to the Cartesian line $2x - 3y = 7$?

A. $[-3, 2]$

B. $[2, -3]$

C. $[3, 2]$

D. $[2, 3]$

Question 4 [1 mark]

Categories: K, A
Which symmetric description represents $\vec{r} = [3, -1, 2] + t[0, 1, -2]$?
A. $x = 3, \quad \frac{y+1}{1} = \frac{z-2}{-2}$
B. $x = 3, \quad \frac{y-1}{1} = \frac{z+2}{-2}$
C. $y = -1, \quad \frac{x-3}{1} = \frac{z-2}{-2}$
D. $z = 2, \quad x = 3 + t, \; y = -1 - 2t$

Question 5 [1 mark]

Categories: T
Classify $L_1 : \vec{r} = [3, -1, 2] + t[2, -1, 3]$ and $L_2 : \vec{r} = [5, -2, 5] + u[-4, 2, -6]$.
A. Distinct parallel lines
B. Perpendicular intersecting lines
C. Skew lines
D. Coincident lines

Question 6 [1 mark]

Categories: A
The directions of two R² lines through $(3, 0)$ are $[1, 1]$ and $[1, 0]$. What is their smaller angle?
A. 90°
B. 60°
C. 45°
D. 0°

Question 7 [1 mark]

Categories: T
What set is described by $\vec{r} = [3, 2, -1] + t[0, 0, 0]$, $t \in \mathbb{R}$?
A. All of R³
B. One point
C. An infinite line
D. A plane

Question 8 [1 mark]

Categories: K, A
A line in R² has direction $[2, 3]$. Which vector is perpendicular to this direction?
A. $[3, -2]$
B. $[2, 3]$
C. $[3, 2]$
D. $[1, 3]$

Part B Reasoning and communication

Question 9 [6 marks]

Categories: A, C
The line passes through $A(3,2)$ and $B(9,6)$. (a) Find a simplified direction. (b) Write vector and parametric equations. (c) Obtain Cartesian form. (d) Verify B using one parameter.
Show your working and state a complete conclusion.

Question 10 [6 marks]

Categories: A, T, C
For $A(3,-1,2)$ and $B(3,3,-6)$, (a) write vector and symmetric equations of AB, (b) explain the fixed coordinate, and (c) determine whether $Q(3,1,-2)$ lies on the line.
Show your working and state a complete conclusion.

Question 11 [6 marks]

Categories: K, A, C
Find k so the R² lines $kx+2y=1$ and $3x+3y=2$ are perpendicular. Show the normal-vector test and justify your conclusion.
Show your working and state a complete conclusion.

Question 12 [6 marks]

Categories: T, C
Let $L:\vec{r}=[3,0,1]+t[1,2,-1]$. Compare it with $M:\vec{r}=[6,6,-2]+u[-2,-4,2]$ and $N:\vec{r}=[6,6,0]+v[-2,-4,2]$. Establish whether each pair is coincident or distinct parallel.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$[2, 3]$

The coefficient of the real parameter is the direction vector. The constant vector locates a starting point.

1 Reasoning. The coefficient of the real parameter is the direction vector. The constant vector locates a starting point.

$$\vec{r} = \vec{r}_0 + t\vec{d}, \qquad \vec{d} = [2, 3]$$

Question 2 solution

Correct answer: C

$(7, 5)$

Multiply both direction components by two, then add the corresponding starting coordinates.

1 Reasoning. Multiply both direction components by two, then add the corresponding starting coordinates.

$$[3, -1] + 2[2, 3] = [7, 5]$$

Question 3 solution

Correct answer: B

$[2, -3]$

The x and y coefficients form a nonzero normal. A direction must be perpendicular to it.

1 Reasoning. The x and y coefficients form a nonzero normal. A direction must be perpendicular to it.

$$\vec{n} = [2, -3], \qquad [2, -3] \cdot [3, 2] = 0$$

Question 4 solution

Correct answer: A

$x = 3, \quad \frac{y+1}{1} = \frac{z-2}{-2}$

The zero x-direction component makes x constant. Only the y and z equations can be divided by nonzero direction components.

1 Reasoning. The zero x-direction component makes x constant. Only the y and z equations can be divided by nonzero direction components.

$$x = 3, \quad y = -1 + t, \quad z = 2 - 2t$$

Question 5 solution

Correct answer: D

Coincident lines

The second direction is negative two times the first. Its starting point is reached on the first line at t=1, so the lines coincide.

1 Reasoning. The second direction is negative two times the first. Its starting point is reached on the first line at t=1, so the lines coincide.

$$[-4, 2, -6] = -2[2, -1, 3], \quad [3, -1, 2] + [2, -1, 3] = [5, -2, 5]$$

Question 6 solution

Correct answer: C

45°

The absolute dot product is one and the magnitudes are square root two and one, giving cosine one over square root two.

1 Reasoning. The absolute dot product is one and the magnitudes are square root two and one, giving cosine one over square root two.

$$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Question 7 solution

Correct answer: B

One point

The zero direction produces no displacement for any parameter. The equation returns only its constant position.

1 Reasoning. The zero direction produces no displacement for any parameter. The equation returns only its constant position.

$$\vec{r} = [3, 2, -1] \quad \text{for every } t$$

Question 8 solution

Correct answer: A

$[3, -2]$

Swapping components and reversing one sign gives a zero dot product.

1 Reasoning. Swapping components and reversing one sign gives a zero dot product.

$$[2, 3] \cdot [3, -2] = 6 - 6 = 0$$

Question 9 solution

Model response

A direction is $[3, 2]$. The equations are $\vec{r} = [3, 2] + t[3, 2]$, $x = 3 + 3t, y = 2 + 2t$, and $2x - 3y = 0$. The point B is reached when $t = 2$.

Marking criteria

1. Obtains a nonzero direction parallel to AB (2 marks).
2. Writes both forms with one shared parameter (2 marks).
3. Derives the correct Cartesian equation (1 mark).
4. Checks both coordinates of B (1 mark).

A direction is $[3, 2]$. The equations are $\vec{r} = [3, 2] + t[3, 2]$, $x = 3 + 3t, y = 2 + 2t$, and $2x - 3y = 0$. The point B is reached when $t = 2$.

1 Direction. Subtract A from B and divide by two.

$$\overrightarrow{AB} = [6, 4] = 2[3, 2]$$

2 Represent. Use A as the starting point.

$$\vec{r} = [3, 2] + t[3, 2], \quad x = 3 + 3t,\ y = 2 + 2t$$

3 Eliminate. Both expressions equal the same parameter.

$$\frac{x - 3}{3} = \frac{y - 2}{2} \Rightarrow 2x - 3y = 0$$

4 Verify. Use the same parameter in both coordinates.

$$t = 2: \quad [3, 2] + 2[3, 2] = [9, 6]$$

Question 10 solution

Model response

Use direction $[0, 1, -2]$. Then $\vec{r} = [3, -1, 2] + t[0, 1, -2]$ and $x = 3,\ (y + 1)/1 = (z - 2)/(-2)$. The x-coordinate stays fixed. Q lies on the line at $t = 2$; B is reached at $t = 4$.

Marking criteria

1. Obtains a valid simplified direction (2 marks).
2. Writes vector and symmetric forms (2 marks).
3. Explains why division by the zero component is invalid (1 mark).
4. Checks Q using one shared parameter (1 mark).

Use direction $[0, 1, -2]$. Then $\vec{r} = [3, -1, 2] + t[0, 1, -2]$ and $x = 3,\ (y + 1)/1 = (z - 2)/(-2)$. The x-coordinate stays fixed. Q lies on the line at $t = 2$; B is reached at $t = 4$.

1 Direction. Simplify the displacement without changing its orientation.

$$\overrightarrow{AB} = [0, 4, -8] = 4[0, 1, -2]$$

2 Vector form. Use A and a nonzero parallel direction.

$$\vec{r} = [3, -1, 2] + t[0, 1, -2]$$

3 Symmetric form. Retain x separately because its direction component is zero.

$$x = 3, \qquad \frac{y+1}{1} = \frac{z-2}{-2}$$

4 Membership. Check all coordinates with t=2.

$$[3, -1, 2] + 2[0, 1, -2] = [3, 1, -2]$$

Question 11 solution

Model response

The normals are $[k, 2]$ and $[3, 3]$. Their dot product must vanish, so $3k + 6 = 0$ and $k = -2$. Both normals remain nonzero, and in R² this establishes perpendicular line directions.

Marking criteria

1. Identifies both normal vectors (2 marks).
2. Uses the correct zero-dot-product condition (2 marks).
3. Solves for k exactly (1 mark).
4. Explains why the test applies to lines in R² (1 mark).

The normals are $[k, 2]$ and $[3, 3]$. Their dot product must vanish, so $3k + 6 = 0$ and $k = -2$. Both normals remain nonzero, and in R² this establishes perpendicular line directions.

1 Normals. Read the coefficients of x and y.

$$\vec{n}_1 = [k, 2], \quad \vec{n}_2 = [3, 3]$$

2 Perpendicularity. For lines in R², perpendicular normals give perpendicular line directions.

$$[k, 2] \cdot [3, 3] = 0$$

3 Solve. Expand and solve the scalar equation.

$$3k + 6 = 0 \Rightarrow k = -2$$

4 Check. Substitute the value into the dot product.

$$3\,(-2) + 6 = 0$$

Question 12 solution

Model response

All directions are parallel because $[-2,-4,2] = -2[1,2,-1]$. The anchor of M lies on L at $t = 3$, so M and L coincide. The same x and y force $t = 3$ for the anchor of N, but L gives $z = -2$ rather than zero. Thus N and L are distinct parallel lines.

Marking criteria

1. Compares all direction vectors (2 marks).
2. Verifies a common point for L and M (2 marks).
3. Shows why the anchor of N fails (1 mark).
4. States both relationships with complete reasoning (1 mark).

All directions are parallel because $[-2,-4,2] = -2[1,2,-1]$. The anchor of M lies on L at $t = 3$, so M and L coincide. The same x and y force $t = 3$ for the anchor of N, but L gives $z = -2$ rather than zero. Thus N and L are distinct parallel lines.

1 Directions. Compare scalar multiples.

$$\vec{d}_M = \vec{d}_N = -2\vec{d}_L$$

2 Shared point. Test the starting point of M.

$$t = 3: \quad [3,0,1] + 3[1,2,-1] = [6,6,-2]$$

3 Failed point. The anchor of N has the wrong z-coordinate.

$$x = 6 \Rightarrow t = 3, \quad z = 1 - 3 = -2 \neq 0$$

4 Classify. Parallel directions require a separate location check.

$$L = M, \qquad L \parallel N, \quad L \neq N$$

5.2 Test 8.2 Plane equations and sketching

Covers Sections 8.4, 8.5, 8.6.

Sections 8.4 to 8.6. An independent unit test with extended written reasoning and complete marking guidance.

12 questions 32 marks Suggested time: 60 min

1. Use exact fractions and radicals unless an approximation is requested.
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Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: K
A plane starts at $A(3,0,1)$. Which condition on its proposed spans $\vec{a},\vec{b}$ ensures they determine a unique plane?

A. $\vec{a}\times\vec{b}=\vec{0}$

B. $\vec{a}=2\vec{b}$

C. Both spans are zero

D. $\vec{a}\times\vec{b}\neq\vec{0}$

Question 2 [1 mark]

Categories: A
The plane $\vec{r}=[3,0,1]+s[1,0,1]+t[0,1,1]$ has which normal?

A. $[1,0,1]$

B. $[0,1,1]$

C. $[-1,-1,1]$

D. $[1,1,1]$

Question 3 [1 mark]

Categories: T
Which equation describes the same plane as $x+2y-z=6$?

A. $x-2y-z=6$

B. $-2x-4y+2z=-12$

C. $-2x-4y+2z=12$

D. $x+2y+z=6$

Question 4 [1 mark]

Categories: A
What is the perpendicular projection of $P(3, -2, 3)$ onto the xy-plane?

A. $(3, -2, 0)$

B. $(0, -2, 3)$

C. $(3, 0, 3)$

D. $(3, -2, -3)$

Question 5 [1 mark]

Categories: K, A
Which equation is the xy-plane trace of $x + 2y + z = 6$?

A. $x + z = 6, y = 0$

B. $2y + z = 6, x = 0$

C. $x = 0, y = 0$

D. $2y + x = 6, z = 0$

Question 6 [1 mark]

Categories: A
Find the x-axis intercept of $x + y + z = 6$.

A. $(0, 0, 6)$

B. $(6, 6, 0)$

C. $(6, 0, 0)$

D. $(0, 6, 0)$

Question 7 [1 mark]

Categories: T
For $x + z = 6$, which coordinate may be chosen freely?

A. No coordinate can vary

B. y

C. x only

D. z only

Question 8 [1 mark]

Categories: K, A
Which plane is perpendicular to $x - y + z = 3$?

A. $x + y = 2$

B. $2x - 2y + 2z = 3$

C. $x - y + z = 0$

D. $x - y = 1$

Part B Reasoning and communication

Question 9 [6 marks]

Categories: A, T, C
The plane contains $A(3,0,1)$, $B(4,1,1)$ and $C(3,1,2)$. Find vector, parametric and Cartesian equations. Explain why the plane is unique and check C.
Show your working and state a complete conclusion.

Question 10 [6 marks]

Categories: A, C
Find vector and Cartesian equations of the plane containing $L : \vec{r} = [3,1,0] + s[2,0,1]$ and $Q(4,2,2)$. Verify both the whole line and Q.
Show your working and state a complete conclusion.

Question 11 [6 marks]

Categories: K, A, C
Find the smaller angle between $x + y + z = 6$ and $x + z = 3$. Show exact magnitudes and an exact cosine ratio before rounding to one decimal place.
Show your working and state a complete conclusion.

Question 12 [6 marks]

Categories: A, T, C
Sketch $2x + 3y + 6z = 24$ using labelled axes. Find all three intercepts and all three coordinate-plane trace equations. Explain why the triangle does not bound the plane.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$\vec{a} \times \vec{b} \neq \vec{0}$

Two independent in-plane directions are required. Their cross product must be nonzero.

1 Reasoning. Two independent in-plane directions are required. Their cross product must be nonzero.

$$\vec{r} = \vec{r}_0 + s\vec{a} + t\vec{b}, \quad \vec{a} \times \vec{b} \neq \vec{0}$$

Question 2 solution

Correct answer: C

$[-1, -1, 1]$

Crossing the independent spans gives a vector perpendicular to both.

1 Reasoning. Crossing the independent spans gives a vector perpendicular to both.

$$[1, 0, 1] \times [0, 1, 1] = [-1, -1, 1]$$

Question 3 solution

Correct answer: B

$-2x - 4y + 2z = -12$

Multiplying the complete equation by the same nonzero constant preserves every solution.

1 Reasoning. Multiplying the complete equation by the same nonzero constant preserves every solution.

$$-2(x + 2y - z) = -2(6)$$

Question 4 solution

Correct answer: A

$(3, -2, 0)$

The xy-plane has z=0. The projection preserves x and y and removes only the z displacement.

1 Reasoning. The xy-plane has z=0. The projection preserves x and y and removes only the z displacement.

$$P(3, -2, 3) \mapsto H(3, -2, 0)$$

Question 5 solution

Correct answer: D

$2y + x = 6, z = 0$

Set z to zero while allowing x and y to vary subject to the remaining linear relation.

1 Reasoning. Set z to zero while allowing x and y to vary subject to the remaining linear relation.

$$z = 0 \Rightarrow x + 2y = 6$$

Question 6 solution

Correct answer: C

$(6, 0, 0)$

Set y=z=0 at an x-axis intercept.

1 Reasoning. Set y=z=0 at an x-axis intercept.

$$y = z = 0 \Rightarrow x = 6$$

Question 7 solution

Correct answer: B

y

The missing y term means translating in the y direction leaves the equation unchanged. X and z remain linked.

1 Reasoning. The missing y term means translating in the y direction leaves the equation unchanged. X and z remain linked.

$$x = 6 - z, \quad y \in \mathbb{R}$$

Question 8 solution

Correct answer: A

$x + y = 2$

The normals [1,-1,1] and [1,1,0] have zero dot product. The other proposed normals are not perpendicular to the reference normal.

1 Reasoning. The normals [1,-1,1] and [1,1,0] have zero dot product. The other proposed normals are not perpendicular to the reference normal.

$$[1, -1, 1] \cdot [1, 1, 0] = 1 - 1 + 0 = 0$$

Question 9 solution

Model response

The spans are $[1,1,0]$ and $[0,1,1]$, with nonzero cross product $[1,-1,1]$. Thus $\vec{r}=[3,0,1]+s[1,1,0]+t[0,1,1]$, $x=3+s, y=s+t, z=1+t$ and $x-y+z=4$. C corresponds to $(s,t)=(0,1)$.

Marking criteria

1. Obtains two valid spans (2 marks).
2. Checks independence and obtains a normal (2 marks).
3. Writes all three requested forms (1 mark).
4. Verifies C and explains uniqueness (1 mark).

The spans are $[1,1,0]$ and $[0,1,1]$, with nonzero cross product $[1,-1,1]$. Thus $\vec{r}=[3,0,1]+s[1,1,0]+t[0,1,1]$, $x=3+s, y=s+t, z=1+t$ and $x-y+z=4$. C corresponds to $(s,t)=(0,1)$.

1 Spans. Subtract A from B and C.

$$\overrightarrow{AB}=[1,1,0], \quad \overrightarrow{AC}=[0,1,1]$$

2 Independence and normal. The cross product is nonzero.

$$[1,1,0]\times[0,1,1]=[1,-1,1]\neq\vec{0}$$

3 Equations. Use the known point, then read components and expand point-normal form.

$$\vec{r}=[3,0,1]+s[1,1,0]+t[0,1,1], \quad x-y+z=4$$

4 Check C. Use one parameter pair and substitute into Cartesian form.

$$(s,t)=(0,1), \qquad 3-1+2=4$$

Question 10 solution

Model response

A connector from the line anchor to Q is $[1,1,2]$. Hence $\vec{r}=[3,1,0]+u[2,0,1]+v[1,1,2]$. A normal is $[-1,-3,2]$, giving $x+3y-2z=6$. The line is obtained when $v=0$; Q is obtained at $(u,v)=(0,1)$.

Marking criteria

1. Finds the connector and checks independence (2 marks).
2. Constructs vector form with two parameters (2 marks).
3. Finds the correct normal and Cartesian equation (1 mark).
4. Verifies the whole line and external point (1 mark).

A connector from the line anchor to Q is $[1,1,2]$. Hence $\vec{r}=[3,1,0]+u[2,0,1]+v[1,1,2]$. A normal is $[-1,-3,2]$, giving $x+3y-2z=6$. The line is obtained when $v=0$; Q is obtained at $(u,v)=(0,1)$.

1 Connector. Subtract the line anchor from Q.

$$\overrightarrow{AQ} = [1, 1, 2]$$

2 Construct. The connector is independent of the line direction.

$$\vec{r} = [3, 1, 0] + u[2, 0, 1] + v[1, 1, 2]$$

3 Cartesian form. Cross the spans and use the anchor.

$$[2, 0, 1] \times [1, 1, 2] = [-1, -3, 2], \quad x + 3y - 2z = 6$$

4 Verify. Substitution reproduces both required objects.

$$v = 0 \Rightarrow L, \quad (u, v) = (0, 1) \Rightarrow Q$$

Question 11 solution

Model response

The normals are $[1, 1, 1]$ and $[1, 0, 1]$. Their dot product is two and their magnitudes are $\sqrt{3}$ and $\sqrt{2}$. Thus $\cos\theta = 2/\sqrt{6}$ and $\theta \approx 35.3°$.

Marking criteria

1. Identifies both normals (2 marks).
2. Calculates the dot product and exact magnitudes (2 marks).
3. Obtains the correct exact cosine ratio (1 mark).
4. Calculates and labels the smaller angle (1 mark).

The normals are $[1, 1, 1]$ and $[1, 0, 1]$. Their dot product is two and their magnitudes are $\sqrt{3}$ and $\sqrt{2}$. Thus $\cos\theta = 2/\sqrt{6}$ and $\theta \approx 35.3°$.

1 Normals. Read coefficients, including the missing y term.

$$\vec{n}_1 = [1, 1, 1], \quad \vec{n}_2 = [1, 0, 1]$$

2 Dot product and magnitudes. Keep the radicals exact.

$$\vec{n}_1 \cdot \vec{n}_2 = 2, \quad \|\vec{n}_1\| = \sqrt{3}, \quad \|\vec{n}_2\| = \sqrt{2}$$

3 Smaller angle. Use the absolute dot product.

$$\cos\theta = \frac{|2|}{\sqrt{3}\sqrt{2}} = \frac{2}{\sqrt{6}}$$

4 Approximate. Use inverse cosine in degree mode only at the end.

$$\theta = \cos^{-1}(2/\sqrt{6}) \approx 35.3°$$

Question 12 solution

Model response

The intercepts are $(12,0,0)$, $(0,8,0)$ and $(0,0,4)$. The traces are $2x+3y=24, z=0$, $2x+6z=24, y=0$ and $3y+6z=24, x=0$. Join the intercepts to indicate orientation and show that the plane extends beyond the triangle.

Marking criteria

1. Finds and labels all intercepts (2 marks).
2. Obtains all three trace equations (2 marks).
3. Draws a coherent labelled plane sketch (1 mark).
4. Explains that the triangle is only a finite patch (1 mark).

The intercepts are $(12,0,0)$, $(0,8,0)$ and $(0,0,4)$. The traces are $2x+3y=24, z=0$, $2x+6z=24, y=0$ and $3y+6z=24, x=0$. Join the intercepts to indicate orientation and show that the plane extends beyond the triangle.

1 Intercepts. Set two coordinates to zero at each axis.

$$A(12,0,0), \quad B(0,8,0), \quad C(0,0,4)$$

2 xy trace. Set z to zero.

$$z=0: \quad 2x+3y=24$$

3 Other traces. Set y=0 and x=0 separately.

$$y=0:\ 2x+6z=24, \qquad x=0:\ 3y+6z=24$$

4 Sketch and interpretation. Draw labelled axes, points and trace lines. The plane is infinite in every in-plane direction.

5.3 Test 9.1 Intersections and systems

Covers Sections 9.1, 9.2, 9.3, 9.4.

Sections 9.1 to 9.4. An independent unit test with extended written reasoning and complete marking guidance.

12 questions 32 marks Suggested time: 60 min

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2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: A
Find the intersection parameter for $\vec{r} = [0, 0, 3] + t[1, 1, 1]$ with $x + y + z = 9$.

A. $t = 1$

B. $t = 3$

C. Every real t

D. $t = 2$

Question 2 [1 mark]

Categories: T
Classify $L : \vec{r} = [3, 0, 0] + t[1, -1, 0]$ relative to $\Pi : x + y = 3$.

A. Parallel and disjoint

B. The line is perpendicular to the plane

C. The line is contained in the plane

D. Exactly one intersection

Question 3 [1 mark]

Categories: T
Classify $L : \vec{r} = [3, 0, 1] + t[1, -1, 0]$ relative to $\Pi : x + y + z = 5$.

A. The line is normal to the plane

B. Parallel and disjoint

C. Contained in the plane

D. Exactly one intersection

Question 4 [1 mark]

Categories: K, T
Classify $L_1 : \vec{r} = [3, 0, 0] + s[1, 0, 0]$ and $L_2 : \vec{r} = [0, 1, 1] + t[0, 1, 0]$.

A. Skew lines

B. Coincident lines

C. Distinct parallel lines

D. Intersecting perpendicular lines

Question 5 [1 mark]

Categories: A
Which vector is parallel to the intersection of $x + y + z = 3$ and $x - y + z = 3$?

A. $[1, 0, 1]$

B. $[0, 1, 0]$

C. $[1, 1, 1]$

D. $[1, 0, -1]$

Question 6 [1 mark]

Categories: A, T
What is the common solution set of $x + y = 3$, $y + z = 3$ and $x + z = 3$?

A. A plane

B. The empty set

C. $\{(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})\}$

D. An infinite line

Question 7 [1 mark]

Categories: T
The equations $x + y + z = 3$ and $2x + 2y + 2z = 7$ reduce to $0 = 1$. What does this imply?

A. The equations describe the same plane

B. There is no common solution

C. One coordinate is free

D. There is a unique point

Question 8 [1 mark]

Categories: T
Consider $x + y + z = 3$, $x - y + z = 3$ and $2x + 2z = 6$. How many independent plane constraints remain?

A. Two, giving an intersection line

B. Three, giving a unique point

C. One, giving an entire plane

D. Zero, giving all of R^3

Part B Reasoning and communication

Question 9 [6 marks]

Categories: A, C
Find the intersection of $L : \vec{r} = [1, 0, 3] + t[2, 1, -1]$ with $\Pi : x + y + z = 8$. Give the point and verify it in both objects.
Show your working and state a complete conclusion.

Question 10 [6 marks]

Categories: A, T, C
Determine whether $L_1 : \vec{r} = [3, 0, 1] + s[1, 2, -1]$ and $L_2 : \vec{r} = [5, 4, 1] + u[0, 1, 1]$ intersect. Solve using separate parameters and check all coordinates. State the full line relationship.
Show your working and state a complete conclusion.

Question 11 [6 marks]

Categories: A, T, C
Find a vector equation of the intersection of $\Pi_1 : x + y + z = 4$ and $\Pi_2 : x - y + z = 2$. Verify its anchor and direction against both planes.
Show your working and state a complete conclusion.

Question 12 [6 marks]

Categories: T, C
For $\Pi_1 : x + y + z = 3$, $\Pi_2 : x - y + z = 3$ and $\Pi_3 : 2x + 2z = k$, find every value of k for which the system is consistent. Describe the complete common solution set in every case.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$t = 2$

Substitute all three coordinate expressions into the plane before solving for the shared parameter.

1 Reasoning. Substitute all three coordinate expressions into the plane before solving for the shared parameter.

$$t + t + (3 + t) = 9 \Rightarrow 3t = 6 \Rightarrow t = 2$$

Question 2 solution

Correct answer: C

The line is contained in the plane

Substitution is an identity for every real t, so every point on the line belongs to the plane.

1 Reasoning. Substitution is an identity for every real t, so every point on the line belongs to the plane.

$$(3 + t) - t = 3$$

Question 3 solution

Correct answer: B

Parallel and disjoint

The parameter cancels, leaving a false statement. The direction is parallel to the plane but the line has no common point with it.

1 Reasoning. The parameter cancels, leaving a false statement. The direction is parallel to the plane but the line has no common point with it.

$$(3 + t) - t + 1 = 4 \neq 5$$

Question 4 solution

Correct answer: A

Skew lines

Their nonparallel directions do not guarantee intersection. The first has z=0 and the second z=1, so they cannot meet.

1 Reasoning. Their nonparallel directions do not guarantee intersection. The first has z=0 and the second z=1, so they cannot meet.

$$z_{L_1} = 0, \quad z_{L_2} = 1, \quad [1, 0, 0] \nparallel [0, 1, 0]$$

Question 5 solution

Correct answer: D

$[1, 0, -1]$

A line in both planes has direction perpendicular to both plane normals.

1 Reasoning. A line in both planes has direction perpendicular to both plane normals.

$$[1, 1, 1] \times [1, -1, 1] = [2, 0, -2] = 2[1, 0, -1]$$

Question 6 solution

Correct answer: C

$\{(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})\}$

Subtracting pairs gives x=y=z. Substituting into any original equation gives twice each coordinate equal to the constant.

1 Reasoning. Subtracting pairs gives x=y=z. Substituting into any original equation gives twice each coordinate equal to the constant.

$$x = y = z, \quad 2x = 3 \Rightarrow x = y = z = \frac{3}{2}$$

Question 7 solution

Correct answer: B

There is no common solution

A contradiction is false for all variable values. The planes have parallel normals but different locations.

1 Reasoning. A contradiction is false for all variable values. The planes have parallel normals but different locations.

$$2(x + y + z) = 6 \neq 7$$

Question 8 solution

Correct answer: A

Two, giving an intersection line

The third equation is the sum of the first two and contributes no new restriction. The first two normals are not parallel.

1 Reasoning. The third equation is the sum of the first two and contributes no new restriction. The first two normals are not parallel.

$$(x + y + z) + (x - y + z) = 2x + 2z = 6$$

Question 9 solution

Model response

Substitution gives $1 + 3 + 2t = 8$, so $t = 2$. The intersection is $P(5, 2, 1)$. Its coordinates come from one shared line parameter and satisfy the plane because $5 + 2 + 1 = 8$.

Marking criteria

1. Forms the correct scalar equation (2 marks).
2. Solves the intersection parameter (2 marks).
3. Calculates all coordinates consistently (1 mark).
4. Verifies the point in both objects (1 mark).

Substitution gives $1 + 3 + 2t = 8$, so $t = 2$. The intersection is $P(5, 2, 1)$. Its coordinates come from one shared line parameter and satisfy the plane because $5 + 2 + 1 = 8$.

1 Substitute. Use all three line coordinates.

$$(1 + 2t) + t + (3 - t) = 8$$

2 Solve. Collect the parameter terms.

$$2t = 4 \Rightarrow t = 2$$

3 Point. Substitute the same value into x,y,z.

$$P = (5, 2, 1)$$

4 Verify. Check the original line and plane.

$$[1, 0, 3] + 2[2, 1, -1] = [5, 2, 1], \quad 5 + 2 + (1) = 8$$

Question 10 solution

Model response

The x equation forces $s = 2$; the y equation then forces $u = 0$. The z values are negative one and positive one, so the lines do not intersect. Their directions are not parallel, so they are skew.

Marking criteria

1. Writes corresponding coordinate equations with separate parameters (2 marks).
2. Solves for the candidate parameter pair (2 marks).
3. Checks the z-coordinate explicitly (1 mark).
4. States and justifies the complete relationship (1 mark).

The x equation forces $s = 2$; the y equation then forces $u = 0$. The z values are negative one and positive one, so the lines do not intersect. Their directions are not parallel, so they are skew.

1 x equation. Use separate parameters for the two lines.

$$3 + s = 5 \Rightarrow s = 2$$

2 y equation. Use the value of s to solve u.

$$2s = 4 + u \Rightarrow u = 0$$

3 z check. Check the third equation rather than assuming agreement.

$$1 - s = -1 \neq 1 + u = 1$$

4 Classify. The directions are not scalar multiples; use the coordinate check to distinguish an intersection from skew lines.

$$L_1 \cap L_2 = \varnothing$$

Question 11 solution

Model response

Subtract the equations to obtain $y = 1$. Then $z = 3 - x$. One intersection line is $\vec{r} = [0, 1, 3] + t[1, 0, -1]$. The anchor satisfies both equations, and the direction has zero dot product with both normals.

Marking criteria

1. Eliminates a variable correctly (2 marks).
2. Introduces a free parameter and obtains all coordinates (2 marks).
3. Writes a correct intersection line (1 mark).
4. Checks both location and direction in both planes (1 mark).

Subtract the equations to obtain $y = 1$. Then $z = 3 - x$. One intersection line is $\vec{r} = [0, 1, 3] + t[1, 0, -1]$. The anchor satisfies both equations, and the direction has zero dot product with both normals.

1 Eliminate. Subtract the second equation from the first.

$$2y = 2 \Rightarrow y = 1$$

2 Free variable. Let x=t and solve z.

$$x = t, \quad y = 1, \quad z = 3 - t$$

3 Vector form. Separate constants from parameter coefficients.

$$\vec{r} = [0, 1, 3] + t[1, 0, -1]$$

4 Verify. The anchor lies on both planes and the direction is perpendicular to both normals.

$$0+1+3=4, \quad 0-1+3=2, \quad [1,0,-1]\cdot[1,\pm1,1]=0$$

Question 12 solution

Model response

Adding the first two equations gives $2x+2z=6$. Thus $k=6$ makes the third equation redundant and the common set is $\vec{r}=[0,0,3]+t[1,0,-1]$. For every other k the system is inconsistent. No value gives exactly one point because the third equation either repeats or contradicts the existing constraint.

Marking criteria

1. Identifies the dependent combination of equations (2 marks).
2. Obtains the exact consistency condition (2 marks).
3. Parameterizes the full common line (1 mark).
4. Classifies all other values and explains why a unique point is impossible (1 mark).

Adding the first two equations gives $2x+2z=6$. Thus $k=6$ makes the third equation redundant and the common set is $\vec{r}=[0,0,3]+t[1,0,-1]$. For every other k the system is inconsistent. No value gives exactly one point because the third equation either repeats or contradicts the existing constraint.

1 Compare equations. Add the first two constraints.

$$(x+y+z)+(x-y+z)=2x+2z=6$$

2 Consistency. The third equation must have the same right side.

$$k=6$$

3 Common line. Subtract the first two equations and parameterize the remaining relation.

$$y=0, \quad z=3-x, \quad \vec{r}=[0,0,3]+t[1,0,-1]$$

4 Other cases. A different right side produces a contradiction.

$$k\neq6 \Rightarrow \Pi_1\cap\Pi_2\cap\Pi_3=\varnothing$$

5.4 Test 9.2 Shortest distances

Covers Sections 9.5, 9.6.

Sections 9.5 to 9.6. An independent unit test with extended written reasoning and complete marking guidance.

12 questions 32 marks Suggested time: 60 min

1. Use exact fractions and radicals unless an approximation is requested.
2. For written responses, show the method, state the geometric conclusion and verify it in the original equations.
3. A labelled sketch is required when requested. It may be completed on paper alongside your typed response.
4. Award written-response marks using the stated criteria after completing the paper. Model answers are examples of valid reasoning.

Question paper

Part A Knowledge and application

Question 1 [1 mark]

Categories: A
Find the distance from $P(0,3)$ to $3x+4y=0$.

A. 3

B. $\frac{9}{5}$

C. 12

D. $\frac{12}{5}$

Question 2 [1 mark]

Categories: A
Find the distance from $P(0,0,3)$ to $L:\vec{r}=t[1,0,0]$.

A. 4

B. 7

C. 3

D. 0

Question 3 [1 mark]

Categories: K
The plane is $2x-y+2z=3$. What is the magnitude of a normal used in the Cartesian point-plane distance formula?

A. $\sqrt{5}$

B. 3

C. 5

D. 9

Question 4 [1 mark]

Categories: K
What is the distance from $P(3,0,0)$ to $x=3$?

A. 0

B. 1

C. 4

D. 5

Question 5 [1 mark]

Categories: T
A student multiplies $x+2y+2z=3$ by five before calculating the distance from a fixed point. What happens to the distance?

A. It becomes five times larger

B. It becomes one fifth as large

C. It becomes zero

D. It remains unchanged

Question 6 [1 mark]

Categories: K, T
For a point outside $\Pi: x+2y+2z=3$, which condition must its shortest connector to the plane satisfy?

A. Its endpoint may be outside the plane

B. Its direction has zero magnitude

C. Its direction is parallel to [1,2,2]

D. Its direction is parallel to the plane

Question 7 [1 mark]

Categories: A
For $P(1,3,3)$ and $L: \vec{r}=t[1,0,0]$, what is the nearest point H on the line?

A. $(0,3,3)$

B. $(1,0,0)$

C. $(0,0,0)$

D. $(1,3,0)$

Question 8 [1 mark]

Categories: A, T
Find the separation of the parallel planes $x+2y+2z=3$ and $2x+4y+4z=18$.

A. 2

B. 4

C. 6

D. 5

Part B Reasoning and communication

Question 9 [6 marks]

Categories: A, T, C
For $P(3,2)$ and $L: 3x+4y=1$, find the exact distance and nearest point H. Verify that H lies on L and that PH is perpendicular to L.
Show your working and state a complete conclusion.

Question 10 [6 marks]

Categories: A, T, C
Find the exact distance from $P(2,3,3)$ to $L: \vec{r}=[1,3,0]+t[1,1,0]$. Use projection or the cross-product area method, then find and verify the perpendicular foot.
Show your working and state a complete conclusion.

Question 11 [6 marks]

Categories: A, C
For $P(3,1,4)$ and $\Pi: x+2y+2z=4$, find the exact distance and the nearest point H. Verify both plane membership and the normal direction of PH.
Show your working and state a complete conclusion.

Question 12 [6 marks]

Categories: T, C
A student claims that the shortest distance from $P(1,3,3)$ to the x-axis is its distance to the origin. Assess the claim by finding the correct foot, comparing the two exact lengths and explaining the geometric condition.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: D

$\frac{12}{5}$

Divide the absolute Cartesian residual by the magnitude of the nonzero normal.

1 Reasoning. Divide the absolute Cartesian residual by the magnitude of the nonzero normal.

$$d = \frac{|3(0) + 4(3)|}{\sqrt{3^2 + 4^2}} = \frac{12}{5}$$

Question 2 solution

Correct answer: C

3

The line is the x-axis. The perpendicular foot is the origin for this particular point.

1 Reasoning. The line is the x-axis. The perpendicular foot is the origin for this particular point.

$$d = \sqrt{0^2 + 3^2} = 3$$

Question 3 solution

Correct answer: B

3

The normal is [2,-1,2], whose squared magnitude is four plus one plus four.

1 Reasoning. The normal is [2,-1,2], whose squared magnitude is four plus one plus four.

$$\|[2, -1, 2]\| = \sqrt{4 + 1 + 4} = 3$$

Question 4 solution

Correct answer: A

0

The point satisfies the plane equation, so it already lies on the plane.

1 Reasoning. The point satisfies the plane equation, so it already lies on the plane.

$$|3 - 3| = 0$$

Question 5 solution

Correct answer: D

It remains unchanged

Both the absolute residual and the normal magnitude gain the same factor of five.

1 Reasoning. Both the absolute residual and the normal magnitude gain the same factor of five.

$$\frac{|5(Ax_0 + By_0 + Cz_0 - D)|}{\|5\vec{n}\|} = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\|\vec{n}\|}$$

Question 6 solution

Correct answer: C

Its direction is parallel to [1,2,2]

The shortest connector is perpendicular to the plane, so it follows a nonzero normal direction and ends on the plane.

1 Reasoning. The shortest connector is perpendicular to the plane, so it follows a nonzero normal direction and ends on the plane.

$$\overrightarrow{HP} = \lambda[1, 2, 2], \qquad H \in \Pi$$

Question 7 solution

Correct answer: B

$(1, 0, 0)$

A foot on the x-axis has y=z=0. Matching x=1 makes the connector perpendicular to [1,0,0].

1 Reasoning. A foot on the x-axis has y=z=0. Matching x=1 makes the connector perpendicular to [1,0,0].

$$H = (1, 0, 0), \quad (P - H) \cdot [1, 0, 0] = [0, 3, 3] \cdot [1, 0, 0] = 0$$

Question 8 solution

Correct answer: A

2

Divide the second equation by two before comparing constants. The constant difference is six and the shared normal magnitude is three.

1 Reasoning. Divide the second equation by two before comparing constants. The constant difference is six and the shared normal magnitude is three.

$$d = \frac{|(9) - 3|}{3} = 2$$

Question 9 solution

Model response

The residual is 16, so $d = \frac{16}{5}$. The foot is $H(\frac{27}{25}, \frac{-14}{25})$. It satisfies $3x + 4y = 1$; $P - H = \frac{16}{25}[3, 4]$ is parallel to the line normal and perpendicular to direction $[4, -3]$.

Marking criteria

1. Calculates the exact distance using a nonzero normal (2 marks).
2. Finds the exact perpendicular foot (2 marks).
3. Checks that H lies on the line (1 mark).
4. Verifies the perpendicular connector (1 mark).

The residual is 16, so $d = \frac{16}{5}$. The foot is $H(\frac{27}{25}, \frac{-14}{25})$. It satisfies $3x + 4y = 1$; $P - H = \frac{16}{25}[3, 4]$ is parallel to the line normal and perpendicular to direction $[4, -3]$.

1 Distance. Use the absolute residual and the normal magnitude.

$$d = \frac{|3(3) + 4(2) - 1|}{5} = \frac{16}{5}$$

2 Foot. Remove the normal displacement from P.

$$H = [3, 2] - \frac{16}{25}[3, 4] = [\frac{27}{25}, \frac{-14}{25}]$$

3 Membership. Substitute the exact foot coordinates.

$$3(\frac{27}{25}) + 4(\frac{-14}{25}) = 1$$

4 Perpendicularity. The connector is a multiple of the normal.

$$[3, 4] \cdot [4, -3] = 12 - 12 = 0$$

Question 10 solution

Model response

The displacement from the anchor to P is $[1, 0, 3]$. Projection gives $t = 1/2$, so $H(3/2, \frac{7}{2}, 0)$. The residual is $[1/2, -1/2, 3]$, perpendicular to the line direction. Its magnitude is $\sqrt{38}/2$.

Marking criteria

1. Obtains the correct anchor displacement and direction (2 marks).
2. Uses projection or cross-product distance correctly (2 marks).
3. Calculates and verifies the perpendicular foot (1 mark).
4. Reports the exact distance with reasoning (1 mark).

The displacement from the anchor to P is $[1, 0, 3]$. Projection gives $t = 1/2$, so $H(3/2, \frac{7}{2}, 0)$. The residual is $[1/2, -1/2, 3]$, perpendicular to the line direction. Its magnitude is $\sqrt{38}/2$.

1 Displacement. Subtract the line anchor from P.

$$\vec{w} = [1, 0, 3], \quad \vec{d} = [1, 1, 0]$$

2 Projection parameter. Project w onto the line direction.

$$t = \frac{\vec{w} \cdot \vec{d}}{\vec{d} \cdot \vec{d}} = \frac{1}{2}$$

3 Foot and check. Use t=1/2 in the line, then check the residual.

$$H = [\tfrac{3}{2}, \frac{7}{2}, 0], \quad [\tfrac{1}{2}, -\tfrac{1}{2}, 3] \cdot [1, 1, 0] = 0$$

4 Exact length. Compute the magnitude without premature rounding.

$$d = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + 9} = \frac{\sqrt{38}}{2}$$

Question 11 solution

Model response

The normal is $[1, 2, 2]$ with magnitude three. The residual is 9, giving $d = 3$. The foot is $H(2, -1, 2)$. Substitution gives four, and $P - H = \frac{9}{9}[1, 2, 2]$.

Marking criteria

1. Identifies the normal and evaluates the residual (2 marks).
2. Obtains the exact distance (2 marks).
3. Computes the perpendicular foot (1 mark).
4. Verifies membership and perpendicularity (1 mark).

The normal is $[1, 2, 2]$ with magnitude three. The residual is 9, giving $d = 3$. The foot is $H(2, -1, 2)$. Substitution gives four, and $P - H = \frac{9}{9}[1, 2, 2]$.

1 Residual and normal. Evaluate the Cartesian expression at P.

$$r = 3 + 2 + 8 - 4 = 9, \quad \|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$$

2 Distance. Take the absolute scalar residual over normal magnitude.

$$d = \frac{9}{3} = 3$$

3 Nearest point. Subtract the required normal displacement.

$$H = [3, 1, 4] - \frac{9}{9}[1, 2, 2]$$

4 Verify. The point lies on the plane and the connector is normal to it.

$$x_H + 2y_H + 2z_H = 4, \quad P - H = \frac{9}{9}\vec{n}$$

Question 12 solution

Model response

The foot is $H(1,0,0)$, giving $PH=\sqrt{18}$. The origin is on the line, but $PO=\sqrt{19}$ is longer. The correct connector has zero dot product with the x-axis direction. A point on the target is not necessarily the nearest point.

Marking criteria

1. Finds the correct perpendicular foot (2 marks).
2. Checks the zero dot product (2 marks).
3. Compares both exact connector lengths (1 mark).
4. Explains why target membership alone is insufficient (1 mark).

The foot is $H(1,0,0)$, giving $PH=\sqrt{18}$. The origin is on the line, but $PO=\sqrt{19}$ is longer. The correct connector has zero dot product with the x-axis direction. A point on the target is not necessarily the nearest point.

1 Foot. A point on the x-axis has y=z=0; choose the same x-coordinate as P.

$$H=(1,0,0)$$

2 Perpendicularity. Check the connector against the line direction.

$$[0,3,3]\cdot[1,0,0]=0$$

3 Compare lengths. Keep both magnitudes exact.

$$PH=\sqrt{18}<\sqrt{19}=PO$$

4 Conclusion. Reject the claim. The shortest connector must meet the target at a right angle, which the connector to the origin does not.

6 Final examination

6.1 Lines and Planes final examination

A 60-mark examination on Sections 8.1 to 8.6 and 9.1 to 9.6, combining representation, sketching, intersections, systems, exact distances, and mathematical communication.

24 questions 60 marks Suggested time: 100 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K
Which statement about a Cartesian line in R^3 is correct?

A. One equation Ax+By+Cz+D=0 always defines a line.

B. It can be represented as the intersection of two non-parallel planes.

C. It requires three independent parameters.

D. It has no direction vector.

Question 2 [1 mark]

Categories: K, C
Convert $\frac{x-2}{3} = \frac{y+1}{-2} = z - 4$ to parametric form.

A. $x = 2 + 3t,\ y = -1 - 2t,\ z = 4 + t$

B. $x = 3 + 2t,\ y = -2 - t,\ z = 1 + 4t$

C. $x = 2 - 3t,\ y = -1 + 2t,\ z = 4 + t$

D. $x = 2 + 3t,\ y = 1 - 2t,\ z = 4$

Question 3 [1 mark]

Categories: A
Which plane contains A(1,0,0), B(0,1,0), and C(0,0,1)?

A. $x + y + z = 0$

B. $x + y + z = 1$

C. $x - y + z = 1$

D. $xyz = 1$

Question 4 [1 mark]

Categories: A
Find the intersection of $\vec{r} = (0, 1, 0) + t(2, -1, 3)$ and $x + y - z = 0$.

A. $(0, 1, 0)$

B. $(2, 0, 3)$

C. $(1, 1/2, 3/2)$

D. No intersection

Question 5 [1 mark]

Categories: T
Classify $2x - y + z = 4$ and $4x - 2y + 2z = 11$.

A. Intersect in a line

B. Coincident

C. Distinct parallel planes

D. Perpendicular planes

Question 6 [1 mark]

Categories: T, C
Which line is the intersection of $x + y + z = 3$ and $x - y + z = 1$?

A. $\vec{r} = (2, 1, 0) + t(-1, 0, 1)$

B. $\vec{r} = (0, 0, 0) + t(1, 1, 1)$

C. $\vec{r} = (3, 1, 1) + t(1, 0, 1)$

D. $\vec{r} = (2, 0, 1) + t(0, 1, 0)$

Question 7 [1 mark]

Categories: A, T
Find the distance from $P(3, 2, 0)$ to $\vec{r} = (1, 1, 1) + t(2, 0, -1)$.

A. 0

B. 1

C. $\sqrt{5}$

D. 5

Question 8 [1 mark]

Categories: C
Substituting a line into a plane simplifies to 0=0. Which conclusion communicates the result completely?

A. The equations are wrong.

B. The line is parallel to the plane, so there is no intersection.

C. Every point on the line satisfies the plane; the line lies in the plane and there are infinitely many intersections.

D. The plane lies on the line.

Question 9 [1 mark]

Categories: K, A
Which three points should be plotted first to sketch $3x + 2y + 6z = 6$ by intercepts?

A. $(2, 0, 0), (0, 3, 0), (0, 0, 1)$

B. $(3, 0, 0), (0, 2, 0), (0, 0, 6)$

C. $(6, 0, 0), (0, 6, 0), (0, 0, 6)$

D. $(1, 0, 0), (0, 1, 0), (0, 0, 1)$

Question 10 [1 mark]

Categories: A
Find the distance from $P(4, 1)$ to $3x - 4y + 2 = 0$.

A. 2

B. $\frac{10}{7}$

C. $\sqrt{5}$

D. 10

Question 11 [1 mark]

Categories: K, A
In $\mathbb{R}^2$, which parametric equations represent $\vec{r} = (2, -1) + t(3, 2)$?

A. $x = 2 + 3t,\ y = -1 + 2t$

B. $x = 3 + 2t,\ y = 2 - t$

C. $x = 2 + 3t,\ y = 1 + 2t$

D. $x = 2 + 3s,\ y = -1 + 2t$, with independent s, t

Question 12 [1 mark]

Categories: K, T
Which statement gives the correct sketching information for the plane $x + y = 0$ in $\mathbb{R}^3$?

A. It is the coordinate plane z=0.

B. It is only a line in the xy-plane.

C. It contains the z-axis, and three distinct non-zero axis intercepts cannot be used to sketch it.

D. It has the three intercepts (1,0,0), (0,1,0), and (0,0,1).

Question 13 [1 mark]

Categories: K, T
Elimination in a system of three linear equations produces the equation $0 = 4$. What does this establish?

A. The system has one solution.

B. The solution set is a line.

C. The solution set is a plane.

D. The system is inconsistent and has no common solution.

Question 14 [1 mark]

Categories: T
The planes are $x + y + z = 2$, $2x - y + z = 1$, and $3x + 2z = 3$. What is their common solution set?

A. Exactly one point

B. A line

C. A plane

D. No common points

Section B: Application

Question 15 [5 marks]

Categories: A, T, C
Two lines intersect at $P = (1, 0, 2)$ and have directions $\vec{d_1} = (1, 1, 0)$ and $\vec{d_2} = (0, 1, 1)$. Construct a Cartesian equation of their plane and verify the result.
Show your working and state a complete conclusion.

Question 16 [4 marks]

Categories: A, C
For the line $3x - 4y + 5 = 0$ in $\mathbb{R}^2$, (a) find an anchor and a direction vector, (b) write vector and parametric equations, and (c) find its exact distance from the origin.
Show your working and state a complete conclusion.

Question 17 [4 marks]

Categories: A, C
A line passes through $A(2, -1, 3)$ and $B(2, 3, 1)$. Write vector and symmetric equations, handling the zero direction component correctly. Then determine whether $P(2, 5, 0)$ lies on the line.
Show your working and state a complete conclusion.

Question 18 [4 marks]

Categories: A, T, C
The plane is $\vec{r} = (0, 1, -1) + s(1, 0, 2) + t(0, 2, -1)$. (a) Write its parametric equations. (b) Find a Cartesian equation. (c) Explain why the two spans define a plane rather than a line.
Show your working and state a complete conclusion.

Question 19 [4 marks]

Categories: A, C
Describe how to sketch $2x + 3z = 6$ in $\mathbb{R}^3$. Give the x- and z-intercepts, explain why there is no y-intercept, and identify the direction in which the plane extends. Give one additional point that helps show this extension.
Show your working and state a complete conclusion.

Question 20 [5 marks]

Categories: A, T, C
Determine the relationship between $L_1 : \vec{r} = (1,-1,2) + t(2,1,-1)$ and $L_2 : \vec{r} = (2,4,-3) + s(1,-1,1)$. Find any intersection, verify all coordinates, and state a complete geometric conclusion.
Show your working and state a complete conclusion.

Question 21 [5 marks]

Categories: A, T, C
Solve the system $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$ by elimination. Show your reduced equations, verify the solution in all three original equations, and interpret the result as an intersection of planes.
Show your working and state a complete conclusion.

Question 22 [5 marks]

Categories: A, T, C
Find the exact distance from $P(4,1,2)$ to the plane $2x - y + 2z = 3$. Also find the perpendicular foot H, then verify that H lies on the plane and that $\overrightarrow{HP}$ is normal to it.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 23 [5 marks]

Categories: K, T, C
A line $\vec{r} = (0,0,0) + t(1,-1,0)$ is compared with the plane $x + y + z = 2$. Explain the relationship using both a vector test and a geometric conclusion.
Show your working and state a complete conclusion.

Question 24 [5 marks]

Categories: T, C
For all real values of k, classify the system $x + y + z = 3$, $x - y + z = 1$, and $2x + ky + 2z = 5$. Give the complete common solution set when one exists, and explain whether a unique solution is possible.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

It can be represented as the intersection of two non-parallel planes.

One scalar linear equation in R³ defines a plane; two independent plane equations can intersect in a line.

1 Dimension check. Each independent scalar constraint reduces dimension by one.

$$\mathbb{R}^3 \xrightarrow{\text{one constraint}} \text{plane} \xrightarrow{\text{second constraint}} \text{line}$$

Question 2 solution

Correct answer: A

$x = 2 + 3t,\ y = -1 - 2t,\ z = 4 + t$

Set every symmetric expression equal to the shared parameter t. Since z-4=t, z=4+t.

1 Set equal to t. Solve each relation.

$$\frac{x-2}{3} = t,\ \frac{y+1}{-2} = t,\ z - 4 = t$$

Question 3 solution

Correct answer: B

$x + y + z = 1$

Each point is a unit intercept, so the intercept equation is x+y+z=1.

1 Spans. Use AB=(-1,1,0) and AC=(-1,0,1).

$$\overrightarrow{AB} \times \overrightarrow{AC} = (1, 1, 1)$$

2 Plane. Use A in point-normal form.

$$(x - 1) + y + z = 0 \Rightarrow x + y + z = 1$$

Question 4 solution

Correct answer: C

$(1, 1/2, 3/2)$

Substitution gives 2t+(1-t)-3t=0, so 1-2t=0 and t=1/2. Evaluating the line gives (1,1/2,3/2).

1 Substitute. Use x=2t, y=1-t, z=3t.

$$2t + 1 - t - 3t = 0 \Rightarrow 1 - 2t = 0 \Rightarrow t = \frac{1}{2}$$

2 Evaluate. Return to the line.

$$P = \left(1, \frac{1}{2}, \frac{3}{2}\right)$$

Question 5 solution

Correct answer: C

Distinct parallel planes

The normals scale by 2, but 2(4)=8≠11. The planes are parallel and distinct.

1 Compare scale. Double the first equation's left side and constant.

$$2(2x - y + z) = 4x - 2y + 2z, \qquad 2(4) = 8 \neq 11$$

Question 6 solution

Correct answer: A

$\vec{r} = (2, 1, 0) + t(-1, 0, 1)$

Subtracting gives y=1; then x+z=2. Let z=t to obtain the listed line.

1 Eliminate. Subtract the equations.

$$2y = 2 \Rightarrow y = 1$$

2 Parameterize. Let z=t in x+z=2.

$$x = 2 - t \Rightarrow \vec{r} = (2, 1, 0) + t(-1, 0, 1)$$

Question 7 solution

Correct answer: B

1

AP=(2,1,-1), AP×d=(-1,0,-2), and both the numerator and denominator have magnitude √5.

1 Apply formula. Use the cross-product distance formula.

$$d = \frac{\|(2, 1, -1) \times (2, 0, -1)\|}{\|(2, 0, -1)\|} = \frac{\|(-1, 0, -2)\|}{\sqrt{5}} = 1$$

Question 8 solution

Correct answer: C

Every point on the line satisfies the plane; the line lies in the plane and there are infinitely many intersections.

An identity means the equation is true for every parameter value, not just one.

1 Interpret the identity. No restriction remains on t.

$$0 = 0 \text{ for all } t \in \mathbb{R} \Rightarrow L \subset \Pi$$

Question 9 solution

Correct answer: A

$(2,0,0), (0,3,0), (0,0,1)$

Set the other two variables to zero for each axis intercept.

1 Find the intercepts. Solve each one-variable equation.

$$x = 2, \qquad y = 3, \qquad z = 1$$

Question 10 solution

Correct answer: A

2

Substitute the point into the numerator and divide by the magnitude of the normal.

1 Use the R² formula. Evaluate the numerator and denominator.

$$d = \frac{|3(4) - 4(1) + 2|}{\sqrt{3^2 + (-4)^2}} = \frac{10}{5} = 2$$

Question 11 solution

Correct answer: A

$x = 2 + 3t,\ y = -1 + 2t$

Each coordinate uses the same parameter. The starting point supplies the constants and the direction supplies the parameter coefficients.

1 Reasoning. Each coordinate uses the same parameter. The starting point supplies the constants and the direction supplies the parameter coefficients.

$$\vec{r} = (2 + 3t, -1 + 2t)$$

Question 12 solution

Correct answer: C

It contains the z-axis, and three distinct non-zero axis intercepts cannot be used to sketch it.

The variable z is unrestricted. Every point (0,0,z) satisfies the equation, so the whole z-axis lies in the plane. Both x- and y-axis intersections occur at the origin.

1 Reasoning. The variable z is unrestricted. Every point (0,0,z) satisfies the equation, so the whole z-axis lies in the plane. Both x- and y-axis intersections occur at the origin.

$$x = y = 0 \Rightarrow 0 = 0 \quad \text{for every } z \in \mathbb{R}$$

Question 13 solution

Correct answer: D

The system is inconsistent and has no common solution.

A contradiction cannot hold for any values of the variables. Since elimination preserves the solution set, the original equations have no simultaneous solution.

1 Reasoning. A contradiction cannot hold for any values of the variables. Since elimination preserves the solution set, the original equations have no simultaneous solution.

$$0 = 4 \quad \text{is false} \Rightarrow \text{no solution}$$

Question 14 solution

Correct answer: B

A line

The third equation is the sum of the first two. Their normals are not parallel, so the two independent constraints determine a line; the third adds no restriction.

1 Reasoning. The third equation is the sum of the first two. Their normals are not parallel, so the two independent constraints determine a line; the third adds no restriction.

$$(x + y + z) + (2x - y + z) = 3x + 2z = 3$$

Question 15 solution

Model response

The lines provide two in-plane directions. Their cross product is (1,1,0) cross (0,1,1)=(1,-1,1), so a normal is n=(1,-1,1). Using P gives (x-1)-y+(z-2)=0, or x-y+z=3. The point P satisfies the equation, and the normal has dot product 0 with each line direction, so the plane is verified.

Marking criteria

1. Uses the intersection point as the plane anchor (1 mark).
2. Finds a non-zero normal using the cross product (1 mark).
3. Builds and simplifies the Cartesian equation (1 mark).
4. Verifies that the anchor satisfies the equation (1 mark).
5. Verifies that both line directions are perpendicular to the normal (1 mark).

Two intersecting lines determine a plane when their direction vectors are not parallel.

1 Find the normal. Cross the line directions.

$$(1, 1, 0) \times (0, 1, 1) = (1, -1, 1)$$

2 Use the intersection point. Build point-normal form.

$$(x - 1) - y + (z - 2) = 0 \Rightarrow x - y + z = 3$$

3 Verify. Check the anchor and perpendicularity.

$$(1,-1,1)\cdot(1,1,0)=0, \quad (1,-1,1)\cdot(0,1,1)=0$$

Question 16 solution

Model response

A normal is $(3,-4)$, so choose $\vec{d}=(4,3)$. The point $(1,2)$ lies on the line because $3-8+5=0$. Thus $\vec{r}=(1,2)+t(4,3)$, with $x=1+4t$, $y=2+3t$, $t\in\mathbb{R}$. The distance from the origin is $d=|5|/\sqrt{3^2+(-4)^2}=1$.

Marking criteria

1. Chooses a direction perpendicular to (3,-4), such as (4,3) (1 mark).
2. Finds an anchor satisfying the equation, such as (1,2) (1 mark).
3. Writes correct vector and parametric equations with one shared parameter (1 mark).
4. Calculates the exact origin-to-line distance as 1 (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Anchor and direction. An anchor satisfies the equation; a direction is perpendicular to its normal.

$$3(1)-4(2)+5=0, \qquad (3,-4)\cdot(4,3)=0$$

2 Represent the line. Use one real parameter for both coordinates.

$$\vec{r}=(1,2)+t(4,3), \qquad x=1+4t, \quad y=2+3t$$

3 Distance. Substitute the origin into the scalar distance formula.

$$d=\frac{|3(0)-4(0)+5|}{\sqrt{3^2+(-4)^2}}=1$$

Question 17 solution

Model response

The displacement is $B-A=(0,4,-2)$, so choose $\vec{d}=(0,2,-1)$. The vector equation is $\vec{r}=(2,-1,3)+t(0,2,-1)$. Symmetric form is $x=2$, $\frac{y+1}{2}=\frac{z-3}{-1}$. For $t=3$, the line gives $(2,5,0)$, so P lies on it.

Marking criteria

1. Finds a valid direction such as (0,2,-1) (1 mark).
2. Writes a correct vector equation (1 mark).
3. Writes x=2 separately and equates only the non-zero direction ratios (1 mark).
4. Uses the same parameter in all coordinates to verify that P lies on the line (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction. Simplify the displacement without changing the line.

$$B - A = (0, 4, -2) = 2(0, 2, -1)$$

2 Equation forms. A zero direction component gives a fixed coordinate, not division by zero.

$$\vec{r} = (2, -1, 3) + t(0, 2, -1), \qquad x = 2, \quad \frac{y+1}{2} = \frac{z-3}{-1}$$

3 Membership. Substitute one shared parameter.

$$t = 3 \Rightarrow (2, -1 + 6, 3 - 3) = (2, 5, 0)$$

Question 18 solution

Model response

The parametric equations are $x = s$, $y = 1 + 2t$, and $z = -1 + 2s - t$. The cross product of the spans is $(-4, 1, 2)$, so a Cartesian equation is $-4x + y + 2z = -1$. The cross product is non-zero, which proves that the spans are not parallel. They provide two independent directions and therefore define a plane.

Marking criteria

1. Writes all three parametric equations correctly (1 mark).
2. Finds a non-zero normal (-4,1,2), or a scalar multiple (1 mark).
3. Obtains -4x+y+2z=-1, or an equivalent equation (1 mark).
4. Uses the non-zero cross product to explain span independence and the plane dimension (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Parametric form. Read each coordinate separately.

$$x = s, \qquad y = 1 + 2t, \qquad z = -1 + 2s - t$$

2 Normal. Cross the two spanning vectors.

$$(1, 0, 2) \times (0, 2, -1) = (-4, 1, 2) \neq \vec{0}$$

3 Cartesian form. Use the anchor (0,1,-1) in point-normal form.

$$-4x + (y - 1) + 2(z + 1) = 0 \Rightarrow -4x + y + 2z = -1$$

Question 19 solution

Model response

The x-intercept is $(3,0,0)$ and the z-intercept is $(0,0,2)$. A y-axis point has $x = z = 0$, which would require $0 = 6$, so there is no y-intercept. Since y does not appear, draw the segment between the intercepts and extend it in both y-directions to represent the infinite plane. For example, $(3,1,0)$ also lies on the plane.

Marking criteria

1. Finds the intercepts (3,0,0) and (0,0,2) (1 mark).
2. Explains that setting x=z=0 gives a contradiction, so no y-axis point lies on the plane (1 mark).
3. States that y is unrestricted and the plane extends in the y-direction (1 mark).
4. Gives a valid additional point such as (3,1,0) and describes extending the intercept segment parallel to the y-axis (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Intercepts. Set the other coordinates to zero.

$$z = 0 \Rightarrow x = 3, \qquad x = 0 \Rightarrow z = 2$$

2 Missing intercept. Test the entire y-axis.

$$x = z = 0 \Rightarrow 0 = 6 \quad \text{which is impossible}$$

3 Extend the sketch. The y-coordinate is free.

$$(3, y, 0) \in \Pi \quad \text{for all } y \in \mathbb{R}$$

Question 20 solution

Model response

The x- and y-equations give $2t - s = 1$ and $t + s = 5$. Their solution is $t = 2$, $s = 3$, giving $(5,1,0)$. The z-coordinates agree because $2 - 2 = -3 + 3 = 0$. The directions are not scalar multiples, so the lines intersect at exactly one point, $(5,1,0)$.

Marking criteria

1. Equates coordinates using distinct parameters (1 mark).
2. Finds t=2 and s=3 (1 mark).
3. Finds the point (5,1,0) (1 mark).
4. Checks the third coordinate as well as the coordinates used to solve (1 mark).
5. Uses the non-parallel directions to conclude that the lines intersect at exactly one point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Equate coordinates. Keep the two parameters independent.

$$1+2t=2+s, \qquad -1+t=4-s, \qquad 2-t=-3+s$$

2 Solve. Add the first two simplified equations.

$$2t-s=1, \quad t+s=5 \Rightarrow 3t=6 \Rightarrow t=2,\ s=3$$

3 Verify and classify. Both lines produce the same three coordinates, and their directions are not parallel.

$$\vec{r}_1(2)=\vec{r}_2(3)=(5,1,0)$$

Question 21 solution

Model response

Subtracting twice the first equation from the second gives $3y+z=9$. Subtracting the first from the third gives $y-2z=-4$. Solving gives $y=2$, $z=3$, then $x=1$. The left sides of the original equations become 6, 3, and 2. Thus the system is consistent with the unique solution $(1,2,3)$, and the three planes meet at this point.

Marking criteria

1. Eliminates x to obtain 3y+z=9 (1 mark).
2. Obtains the independent equation y-2z=-4 (1 mark).
3. Solves y=2 and z=3 (1 mark).
4. Finds x=1 (1 mark).
5. Checks all three original equations and identifies one common point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate x. Use equation 1 to eliminate x from equations 2 and 3.

$$E_2-2E_1:\ -3y-z=-9, \qquad E_3-E_1:\ y-2z=-4$$

2 Solve the reduced system. Substitute z=9-3y into the second reduced equation.

$$y-2(9-3y)=-4 \Rightarrow 7y=14 \Rightarrow y=2,\ z=3,\ x=1$$

3 Verify. Check all three original equations.

$$1+2+3=6, \qquad 2(1)-2+3=3, \qquad 1+2(2)-3=2$$

Question 22 solution

Model response

The normal is $\vec{n} = (2, -1, 2)$, with magnitude 3. The residual at P is $2(4) - 1 + 2(2) - 3 = 8$, so $d = 8/3$. The foot is $H = P - \frac{8}{9}\vec{n} = (20/9, 17/9, 2/9)$. It satisfies $2(20/9) - 17/9 + 2(2/9) = 3$. Also $\overrightarrow{HP} = P - H = \frac{8}{9}(2, -1, 2)$, so the connector is perpendicular to the plane.

Marking criteria

1. Calculates the signed residual as 8 (1 mark).
2. Uses the normal (2,-1,2) with magnitude 3 (1 mark).
3. Obtains the exact distance 8/3 (1 mark).
4. Finds H=(20/9,17/9,2/9) using the normal projection (1 mark).
5. Verifies plane membership and that HP is a scalar multiple of the normal (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Distance. Divide the absolute residual by the normal magnitude.

$$d = \frac{|2(4) - 1 + 2(2) - 3|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{8}{3}$$

2 Perpendicular foot. Remove the normal component from P.

$$H = (4, 1, 2) - \frac{8}{9}(2, -1, 2) = \left(\frac{20}{9}, \frac{17}{9}, \frac{2}{9}\right)$$

3 Verify. Check both membership and the normal direction of the connector.

$$2\left(\frac{20}{9}\right) - \frac{17}{9} + 2\left(\frac{2}{9}\right) = 3, \qquad P - H = \frac{8}{9}\vec{n}$$

Question 23 solution

Model response

The plane normal is n=(1,1,1). The dot product n dot d is 1-1+0=0, so the line direction is parallel to the plane. The anchor (0,0,0) does not satisfy x+y+z=2. Therefore the entire line is outside the plane, the line is parallel to it, and there are no intersection points.

Marking criteria

1. Identifies the plane normal (1 mark).
2. Calculates the normal-direction dot product as zero (1 mark).
3. Explains that this establishes a direction parallel to the plane (1 mark).
4. Checks that the line anchor is outside the plane (1 mark).
5. Concludes that the line is parallel and disjoint, with no intersection (1 mark).

A zero normal-direction dot product gives parallel direction. The anchor test then separates a contained line from a disjoint one.

1 Vector test. Use the plane normal and line direction.

$$(1,1,1)\cdot(1,-1,0)=0$$

2 Anchor test. Evaluate the plane at the origin.

$$0+0+0\neq 2$$

3 Conclusion. The line is parallel and disjoint.

$$L\parallel\Pi,\qquad L\cap\Pi=\varnothing$$

Question 24 solution

Model response

Subtracting the second equation from the first gives $y=1$, and then $x+z=2$. The third equation becomes $2(x+z)+k=5$, or $4+k=5$. For $k=1$, it adds no new restriction, so the solution set is $\vec{r}=(2,1,0)+t(-1,0,1)$, $t\in\mathbb{R}$. For $k\neq 1$, the third equation contradicts the first two, so the system is inconsistent. No value of k produces a unique point.

Marking criteria

1. Uses the first two equations to find y=1 (1 mark).
2. Obtains x+z=2 (1 mark).
3. Derives the compatibility condition k=1 from the third equation (1 mark).
4. Gives the intersection line r=(2,1,0)+t(-1,0,1) when k=1 (1 mark).
5. Concludes that k not equal to 1 is inconsistent and that no k gives a unique point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 First two planes. Subtract, then substitute y=1.

$$2y=2\Rightarrow y=1,\qquad x+z=2$$

2 Third-plane condition. Its restriction is independent of the free line parameter.

$$2(x+z)+ky=5\Rightarrow 4+k=5\Rightarrow k=1$$

3 Classify. Let z=t in the consistent case.

$$k=1:\ \vec{r}=(2,1,0)+t(-1,0,1),\ t\in\mathbb{R};\qquad k\neq 1:\ \varnothing$$

7 Formula reference

Lines

Vector

$\vec{r} = \vec{r}_0 + t\vec{d}$

d ≠ 0

Parametric

$x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$

one shared parameter

Symmetric

$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

separate zero-direction coordinates

R² scalar

$Ax + By + C = 0$

normal (A,B)

Planes

Vector

$\vec{r} = \vec{r}_0 + s\vec{a} + t\vec{b}$

a and b independent

Point-normal

$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$

n perpendicular to plane

Cartesian

$Ax + By + Cz + D = 0$

normal (A,B,C)

Normal from spans

$\vec{n} = \vec{a} \times \vec{b}$

cross product non-zero

Intersections and distance

Line-plane parameter

$t = \frac{k - \vec{n} \cdot \vec{r}_0}{\vec{n} \cdot \vec{d}}$

when n·d ≠ 0

Two-plane direction

$\vec{d} = \vec{n}_1 \times \vec{n}_2$

non-parallel normals

Point-line R²

$d = \frac{|Ax_P + By_P + C|}{\sqrt{A^2 + B^2}}$

scalar line form

Point-line R³

$d = \frac{\|\overrightarrow{AP} \times \vec{d}\|}{\|\vec{d}\|}$

A lies on line

Angles and system checks

Line-plane angle

$\sin\theta = \frac{|\vec{d} \cdot \vec{n}|}{\|\vec{d}\|\|\vec{n}\|}$

acute angle from line to plane

Plane-plane angle

$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\|\vec{n}_2\|}$

acute angle between normals

Point-plane distance

$d = \frac{|Ax_P + By_P + Cz_P + D|}{\sqrt{A^2 + B^2 + C^2}}$

plane in Cartesian form

Consistency

$0 = 0 \Rightarrow$ redundant equation, $\quad 0 = c \neq 0 \Rightarrow$ no solution

a consistent system with a free variable has infinitely many solutions

8 Classroom sources and online resources

The lesson sequence follows Sections 8.1 to 8.6 and 9.1 to 9.6 identified in the supplied classroom materials. Explanations, worked examples, diagrams and assessment solutions in this coursebook are typed course content. The classroom slide page references in the lessons use PDF page numbering.

1. Chapter8 Slides(1).pdf. Classroom slide deck, 58 PDF pages. Section 8.6 is listed in the chapter objectives; the coursebook includes the required plane-sketching lesson.
2. 09 - Chapter 9, Relationships Between Points Lines and Planes(1).pdf. Classroom slide deck, 54 PDF pages.
3. CCT Guidlines July 2026 (1)(1).pdf. MCV4U Course Culminating Task Guidelines. Topic H identifies Lines and Planes. The required submission includes the website link and a PDF version.
4. CCT Rubric 2026.pdf. Assessment criteria for accuracy, sequencing, homework, explanation, organization, delivery, assessments and project conference.

Original teaching video recordings are prepared and uploaded by the group. At least two instructional videos must each run for eight minutes or longer. The website provides access to the recordings as they are published.

Online website

https://mcv4u-lines-and-planes.hozantaxprivate.chatgpt.site